SINAMICS G120

CU230P-2 Control Units

Operating Instructions· Edition 03/2009 · FW 4.2

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Answers for industry.

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SINAMICS G120

Control Units
Control Units, CU230P-2

Operating Instructions

03/2009 Edition

03/2009, FW 4.2
A5E02430659B AA

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

⚠DANGER
indicates that death or severe personal injury **will** result if proper precautions are not taken.

⚠WARNING
indicates that death or severe personal injury **may** result if proper precautions are not taken.

⚠CAUTION
with a safety alert symbol, indicates that minor personal injury can result if proper precautions are not taken.

CAUTION
without a safety alert symbol, indicates that property damage can result if proper precautions are not taken.

NOTICE
indicates that an unintended result or situation can occur if the corresponding information is not taken into account.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The device/system may only be set up and used in conjunction with this documentation. Commissioning and operation of a device/system may only be performed by **qualified personnel**. Within the context of the safety notes in this documentation qualified persons are defined as persons who are authorized to commission, ground and label devices, systems and circuits in accordance with established safety practices and standards.

Proper use of Siemens products

Note the following:

⚠WARNING
Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be adhered to. The information in the relevant documentation must be observed.

Trademarks

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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A5E02430659B AA
Ⓟ 04/2009

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Note

The term "speed" is used in this manual instead of and with the equivalent meaning to the term "frequency". The rare cases in which the term "speed" actually means "speed in rev/min" can be deduced from the context.

The actual speed depends on the number of pole pairs of the connected motor.

1.1 Customer documentation and product support on the Internet

Manual collection and online document support for standard drive units

Collection of manuals for standard drive units

The collection of manuals for standard drive units is an extensive compilation of all documentation for standard drive units which covers the entire range of standard drive products including inverters, motors and geared motors. It can be ordered as a DVD which runs on a special Java-controlled HTML interface.

The order number of the manual collection for standard drive units is:
6SL3298-0CA00-0MG0

Online documentation

You will find the documentation for all standard drive units on the Internet:
(http://support.automation.siemens.com/ww/view/en/4000024)

All documents, including operating instructions and parameter lists, are available for downloading.

Generic station description files (GSD)

Generic station description files (GSD) are used to integrate inverters into higher-order control units, e.g. SIMATIC S7 via **PROFIBUS DP**. You can download the GSD files from the Internet under: (http://support.automation.siemens.com/ww/view/en/23450835)

Electronic Data Sheets (EDS file)

The EDS files are used to integrate an inverter via CAN into the particular system configuration. You can download the EDS files from the Internet under:
(http://support.automation.siemens.com/WW/view/en/35209032)

Manuals for inverters with CU230P-2 Control Unit

The following manuals are available for inverters with CU230P-2 Control Units:

- For the CU230P-2
  - Getting Started
  - Operating Instructions
  - Function Manual
  - List Manual
- For the Power Module
  - Getting Started
  - Hardware Installation Manual

General product information

Comprehensive information and support tools are available at the following addresses for the frequency inverters: (http://www.siemens.com/sinamics-g120)

Application examples

You will find application examples and useful notes on the application of frequency inverters under the following link:
(http://support.automation.siemens.com/WW/view/en/20208582/136000)

2 Safety instructions

Safety instructions

The warnings, safety information and remarks that follow are intended to be used as both safety measures for users and measures that can be put in place to avoid damage to the product or to components of the connected machines. The following section provides an overview of warnings, safety information and remarks that are generally applicable to any work involving the inverter. These are divided into general instructions and instructions for transportation and storage, commissioning, operation, repair, disassembly, and disposal.

Special warnings, information and notes relating to specific activities/work are listed at the beginning of the respective sections of this manual and are repeated or expanded at critical points in these sections.

Please read this information carefully, it has been included for your personal safety and to help you extend the service life of your inverter and other devices connected to it.

General information

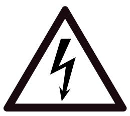
WARNING

These devices are at hazardous voltages and control rotating mechanical parts which can be potentially dangerous. Non-observance of the warnings or non-compliance with the instructions in this manual can lead to danger to life, serious injury or substantial damage to property.

Protection by means of SELV/PELV in case of direct touching is permitted only in areas with equipotential bonding and in dry interior spaces. If these conditions are not fulfilled, other protective measures against electric shock are to be taken, e.g., protective insulation.

Only suitably qualified personnel who have previously familiarized themselves with all the instructions regarding safety, installation, operating and maintenance as set out in this manual are permitted to work on these devices. Successful and safe operation of these devices depends on their proper handling, installation, operation and maintenance.

Since the residual current for this product is greater than 3.5 mA AC, a fixed ground connection is required and the minimum size of the protective conductor must comply with local safety regulations for equipment with a high leakage current.



The power supply, direct-current and motor terminals as well as the brake cables and thermistor cables can carry hazardous voltages even when the inverter is out of service. Once the power supply has been disconnected, wait at least 5 minutes until the device has discharged itself. Only then can you start the installation work. Also carefully ensure that no dangerous voltages are present at the Control Unit.

It is strictly forbidden to isolate the device from the supply on the motor side; isolation from the supply must always be carried out on the supply side of the inverter.

Before the power supply for the inverter is connected, ensure that the terminal box of the motor is connected.

This device is designed to ensure internal motor overload protection in accordance with UL508C. See P0610 and P0335; i^2t has been set to ON as the standard setting.

If an LED or similar indicator does not light up or is not active when a function is switched from ON to OFF, this does not mean that the unit has been switched off or is current-free.

The inverter must always be properly grounded.

The device must be isolated from the power supply before any cables, plugs or wires are connected to the device or altered.

Make sure that the inverter has been configured for the correct supply voltage. It must be ensured that the inverter is not connected to a higher supply voltage.

Static discharge on surfaces or at interfaces which are not generally accessible (e.g. terminals or connector pins) can cause malfunctions or defects. The ESD protective measures should therefore be observed during work with inverters or inverter components.

The general and regional installation and safety regulations for working on equipment carrying hazardous voltages (e.g. EN 50178) as well as the relevant stipulations regarding the correct use of tools and personal protective equipment (PPE) are especially to be observed.

⚠ CAUTION
Children and other unauthorized persons must be forbidden access to the devices! It is only permissible to use these devices for the purpose indicated by the manufacturer. Unauthorized changes and the use of spare parts and accessories which are not sold or recommended by the manufacturer of the device can lead to fires, electric shock and injuries.

NOTICE
This manual is to be kept somewhere close to the devices and must be easily accessible for all users. If measurements or tests have to be carried out on the live device, the stipulations of safety regulation BGV A2 are to be complied with, especially § 8 "Permissible deviations during work on live parts". Suitable electronic tools are to be used. Before installation and commissioning, please read this safety information and the warnings carefully as well as the warning signs fitted to the devices. It must be ensured that the warning signs are always legible; any signs that are damaged or missing are to be replaced.

Transport and storage

⚠ WARNING
Appropriate transport and storage, as well as careful operation and maintenance are essential for the correct and safe operation of the devices.

⚠ CAUTION
During transport and storage, the device must be protected against mechanical shocks and vibrations. It is important to protect the unit against water (rain) and against excessively high/excessively low temperatures.

Commissioning

⚠ WARNING
Work performed on the devices by unqualified personnel or failure to observe warnings can cause serious bodily injury or considerable property damage. Work on the devices must only be carried out by qualified personnel who are familiar with the design, installation, commissioning and operation of the devices.

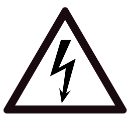
CAUTION

Cable connection

The control cables must be routed separately from the supply cables. If this is not possible, we recommend the use of shielded cables. The connection must be performed according to the instructions in the "Installation" section of this manual, in order to prevent inductive and capacitive interference from affecting smooth operation of the plant.

During operation

WARNING



SINAMICS G120 inverters operate with high voltages.

When operating electrical devices, hazardous voltages in certain parts of the devices cannot be avoided. Therefore, emergency off equipment in accordance with EN 60204, IEC 204 (VDE 0113) must be functional in all operating modes of the control equipment. Switching off an emergency off device must not lead to an uncontrolled or undefined restart of the plant.

Certain parameter settings (for example, the functions for automatic restart) can cause the SINAMICS G120 inverter to automatically restart following a power supply failure.

For the areas of the control equipment in which faults can cause considerable damage to property or even serious injury, additional external precautions must be taken, or devices installed to ensure safe operation even if a fault occurs (for example, independent limit switches, mechanical interlocks, etc.).

The motor parameters must be precisely configured so that the motor overload protection functions correctly.

The Control Unit can also have dangerous voltages even if the Power Module power supply is shut down. Therefore carefully ensure that the 230 V AC power supply voltage for the relay outputs on the Control Unit are also shut down before you start any work on the unit.

This device is designed to ensure internal motor overload protection in accordance with UL508C.

You must only use Control Units with fail-safe functions as an "emergency off device" (see EN 60204, Section 9.2.5.4).

The use of mobile radio equipment with a transmitter power of > 1 W close to the units (< 1.5 m) can have a negative impact on the function of the units.

Repairs

⚠ WARNING
Only Siemens customer service, repair centers that have been authorized by Siemens or authorized personnel may repair drive equipment. Al of the persons involved must have in-depth knowledge of all of the warnings and operating instructions as listed in this Manual. All damaged parts or components must be replaced only using parts and components that are listed in the relevant spare parts list. Before opening the device, disconnect the supply voltage in order to gain access to the interior components.

Disassembly and disposal

CAUTION
The packaging of the inverter is re-usable. Store the packaging carefully for re-use. The packaging can be easily dismantled into its individual parts using removable screw and snap catches. The parts can be recycled, disposed of in accordance with local regulations, or returned to the manufacturer.

3 Description

CU230 field of application

The CU230P-2 is a Control Unit that has been optimized for pumps and fans. It can be operated with all power units of the PM240 and PM250 series.

The following versions of CU230P-2 are available:

- CU230P-2 HVAC with RS485 interface for USS and Modbus RTU
- CU230P-2 CAN with CANopen interface
- CU230P-2 DP with PROFIBUS DP interface

They can be commissioned either using the STARTER commissioning software or using the optional IOP operator panel (Intelligent Operator Panel).

You can save all the settings you enter during commissioning and operation to a memory card.

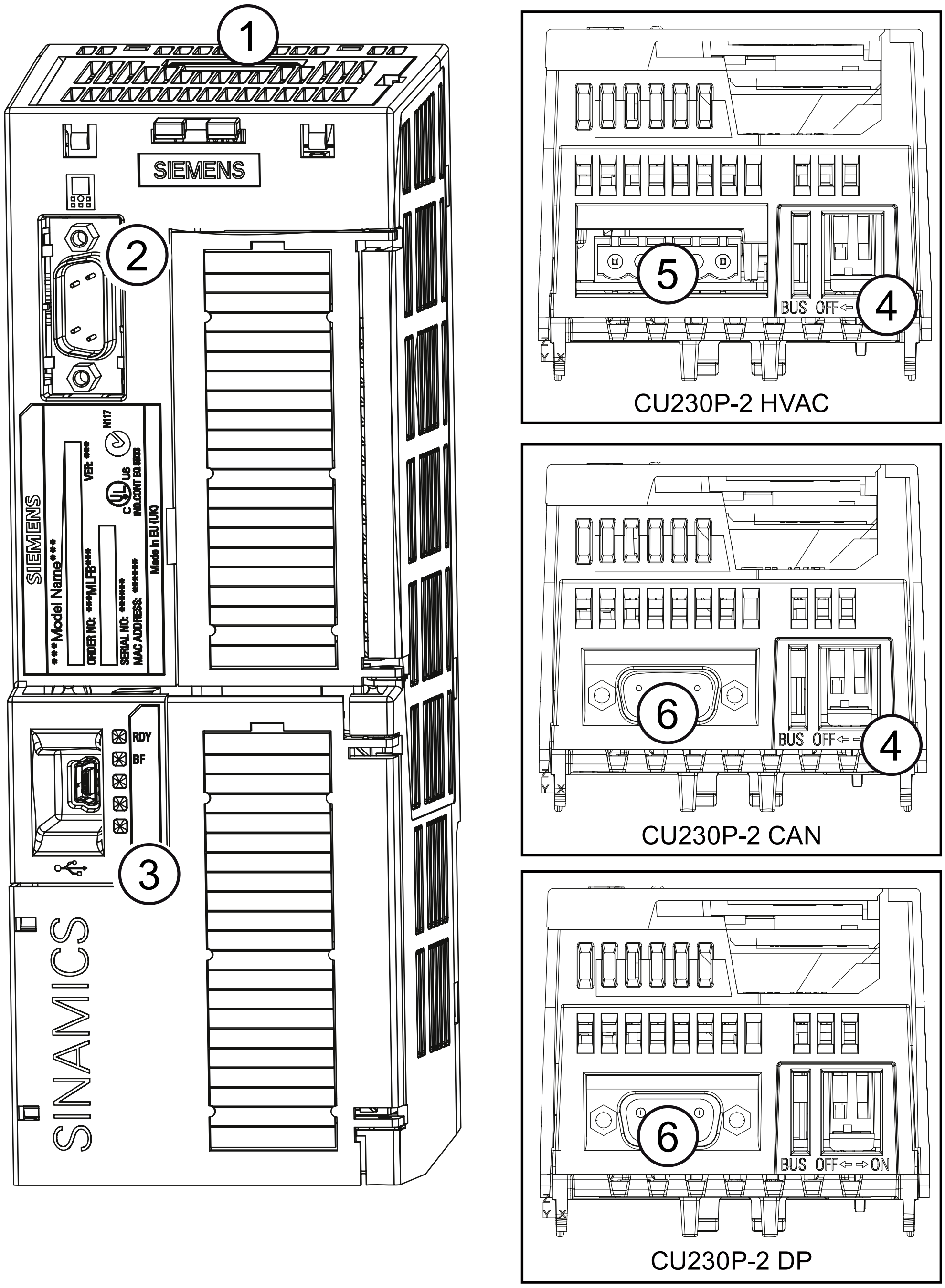
3.1 Interfaces, connectors, switches, control terminals and LEDs of the Control Unit

Overview

The Control Unit is equipped with

- Interfaces
  - the commissioning tools, Starter or IOP,
  - the input and output signals,
  - for bus communication with controls
- DIP switches
  - for configuring the motor temperature sensor,
  - for configuring analog inputs
  - for setting bus addresses (CU230P-2 DP / CU230P-2 CAN)
- Memory card slot

The factory setting for the command and setpoint sources depends on the Control Unit version.



① MMC slot
② IOP- / IOP handheld interface
③ USB interface for STARTER
④ Bus terminating switch
⑤ RS485 connector
⑥ SUB-D connector

Figure 3-1 Control Unit CU230P-2, doors closed

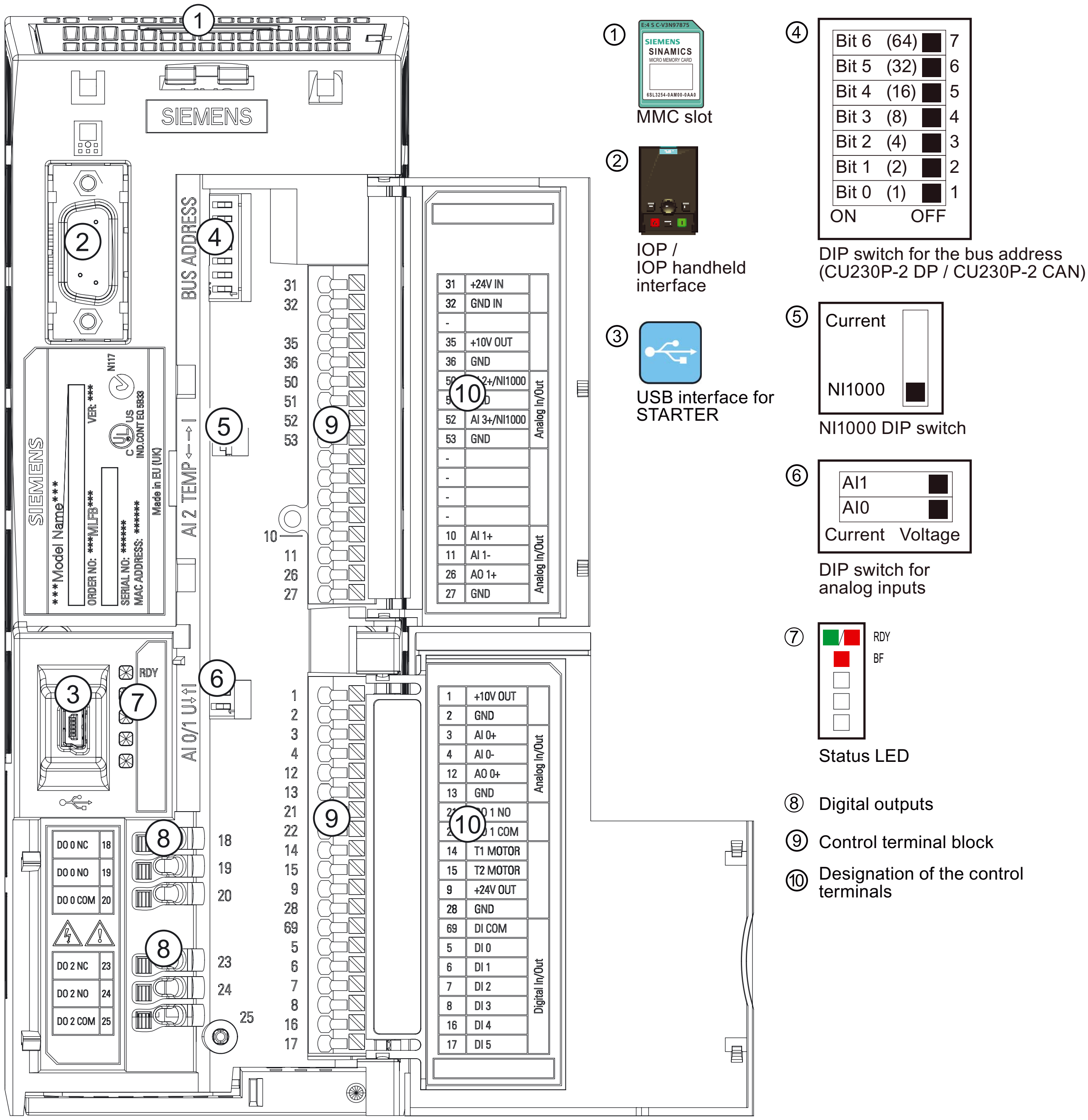


Figure 3-2 Control Unit CU230P-2, doors open

4 Installation

4.1 Procedure for installing the inverter

Prerequisites for installing the inverter

Check that the following prerequisites are fulfilled before you install the inverter:

- Are the components, tools and small parts required for installation available?
- Are the ambient conditions permissible? See: Technical data (Page 165)
- Have the cables and conductors been routed in accordance with the applicable regulations?
- Are the minimum distances from other equipment complied with? (Cooling sufficient?)

Installation sequence

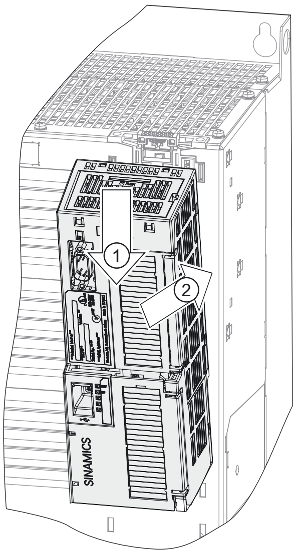
1. Install the Power Module (for detailed instructions, see the Hardware Installation Manual for the Power Module)
   - Remove the terminal covers - where applicable
   - Connect motor cable and power cable
   - Terminate the shield over a large area, if necessary using a shield connection set
   - Refit the terminal covers
2. Mount the Control Unit
   - Open the terminal covers of the Control Unit
   - Connect the control lines to the terminals
   - Terminate the shield over a large area, if necessary using a shield connection set
   - Close the terminal covers again
3. Control Unit – for operation in a higher-level control – connect to the fieldbus
   - For PROFIBUS DP and CANopen connect it via the 9-pin sub D connector
   - For RS485, connect it via the two-part bus connector
4. To commission the drive unit, either plug-in the operator control/display instrument (operator panel) or connect the inverter to the PC using the PC connection kit.

Installation has now been completed and you can begin commissioning.

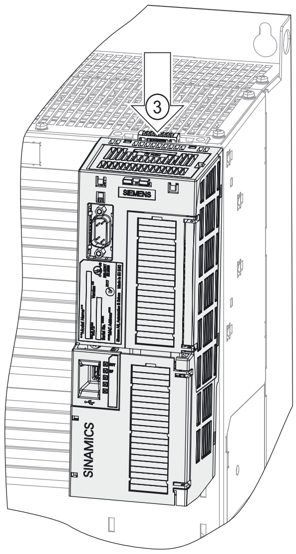
4.2 Snap the Control Unit onto the Power Module

Snap the Control Unit onto the Power Module

Plug the Control Unit onto the Power Module as shown. All the necessary electrical connections are made between the two components.



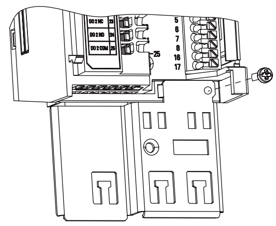
If you want to remove the Control Unit, press the unlatching button on the Power Module ③. You can then tilt the Control Unit forwards and remove it from the Power Module.



Screening kit for the Control Unit

A screening kit is available for the Control Unit. This allows easy bonding of control cable shields for connection to the Control Unit potential. The set must be attached to the Control Unit by means of an M3 screw (supplied with the kit).

Regardless of whether or not a shield connection kit is used, the Control Unit is connected to the reference potential of the Power Module as soon as it is snapped onto the module.



Retain the shield connection kit

⚠ WARNING
If 230 V relays are controlled from the digital outputs DO0 and DO2, then these must always be switched into a no-voltage condition before the Control Unit is removed from the Power Module. The reason for this is that when the Control Unit is removed, the connection to the protective conductor is interrupted.

4.3 Control terminals of the Control Units

Arrangement and functions of the control terminals on the CU230P-2 Control Units

All Control Units have the same control terminals. The factory presettings for certain terminals differ, however, depending on the CU variant.

Color coding of the terminals

- Gray: Analog inputs and outputs
- Light gray: Digital inputs and outputs
- White: Other inputs and outputs

Terminals	Terminal designations		Explanations
31	+24V IN		External 24 V
32	GND		Reference potential for terminal 31
-	-		---
35	+10V OUT		10V voltage output
36	GND		Reference potential for terminal 35
50	NI1000/AI2+		Input for NI1000 or AI2+ temperature sensor
51	GND		Reference potential for terminal 50
52	NI1000/AI3+		Input for NI1000 or AI3+ temperature sensor
53	GND		Reference potential for terminal 52
-	-		---
-	-		---
-	-		---
-	-		---
10	AI1+	Analogue In/Out	Analog input 1, positive*
11	AI1-	Analogue In/Out	Analog input 1, negative*
26	AO1+	Analogue In/Out	Analog output 1, positive (0 V … 10 V, 0 mA … 20 mA, max. 500 Ω)
27	AO GND	Analogue In/Out	Reference potential for terminal 26
1	+10V OUT		10 V output without electrical isolation, max. 10 mA
2	GND		Reference potential for terminal 1
3	AI 0+	Analogue In/Out	Analog input 0, positive*
4	AI 0-	Analogue In/Out	Analog input 0, negative*
12	AO 0+	Analogue In/Out	Analog output 0, positive (0 V … 10 V, 0 mA … 20 mA, max. 500 Ω)
13	AO GND	Analogue In/Out	Reference potential for terminal 12
21	DO 1 NO		Digital output 1, NO contact, 0.5 A, 30 V DC
22	DO 1 COM		Digital output 1, common contact, 0.5 A, 30 V DC
14	PTC+		Motor temperature sensor (PTC, KTY84-130 or bimetal NC contact)
15	PTC-		Motor temperature sensor (PTC, KTY84-130 or bimetal NC contact)
9	+24V OUT		Isolated 24 V output, max. 100 mA
28	GND		Reference potential for terminal 9
69	DI COM	Digital In/Out	DI COM
5	DI 0	Digital In/Out	Digital input 0, isolated
6	DI 1	Digital In/Out	Digital input 1, isolated
7	DI 2	Digital In/Out	Digital input 2, isolated
8	DI 3	Digital In/Out	Digital input 3, isolated, FF setpoint 0
16	DI 4	Digital In/Out	Digital input 4, isolated, FF setpoint 1
17	DI 5	Digital In/Out	Digital input 5, isolated, FF setpoint 2
18	DO 0 NC		Digital output 0, NC contact; 5 A, 30 V DC / 2 A, 230 V AC
19	DO 0 NO		Digital output 0, NO contact; 5 A, 30 V DC / 2 A, 230 V AC
20	DO 0 COM		Digital output 0, common contact; 5 A, 30 V DC / 2 A, 230 V AC
23	DO 2 NC		Digital output 2, NC contact; 5 A, 30 V DC / 2 A, 230 V AC
24	DO 2 NO		Digital output 2, NO contact; 5 A, 30 V DC / 2 A, 230 V AC
25	DO 2 COM		Digital output 2, common contact; 5 A, 30 V DC / 2 A, 230 V AC

* The analog inputs AI0 and AI1 can be used as additional digital inputs.

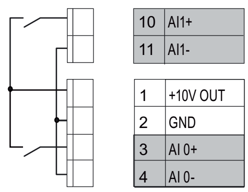


Figure 4-1 Overview of the control terminals

Connecting to the control cables

Note

The control cables must be routed separately from the supply cables. If this is not possible, we recommend the use of shielded cables.

When you have plugged the Control Unit into the Power Module, you must connect the control lines according to your connection diagram. Open up the upper and lower hinged flaps towards the right. The control terminals are now accessible. They are spring-loaded terminals.

Solid or flexible cables are permitted as control lines. Wire end ferrules must not be used for the spring-loaded terminals.

The permissible cable cross section ranges from 0.5 mm^2 to 1.5 mm^2. In the case of full wiring, we recommend the use of cables with a cross section of 1mm^2, cable type:32/0.2, e.g. Farnell, order number 1173019.

Route the control lines in the Control Unit as shown in the figure. If you are using shielded lines, you can either connect the shield to the optional screening kit of the Control Unit or the Power Module or - as close to the inverter as possible - directly to the control cabinet over a large area.

Labels are attached to the inside of the hinged flaps that show the terminal numbers and short designations. Blank labels are attached to the outside of the flaps for the customer's own inscription.

General rules for cable routing

Note the following restrictions with respect to wiring:

- Cables must not be kinked, stretched, crushed or twisted.
- The rules relating to electromagnetic compatibility must be complied with.

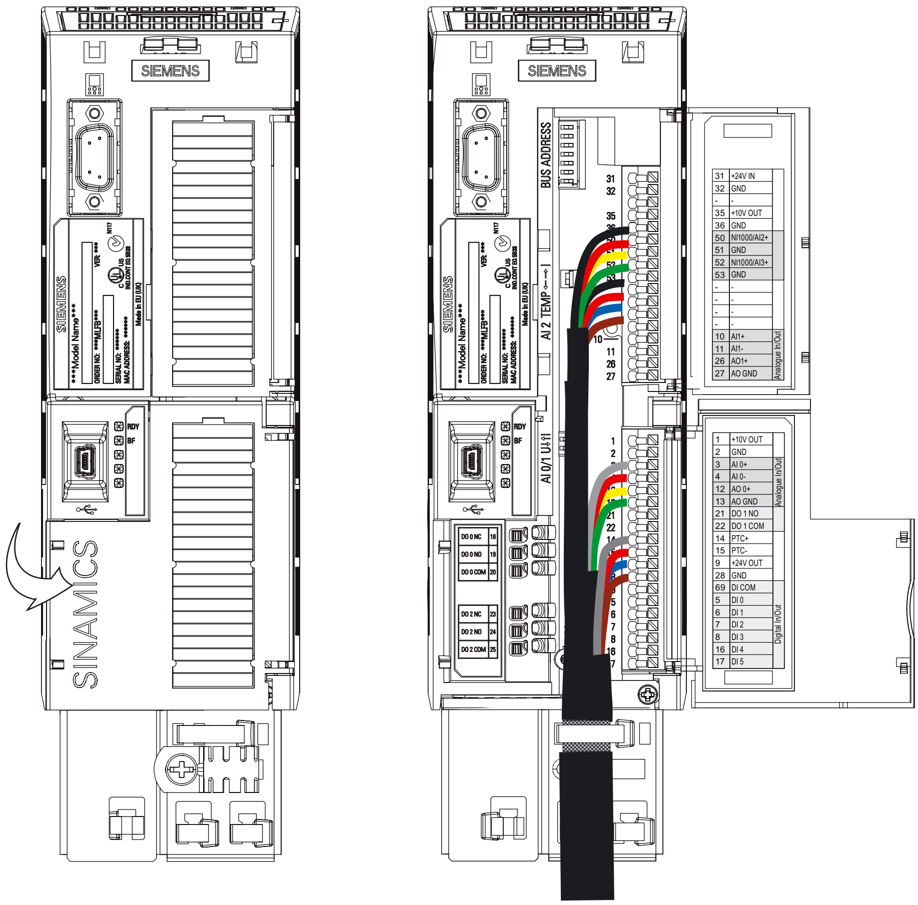


Figure 4-2 Accessing the control terminals and connecting the control lines

⚠ WARNING
Relays with 230 V AC can be switched via DO0 and DO2. These terminals can also be at a certain voltage if the power unit power supply is shut down.

Note

If 230 V is switched via the relay outputs DO0 or DO2, then the associated protective conductor must be connected through the Power Module. You will find details in the installation instructions for the Power Module.

When removing the Control Unit from the Power Unit, under all circumstances it must be ensured that the 230 V connections are in a no-voltage condition.

Wiring options for a CU230P-2 HVAC or CU230P-2 CAN with the factory settings

The following diagram shows the wiring options for a CU230P-2 HVAC or CU230P-2 CAN with the factory settings. The factory settings can be reparameterized to suit the plant conditions. For further details, see Section "Quick commissioning (Page 46) ", or the List Manual.

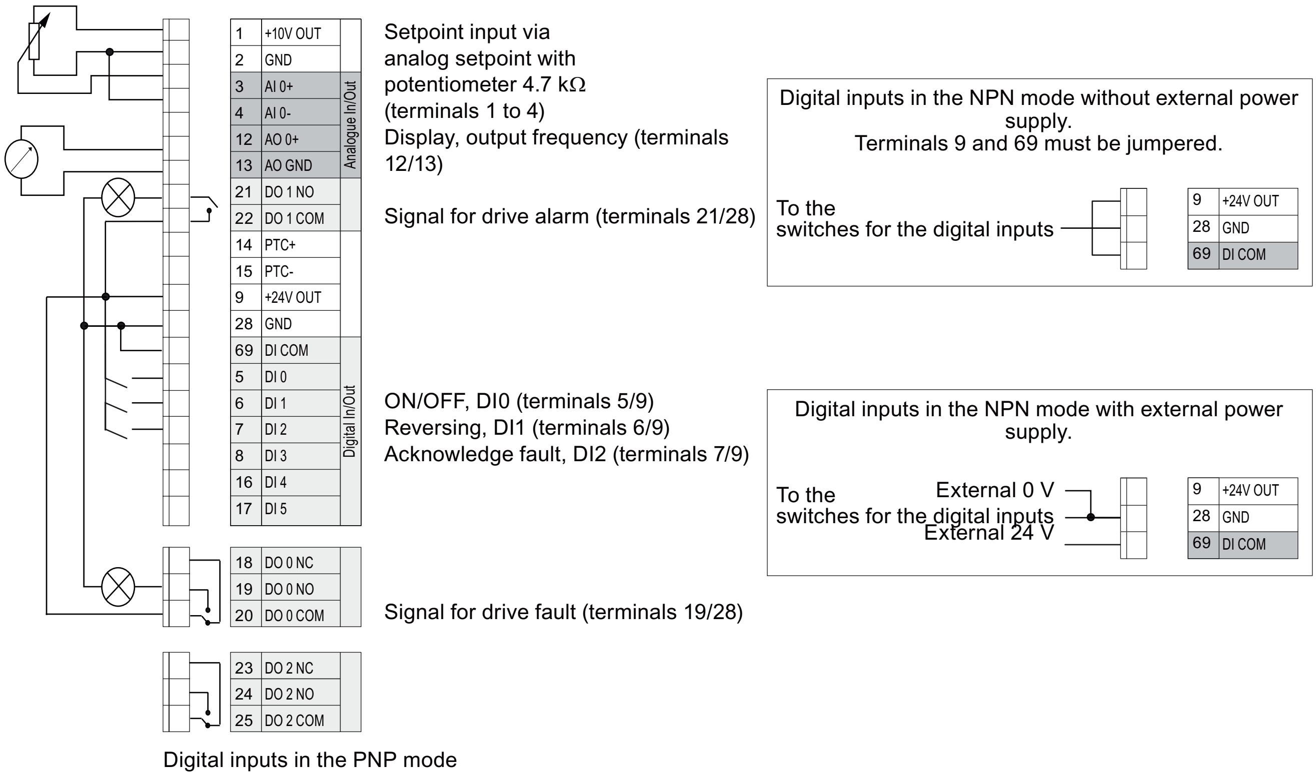


Figure 4-3 Wiring possibilities for a CU230P-2 HVAC with factory settings

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Note

In the NPN mode, a ground fault between the customer contact and digital input may undesirably control the drive input.

4.4 Control-Unit-dependent factory settings

Factory settings for the command and setpoint sources

The factory setting for the command and setpoint sources depends on the type of Control Unit.

CU230P-2 HVAC	Commands and setpoints via terminals
CU230P-2 DP	Commands and setpoints from the controller over the fieldbus interface
CU230P-2 CAN	Commands and setpoints via terminals

The command source can be changed using p0700, and the setpoint source can be changed using p1000.

Table 4- 1 Commands and their sources as set in the factory

Command	CU230P-2 HVAC (p0700 = 2) CU230P-2 CAN (p0700 = 2)		CU230P-2 DP (p0700 = 6)	
	Source	Parameter	Source	Parameter
ON/OFF1	DI0	p0701 = 1	STW1, bit 0	r2090.00
Direction of rotation reversal	DI1	p0702 = 12	STW1, bit 11	r2090.11
Error acknowledgment	DI2	p0703 = 9	STW1, bit 7	r2090.07

Table 4- 2 Setpoint source as set in the factory

Setpoint	CU230P-2 HVAC (p1000= 2) CU230P-2 CAN (p1000 = 2)		CU230P-2 DP (p1000 = 6)	
	Source	Parameter	Source	Parameter
Main setpoint	AI0+ / AI0-	p1070 = 755	PZD from the fieldbus	r2050.1 [Hex]

Factory setting for fault messages and alarms

Faults and alarms are output via parameters r0052.3 and r0052.7. In the factory setting, these parameters are linked to digital outputs DO0 and DO1.

Table 4- 3 Output of faults and alarms with the factory setting

Message	Source	Parameter
Fault	DO0	p0730 = 52.3
Alarm	DO1	p0731 = 52.7

4.5 Subsequent parameterization when switching over p0700

The subsequently listed parameters are re-assigned when switching over parameter p0700. The individual parameters can be subsequently set back to all of the values permissible in the particular parameters.

Note

Please note that for a CU230P-2 DP, the switchover from p0700 = 6 to p0700 = 2 is only possible if p0922 was previously set to 999.

p0700 = 2: => Commands via terminals

p0700 = 6: => Commands via fieldbus

Parameter		Pre-assignment when switching over p0700 =	
		2	6
p0701[x]	Digital input 0	1	0
p0702[x]	Digital input 1	12	0
p0703[X]	Digital input 2	9	9
p0704[x]	Digital input 3	15	15
p0705[X]	Digital input 4	16	16
p0706[x]	Digital input 5	17	17
p0712[x]	Digital input 6	0	0
p0713[x]	Digital input 7	0	0
p0840[x]	BI: ON/OFF1	r0722.0	r2090.0
p0844[x]	BI: 1. OFF2	1	r2090.1
p0848[x]	BI: 1. OFF3	1	r2090.2
p0852[x]	BI: Enable operation	1	r2090.3
p1140[x]	BI: Ramp-function generator	1	r2090.4
p1141[x]	BI: Start ramp-function generator	1	r2090.5
p1142[x]	BI: Enable speed setpoint	1	r2090.6
p2103[x]	BI: 1. Acknowledge faults	0	r2090.7
p0854[x]	BI: Master control by PLC	1	r2090.10
p1113[x]	BI: Setpoint inversion	r0722.1	r2090.11
p1035[x]	BI: Motorized potentiometer, setpoint, raise	0	r2090.13
p1036[x]	BI: Motorized potentiometer, setpoint, lower	0	r2090.14
p2080.0	BI: Binector-connector converter, status word 1	---	p0899.0
p2080.1	BI: Binector-connector converter, status word 1	---	p0899.1
p2080.2	BI: Binector-connector converter, status word 1	---	p0899.2
p2080.3	BI: Binector-connector converter, status word 1	---	p2139.3
p2080.4	BI: Motorized potentiometer, setpoint, lower	---	p0899.4
p2080.5	BI: Binector-connector converter, status word 1	---	p0899.5
p2080.6	BI: Binector-connector converter, status word 1	---	p0899.6
p2080.7	BI: Binector-connector converter, status word 1	---	p2139.7
p2080.8	BI: Binector-connector converter, status word 1	---	p2197.7

Parameter		Pre-assignment when switching over p0700 =	
		2	6
p2080.9	BI: Binector-connector converter, status word 1	---	p0899.9
p2080.10	BI: Motorized potentiometer, setpoint, lower	---	p2199.1
p2080.11	BI: Binector-connector converter, status word 1	---	p1407.7
p2080.12	BI: Binector-connector converter, status word 1	---	p0899.12
p2080.13	BI: Binector-connector converter, status word 1	---	p2135.14
p2080.14	BI: Binector-connector converter, status word 1	---	p2197.3
p2080.15	BI: Binector-connector converter, status word 1	---	p2135.15
p2051.0	CI: PROFIdrive PZD send word	---	p2089.0
p2088	Binector-connector converter, invert status word	---	43008
p2104[x]	BI: 2. Acknowledge faults	r0722.2	---
p1020[x]	BI: Fixed speed setpoint selection bit 0	r0722.3	r0722.3
p1021[x]	BI: Fixed speed setpoint selection bit 1	r0722.4	r0722.4
p1022[x]	BI: Fixed speed setpoint selection bit 2	r0722.5	r0722.5
p2220[x]	BI: Technology controller fixed value selection bit 0	r0722.3	r0722.3
p2221[x]	BI: Technology controller fixed value selection bit 1	r0722.4	r0722.4
p2222[x]	BI: Technology controller fixed value selection bit 2	r0722.5	r0722.5

"---" no change

4.6 Connecting the CU230P-2 HVAC via the RS485 interface

Description

This section describes how the frequency inverter is physically connected to a serial bus system over the RS485 interface. The communication settings are described in the sections "Communication over USS (Page 89)" and "Communication over Modbus RTU (Page 103)".

The Control Unit has a two-part terminal strip underneath the Control Unit which allows the inverter to be integrated into a serial bus system over the RS485 interface. This connector has short-circuit-proof, floating pins. You will find the terminal assignments in the following table.

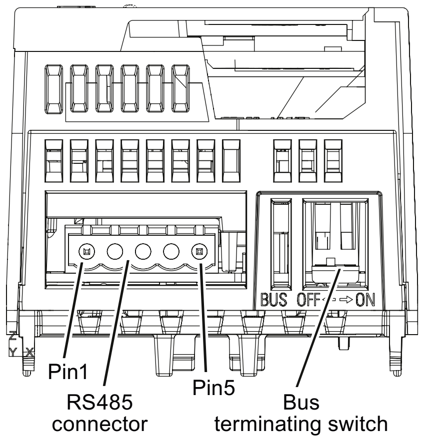


Table 4- 4 Assignments for the terminal strip of the RS485 interface

Contact	Designation	Description
1	0 V	Reference potential
2	RS485P	Receive and send signal (+)
3	RS485N	Receive and send signal (-)
4	Shield	Cable shield
5	---	---

Communication settings

Parameter	USS		Modbus RTU	
	P value	Baud rate	P value	Baud rate
p2020 baud rate	4 5 6 7 8 9 10 11 12 13	2400 4800 9600 19200 38400 57600 76800 93750 115200 187500	 5 6 7	 4800 9600 19200
p2021 address	0 … 30, factory setting = 8, maximum 31 Slaves		1 … 247, factory setting = 7, maximum 247 Slaves	
Maximum cable length 1200 m (3281 ft)				

General specifications and requirements for fault-free communication

NOTICE
When the bus is operating, the first and last bus station must be continuously connected to the supply.

Note

Communication with the controller, even when the supply voltage on the Power Module is switched off

You will have to supply the Control Unit with 24 V DC on terminals 31 and 32 if you require communication to take place with the controller when the line voltage is switched off.

For the first and last stations, you must connect the bus terminating resistor using the DIP switches to the right of the RS485 terminal strip.

You can disconnect one or more slaves from the bus (by unplugging the bus connector) without interrupting the communication for the other stations, but not the first or last.

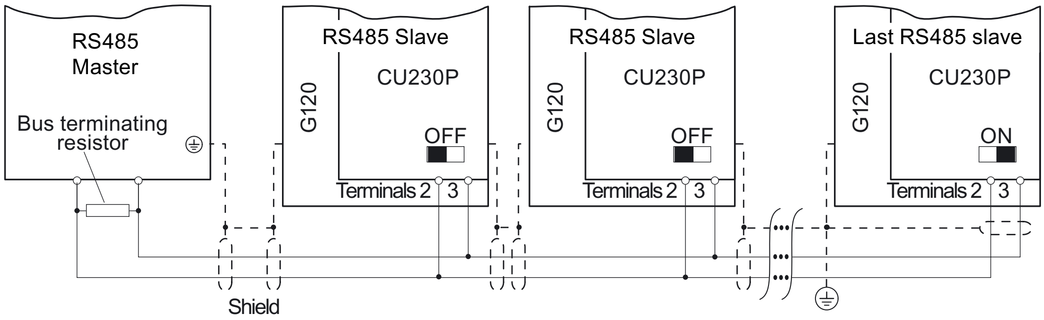


Figure 4-4 Communication network via RS485

Note

For Modbus RTU

Two 100 kΩ resistors are provided to polarize the receive and send cables.

4.7 Connecting CU230P-2 DP to the PROFIBUS DP network

Description

This section describes how the frequency inverter is physically connected to a PROFIBUS DP bus system. The communication settings are described in section "Communication over PROFIBUS DP (Page 68)".

The Control Unit CU230P-2 DP has a 9-pin sub D connector to the PROFIBUS standard underneath the Control Unit for integrating the inverter into the PROFIBUS DP fieldbus system. This connector has short-circuit-proof, floating pins. You will find the pin assignments in the following table.

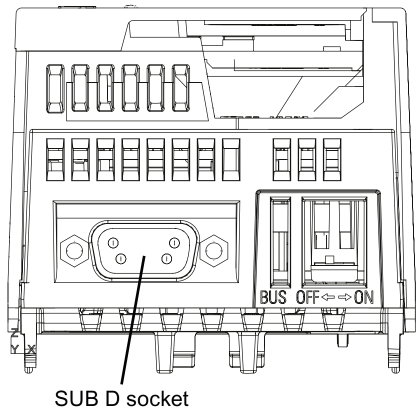
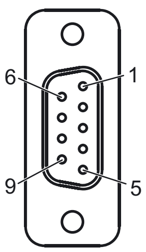


Table 4- 5 Pin assignment of the 9-pin sub D connection (socket) on the Control Unit

	Contact	Designation	Description
	1	PE / shield	Ground connection
	2	---	---
	3	DPB	Data P receive/transmit (B/B’)
	4	RTS	Control signal
	5	0 V	Reference potential for PROFIBUS data (C/C’)
	6	5V	Supply voltage plus
	7	---	---
	8	DPA	Data N receive/transmit (A/A’)
	9	---	---
	Enclosure	PE / shield	Cable shield

PROFIBUS DP cable connector and permissible cable lengths

For connecting the SINAMICS G120 frequency inverter, we recommend the following PROFIBUS DP cable connectors:

- 6GK1 500-0FC00
- 6GK1 500-0EA02

They are equipped with a switch through which the bus terminating resistor can be connected.

Permissible cable lengths

The permissible cable lengths are dependent on the baud rate and the PROFIBUS cable. For further information, please referto:
(http://www.automation.siemens.com/net/html_76/support/printkatalog.htm)

General specifications and requirements for fault-free communication

You can integrate up to 126 stations into a PROFIBUS DP network. These must be subdivided into segments of up to 32 stations. You must activate the bus terminating resistor for the first and last stations of each segment.

You can disconnect one or more slaves from the bus (by unplugging the bus connector) without interrupting the communication for the other stations. In this case you must ensure that the first and last stations are connected with the bus terminating resistor.

Note

Communication with the controller, even when the line voltage is switched off

You will have to supply the Control Unit with 24 V DC on terminals 31 and 32 if you require communication to take place with the controller when the line voltage is switched off.

Setting the PROFIBUS DP address

Before the PROFIBUS DP interface is used, the address of the node point (inverter) must be set.

The following methods are available for setting the PROFIBUS DP address:

- Using the address switch on the Control Unit
- Using parameter p0918

Note

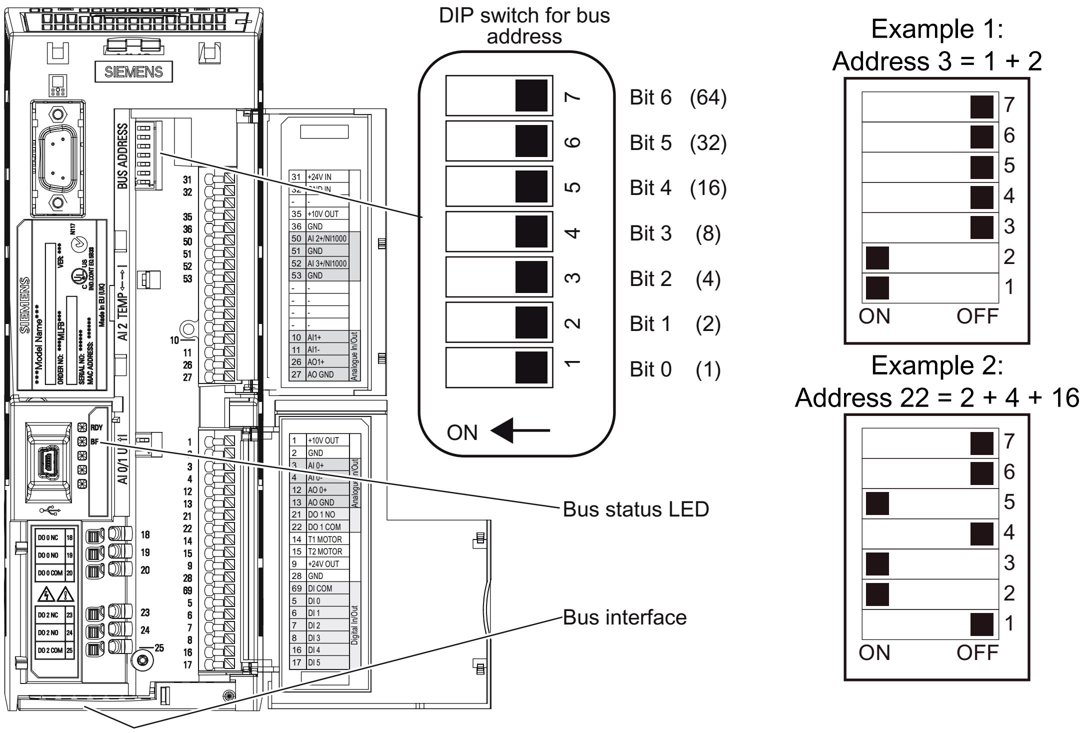
Important notes for setting the PROFIBUS address

The address setting on the DIP switch takes priority over the p0918 settings.

The PROFIBUS DP address can only be set using p0918 when the address 0 is set on the DIP switches of the Control Unit (factory setting). When the address switch is set to a value ≠ 0, the setting in p0918 is ignored.

Valid address range for Siemens controllers: 1 … 125.

Arrangement of the DIP switches on the Control Unit and address examples



Note

A newly set PROFIBUS DP address will only come into effect after switching off and on again. It is particularly important that any external 24 V supply is switched off.

4.8 Connecting CU230P-2 CAN to CAN bus

Description

This section describes how the frequency inverter is physically connected to the CAN bus. The communication settings are described in section "Communication settings for CANopen (Page 113)".

The Control Unit CU230P-2 CAN has a 9-pin sub D connector underneath the Control Unit for integrating the inverter into the CANopen fieldbus system. This connector has short-circuit-proof, floating pins. You will find the pin assignments in the following table.

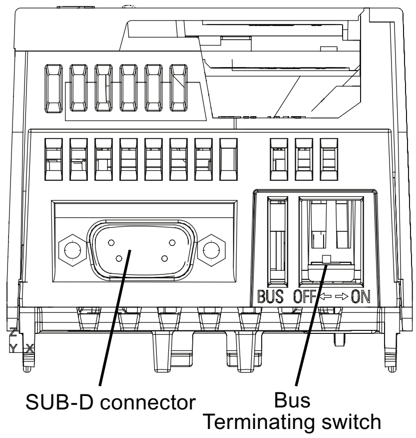
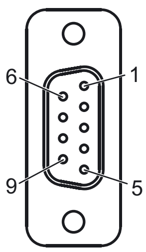


Table 4- 6 Pin assignment of the 9-pin sub D connection (connector) on the Control Unit

	Contact	Designation	Description
	1	Reserved	---
	2	CAN_L	CAN signal (dominant low)
	3	CAN_GND	CAN ground
	4	reserved	---
	5	(CAN _SHLD)	Optional shield
	6	(GND)	Optional CAN ground
	7	CAN_H	CAN signal (dominant high)
	8	reserved	---
	9	ISO	5 V

CANopen cable connector

For setting up a CANopen network, you can use cable for serial 9-pin connections with sub D connectors.

General specifications for CANopen and requirements for fault-free communication

You can integrate up to 126 stations into a CANopen network. These must be subdivided into segments of up to 32 stations. For the first and last station of each segment, you have to activate the bus terminating resistor via the DIP switch on the right next to the SUB-D socket.

You can disconnect one or more slaves from the bus (by unplugging the bus connector) without interrupting the communication for the other stations, but not the first or last.

NOTICE
When the bus is operating, the first and last bus station must be continuously connected to the supply.

Note

Communication with the controller, even when the line voltage is switched off

You will have to supply the Control Unit with 24 V DC on terminals 31 and 32 if you require communication to take place with the controller when the line voltage is switched off.

Setting the CANopen address

In order to integrate an inverter into a CANopen network, you need to set the address. The following possibilities exist:

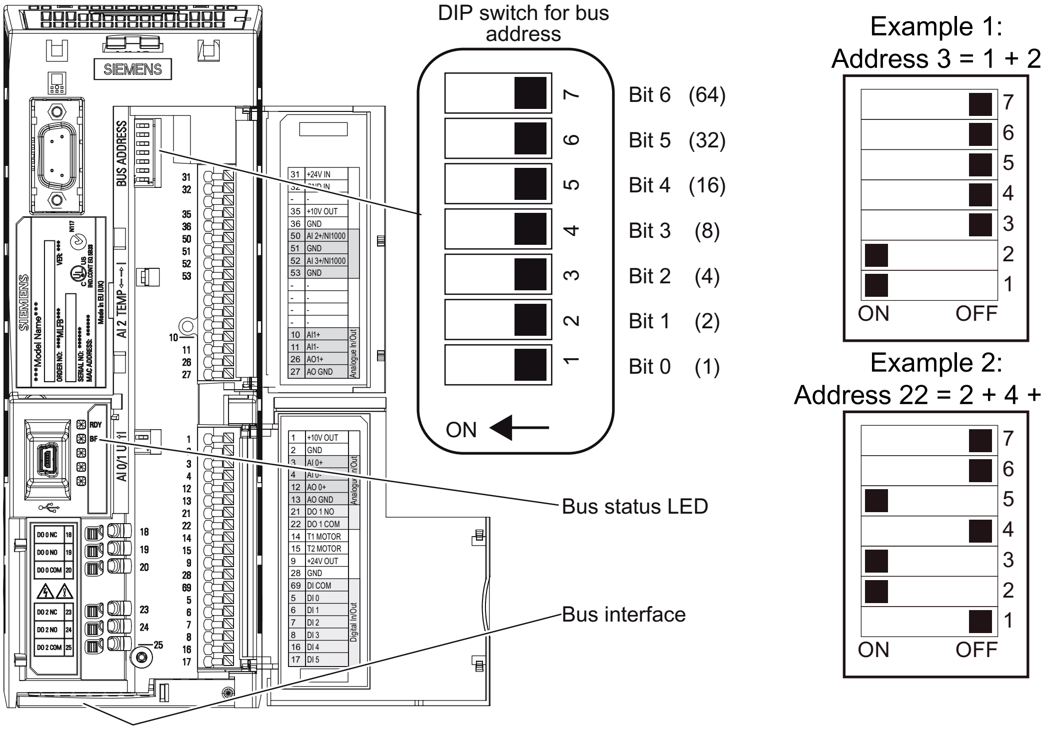
- Using the address switch on the Control Unit
- Via parameter p8620

Note

Important note for setting the CANopen address

The address can only be set via p8620 if the address 0 or 127 is set on the DIP switches of the Control Unit. If the address switches are set to a value ≠ 0 or 127, this address is always active and p8620 is read-only.

Arrangement of the DIP switches on the Control Unit and address examples



CAUTION
A newly set CANopen address will only come into effect after switching off and on again. It is particularly important that any external 24 V supply is switched off.

Further information is available on the CANopen Internet pages (http://www.can-cia.org/index.php?id=47).

4.9 Connecting up the CU230P-2 for commissioning via STARTER on a PC

Description

This section describes how to connect up the inverter via the USB or RS232 interface for commissioning with STARTER. The RS232 and USB interfaces are both located on the front of the Control Unit.

You cannot use both interfaces at the same time.

Connection

- RS232 interface: Null modem cable
- USB interface: Connector type Mini-B 5-pin

Communication settings

- p2010 baud rate: Possible setting values for 4 ≙ 2400 … 12 ≙ 115200 (factory setting = 12), see parameter list for details
- p2011 commissioning address: Possible values 0 … 31, (factory setting = 2)

Points to note about commissioning via the USB interface

In order to commission the inverter via the USB interface, the "Sinamics G120 Control Unit (USB Bridge) driver" must be installed in the address area COM1 to COM8 on the computer.

Installing the USB driver for the Control Unit

Extract the file "Sinamics G120 USB Bridge Driver.zip" to a local directory and start the installation process by double clicking on "SinamicsG120VCPInstaller.exe" and then follow the instructions. If you have already installed the driver, a corresponding message will be displayed.

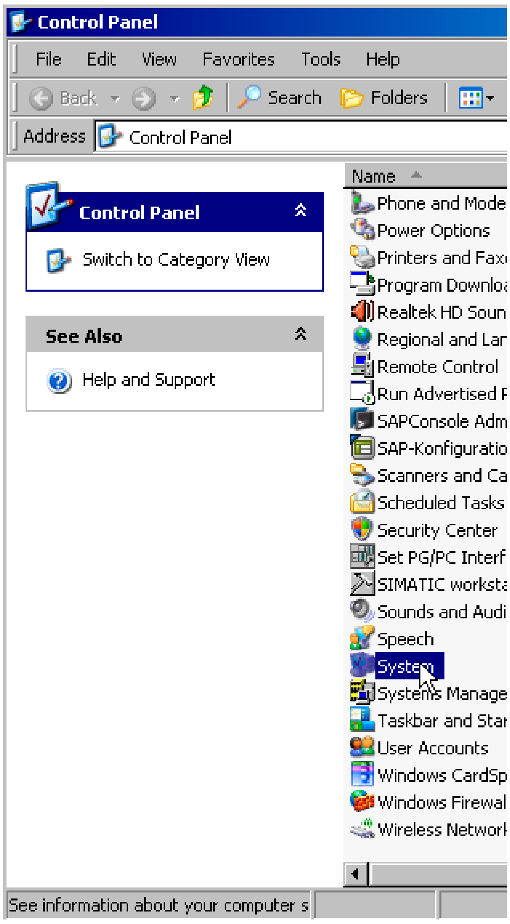
The USB interface is now set up and you can start commissioning the inverter.

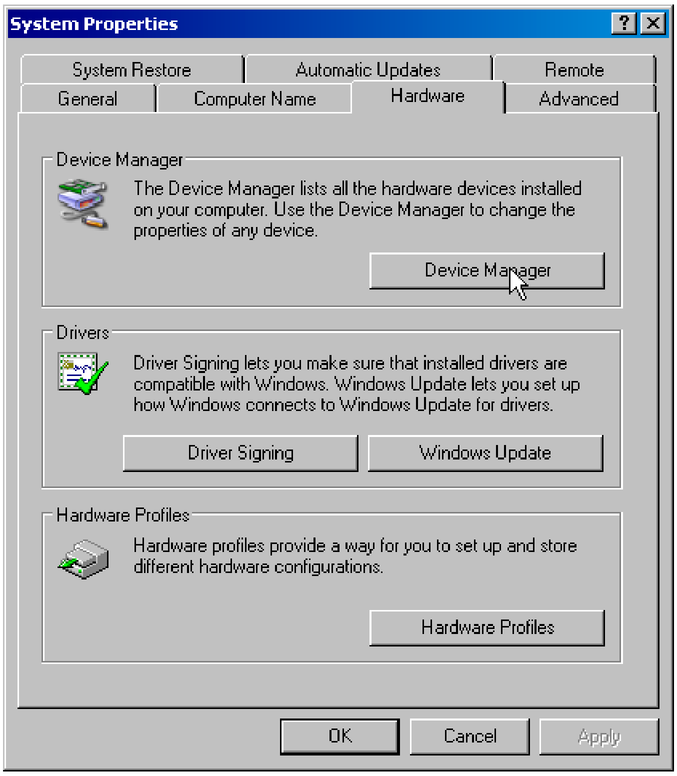
If the connection cannot find the inverter …

If your computer cannot find the inverter even though the inverter is switched on and the USB cable is correctly inserted, it means that the USB driver has been installed on a COM interface with a number higher than 8. In this case, you must manually assign an address from the range COM1 … COM8 to the inverter.

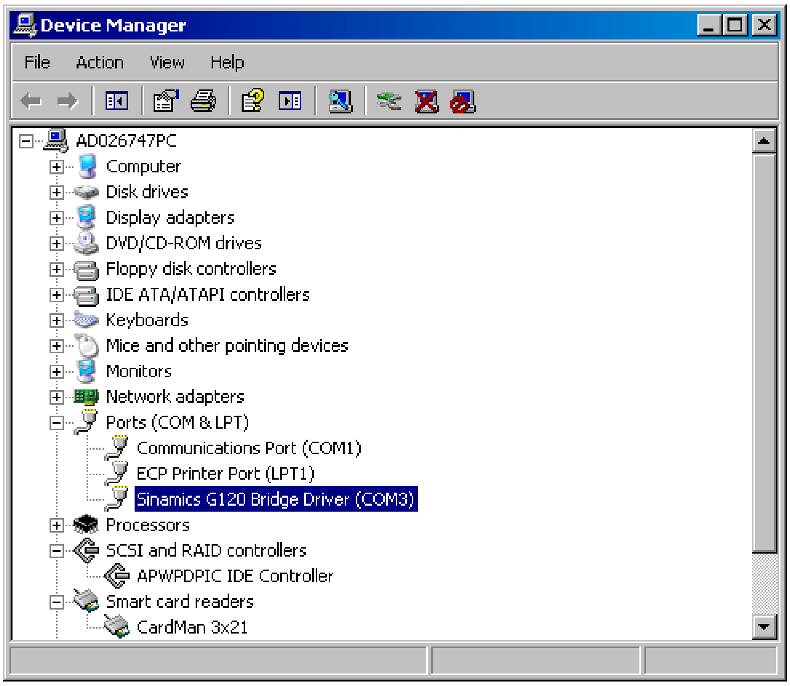
To do so, proceed as follows:

- Click on "Control Panel/Performance and Maintenance (displayed only in the Category View/System"), click on "System", select tab "Hardware" and then button "Device Manager".


Control Panel
File Edit View Favorites Tools Help
Back Search Folders
Address Control Panel
Control Panel
Switch to Category View
See Also
Help and Support
Name
Phone and Mode
Power Options
Printers and Fax
Program Downlo
Realtek HD Soun
Regional and Lar
Remote Control
Run Advertised P
SAPConsole Adm
SAP-Konfiguratio
Scanners and Ca
Scheduled Tasks
Security Center
Set PG/PC Interf
SIMATIC worksta
Sounds and Audi
Speech
System
Systems Manage
Taskbar and Star
User Accounts
Windows CardSp
Windows Firewal
Wireless Networl
See information about your computer s


System Properties
System Restore
Automatic Updates
Remote
General
Computer Name
Hardware
Advanced
Device Manager
The Device Manager lists all the hardware devices installed on your computer. Use the Device Manager to change the properties of any device.
Device Manager
Drivers
Driver Signing lets you make sure that installed drivers are compatible with Windows. Windows Update lets you set up how Windows connects to Windows Update for drivers.
Driver Signing
Windows Update
Hardware Profiles
Hardware profiles provide a way for you to set up and store different hardware configurations.
Hardware Profiles
OK
Cancel
Apply

- In this screen (Device Manager), click on "Ports(COM & LPT)" where you will see the item "Sinamics G120 USB Bridge Driver" and behind it in parenthesis the COM address. Double click on "Sinamics G120 USB Bridge Driver" to open the properties box for this port and look under "Port Settings" / "Advanced" to find the COM port number. Use the pull-down menu to select an address in the range COM1 to COM8 and exit the dialog box by selecting OK.



- If you find that all addresses in the 1 to 8 range are already assigned, you can assign – as described above – a higher address to one of these devices. Now assign the free address to the "Sinamics G120 USB Bridge Driver".
- Leave all other settings unchanged.

Commissioning 5

5.1 Installation checklist

Installation checklist

Before you apply the voltage, check whether you have performed the following steps and whether the necessary conditions are met.

	Check item	✓
1	The ambient conditions specified for the motor and inverter have been complied with.	
2	The inverter and motor are firmly installed.	
3	The inverter and motor are correctly installed and the cooling measures are sufficient.	
4	The motor and the associated application are ready for operation, i.e. in a safe state, the motor can rotate.	
5	The inverter is correctly grounded.	
6	The supply voltage corresponds to the input voltage of the inverter.	
7	The line fuses meet requirements and are correctly installed.	
9	The motor and line connections are established and tightened according to guidelines.	
10	The direction of rotation of the motor has been checked without it coupled to a machine - an incorrect direction of rotation can result in serious material damage.	
11	The motor cable is routed separately away from other cables (if required, shielded motor and control cables should be used).	
12	Control connections must be established corresponding to the regulations.	
13	Tools or other objects that could cause damage to the system have been removed.	
14	The inverter is the only current source for the motor.	

Overview

The frequency inverter is commissioned by changing the inverter settings (parameterization) to ensure that the motor is optimally controlled for the drive task (open-loop and closed-loop control).

Types of commissioning

Both individual and standard commissioning are possible.

Individual commissioning

In the case of individual commissioning, the parameter settings are directly changed in the inverter using either the IOP or STARTER. It can be subdivided as follows:

- Basic commissioning
  - Quick commissioning
  - Calculating motor data
- Optimizing the closed-loop speed control
- Commissioning the application

Standard commissioning

With standard commissioning, a complete data set is written to an inverter. This can be used

- When several inverters have to be set up for the same drive task, e.g. when an inverter is installed in a machine that is manufactured as a series product.
- When a defective component (Power Module or Control Unit) is replaced.

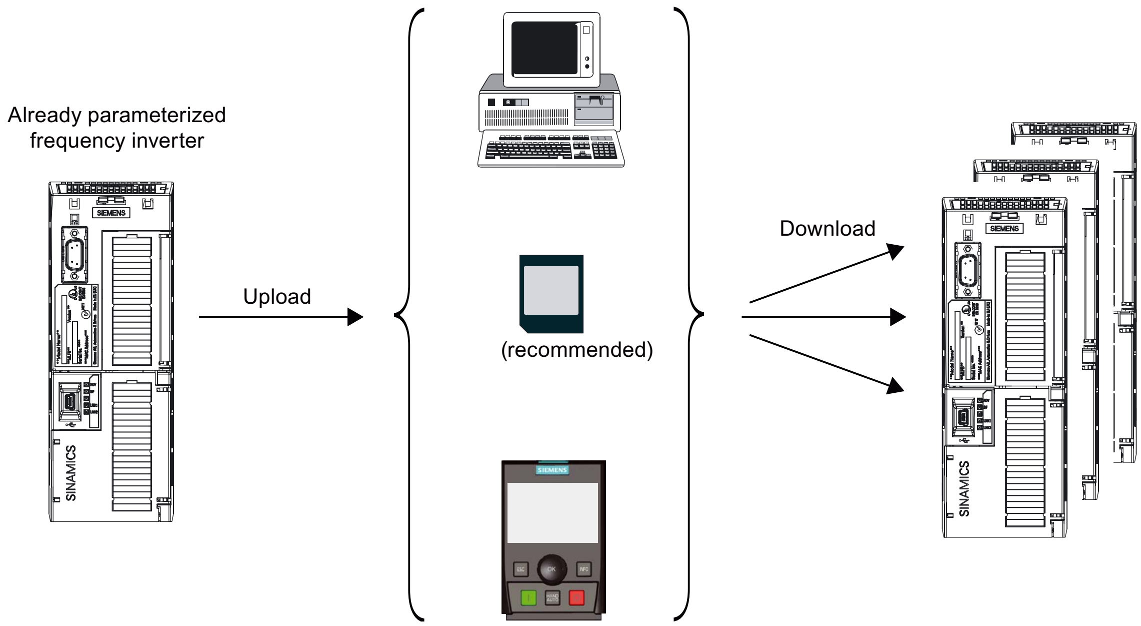


Figure 5-1 Overview of standard commissioning

5.2 Initial switch on

Description

Following switch on, the frequency inverter takes a few seconds to run up. It then loads the parameter settings and values from one of the following sources to its work memory:

• Standard data set on the memory card	Whenever a memory card with a standard data set is inserted
• Non-volatile memory of the Control Unit	If a memory card is not inserted or a standard data set is not available on the memory card

Note

Runup is indicated by the LED as well as by parameter r3996 (1 to 100). When runup has completed, r3996 is set to 0 again and further inputs are possible.

First switch on after installation

When you switch on the inverter for the first time following installation, commissioning is necessary so that the factory settings can be adapted to the drive task. Either standard commissioning (Page 63) or individual commissioning (Page 46) can be performed.

If you have an inverter that has already been parameterized for the drive task, you can perform commissioning using steps 2 to 5 of standard commissioning.

If an inverter with the relevant settings is not available, you must adapt the inverter to the drive task using the procedure for individual commissioning. All the changes that you make will be saved in the non-volatile memory of the Control Unit. We recommend that the data is always saved in addition as a standard data set on a memory card, because you can then perform commissioning with Steps 4 and 5 of standard commissioning following failure of a Power Module or a Control Unit.

⚠WARNING

When commissioning, it must be carefully ensured that dangerous loads do not represent any danger for man or machine.

5.3 Individual commissioning

The individual commissioning process is divided into the following steps:

- Basic commissioning
  - Quick commissioning
  - Possibilities for calculating motor data
  - Possibilities for identifying motor data
- Application commissioning
  - Motor temperature inputs
  - Setting the command sources
  - Setting the frequency setpoint source
  - Assigning the digital outputs
  - Setting the ramp times
- Completing the individual commissioning process

Note

During commissioning, the settings are made in the factory setting of the inverter in the volatile memory (RAM). They are only transferred into the non-volatile memory (EEPROM) after commissioning has been completed using p0971 = 1 (data is transferred from the RAM into the EEPROM).

This means that if you switch-off the inverter while commissioning, the changes that you made will be lost.

5.3.1 Basic commissioning

5.3.1.1 Quick commissioning

Overview

During quick commissioning, the settings are made that are required by every inverter regardless of the closed-loop control mode and communications interface. The following section lists the individual parameters with their most commonly used settings. For further details, refer to the List Manual.

Note

Most parameters are provided with more setting options than specified here. You will find the full range of possible settings in the parameter list.

NOTICE
When commissioning is interrupted by a supply failure, you must repeat the process after a "Reset to factory settings (Page 64)".

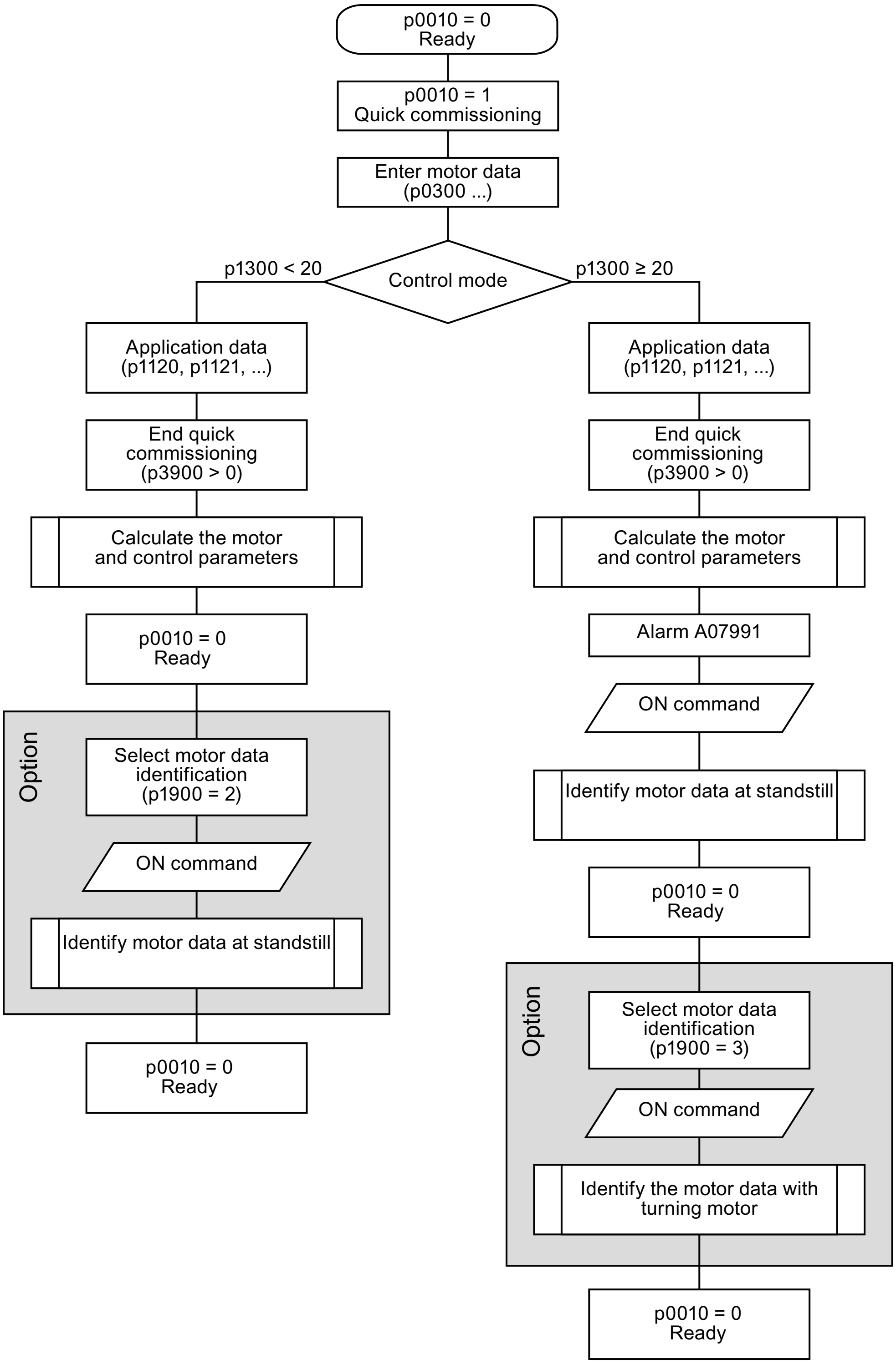


Figure 5-2 Overview of commissioning options

Table 5- 1 Setting parameter filters

Parameter	Description
p0010 = 1	Commissioning parameter filter 0: No filter (factory setting) 1: Quick commissioning 3: Motor commissioning 30: Factory setting

Motor type plate data

The inverter is set up at the factory for applications with a 4-pole, 3-phase induction motor that is compatible with the performance data of the inverter.

When commissioning is performed using STARTER and a Siemens motor is used, it is sufficient to enter the order number - otherwise the motor data must be entered. The entered rating plate data must be in accordance with the connection mode of the motor (star/delta), i.e. in the case of delta connection of the motor, the delta rating plate data must be entered.

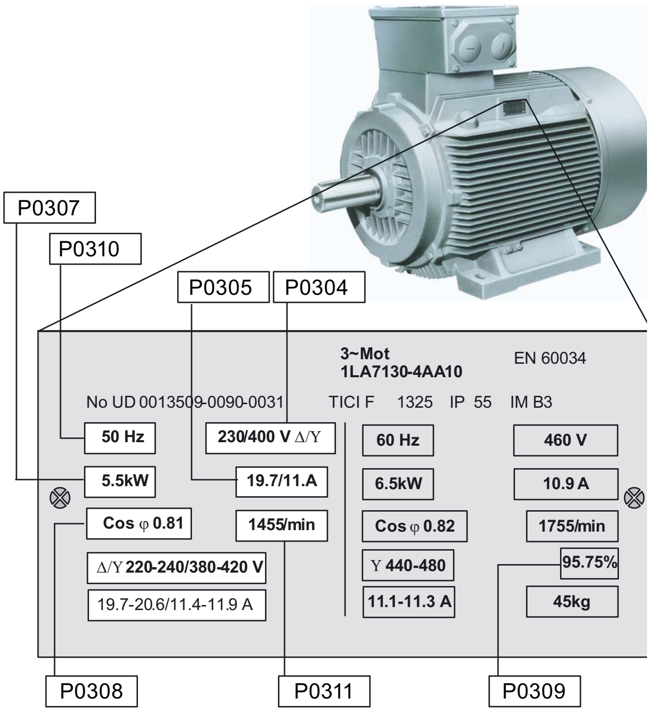


Table 5- 2 Parameters for the motor data

Parameters	Description
p0100 = 0	Europe/North America (enter motor frequency) 0: Europe [kW], frequency 50 Hz as default (factory setting) 1: North America [hp], frequency, as default: 60 Hz (as of FW4.3) 2: North America [kW], frequency, as default: 60 Hz (as of FW4.3)
p0230 = 0	Setting the output filter 0: No filter (factory setting)
p0233 = ...	Inductance of the output filter 0.000 (factory setting)
p0234 = ...	Capacitance of the output filter 0.000 (factory setting)
p0300 = ...	Selecting the motor type (factory setting = 0) 1 = Rotary induction motor (factory setting) 2 = Rotary synchronous motor 17 = 1LA7 standard induction motor: This means that parameters p0335, p0626, p0627 and p0628 are preset as a function of p0307 and p0311 to the optimum values for a 1LA7 motor.

Parameters	Description
p0304 = …	**Rated motor voltage** (enter value in accordance with the motor rating plate in volts) The entered motor rating plate data must be compatible with the motor circuit (star/delta). This means that when the motor is connected in delta, the rating plate data for delta connection must be entered.
p0305 = …	**Rated motor current** Enter the value from the motor rating plate in amperes
p0307 = …	**Rated motor power** Enter the value from the motor rating plate in kW or hp If p0100 = 0 or 2, => Power setting is in kW If p0100 = 1 => Power setting is in hp.
P0308 = …	**Power factor** Enter the value from the rating plate. If you do not enter a value (0.000), the power factor will be computed internally. Display via r0332.
p0310 = …	**Rated motor frequency** Enter the value in Hz in accordance with the data of the motor rating plate When the parameter is changed, the number of pole pairs of the motor is automatically recalculated.
p0311 = …	**Rated motor speed** Enter the value in rpm in accordance with the data of the motor rating plate If p0311 = 0, the value will be calculated internally (necessary in the case of vector control). To function correctly, the slip compensation for V/f control requires the rated motor speed.

Table 5- 3 Selection options for command and frequency setpoint sources

Parameter	Description
p0700 = …	**Select the command source** 2: Terminal (for CU230P-2 HVAC and CU230P-2 CAN cannot be changed) 6: Fieldbus
p1000 = …	**Selection of the frequency setpoint** 0: No main setpoint 1: MOP setpoint 2: Analog setpoint (factory setting for CU230P-2 HVAC and CU230P-2 CAN) 3: Fixed frequency 6: Fieldbus (factory setting for CU230P-2 DP), not available for CU230P-2 CAN 7: Analog setpoint 2

Note

p0922 = 999 must be set for PROFIBUS Control Units (CU230P-2 DP) to allow the command and setpoint sources to be changed.

Table 5- 4 Parameters that must be set in every application

Parameter	Description
p1080 = …	**Minimum frequency** Enter the lowest motor frequency in Hz up to which the motor operates independently of the frequency setpoint. This value applies for both directions of rotation.
p1082 = …	**Maximum frequency** Enter the highest frequency in Hz to which the motor is limited independently of the frequency setpoint. This value applies for both directions of rotation.
p1120 = …	**Ramp-up time** Enter the time in seconds during which the motor should accelerate from standstill up to the maximum motor frequency p1082. A very short ramp-up time can cause shutdown on overcurrent (F30001).
p1121 = …	**Ramp-down time** Enter the time in seconds during which the motor should be braked from the maximum frequency p1082 down to a standstill. A very short ramp-down time can cause shutdown on overcurrent (F30001) or overvoltage (F30002).

Table 5- 5 Possible settings for V/f control

Parameter	Description
P1300 = …	**Control mode** 0: V/f with linear characteristic (factory setting) 1: V/f with FCC 2: V/f with parabolic characteristic 3: V/f with programmable characteristic 4: V/f with linear characteristic and ECO mode 5. V/f multi-point drive 6. V/f multi-point drive and FCC 7: V/f with parabolic characteristic and ECO mode 19. V/f with independent voltage setpoint

Table 5- 6 Possible settings with vector control

Parameter	Description
p1300 = …	**Control mode** 20: Sensorless vector control 22: Sensorless torque vector control

Table 5- 7 End quick commissioning

Parameters	Description
p3900 = …	**End quick commissioning (QC)** 0: Abort quick commissioning (no motor calculations, changes are not accepted) 1: Motor calculation and reset of parameters that were not changed during quick commissioning. 2: Motor calculation and reset of all I/O settings to factory values. 3: Motor calculation only - no other parameters are reset. Note: For p3900 = 1, 2 or 3, p0340 is set to 1 and the value of p1082 is written to p2000. The relevant motor data is calculated.

Note

While the calculations for ending quick commissioning with p3900 > 0 are in progress, you will not be able to input any data. This state is indicated by the LED as well as by parameter r3996 (1 to 100). When the calculations are completed, parameters p0010, p3900 and r3996 are reset to 0 and other inputs are possible. The remaining procedure is dependent on the control mode.

Ending quick commissioning with V/f control

With V/f control mode, quick commissioning with motor data calculation is ended according to the setting of p3900 and the inverter switches to the ready for operation state (p0010 = 0). In order to improve the control quality, however, it is possible to activate identification of the motor and control parameters at standstill (p1900 = 1 or 2).

Ending quick commissioning with vector control

In vector control mode, identification of the motor and control parameters at standstill - in addition to calculation of the motor data - is an integral component of the quick commissioning process. After the motor data have been calculated, the inverter therefore issues alarm A07991, signaling to the user that identification of the motor and control parameters at standstill will be launched with the next ON command. To improve the control quality, identification of the motor and control parameters while the motor is turning (p1900 = 3) can subsequently be activated as an additional process.

5.3.1.2 Options for calculating the motor and closed-loop control data

Overview

The following options are available for calculating the motor and closed-loop control data:

- Automatic calculation at the end of quick commissioning with p3900 > 0 (corresponds to p00340 = 1)
- Automatic calculation at the end of motor data identification (corresponds to p00340 = 3)
- Manual calculation with p0340.
  This method is especially appropriate in cases where the equivalent circuit diagram data and moments of inertia are already known and have been entered in the relevant parameters (p0341 … p0369).

You can select the following settings for the calculation using p0340:

0	No calculation (factory setting)
1	Complete calculation (includes calculations 2,3,4,5)
2	Calculation of the equivalent circuit data
3	Calculation of closed-loop control data (V/f and vector) (includes calculations 4, 5)
4	Calculation of the controller data
5	Calculation of technological limits and thresholds

Note

For complete parameterization (p0340=1), not only the motor and controller parameters are preset but also the parameters which are related to the rated motor data (e.g. torque limit values and reference quantities for interface signals). The List Manual contains a full list of all parameters depending on p0340.

Manual calculation of the motor and closed-loop control data using p0340

Standard calculation

If you do not know the equivalent circuit diagram data, you will need to calculate all the motor parameters. Start the calculation process as indicated below:

Parameter	Description
p0340 = 1	**Calculation of motor parameters** 1: Complete parameterization (factory setting)

Calculation of the motor data if the equivalent circuit diagram data are available

If you already have the data for parameters p0341 to p0369, you can use them as a basis for calculating the motor data. In this case, enter the data as indicated in the table below and start the motor parameterization process by setting p0340 = 3.

Parameter	Description
p0341 = …	**Motor moment of inertia [kg*m^2]**
p0342 = …	**Moment of inertia ratio total/motor**
p0344 = …	**Motor weight** (enter in kg)
p0350 = …	**Stator resistance (phase)** (enter in Ω). Stator resistance in Ω of the connected motor (phase). This parameter value does not contain the line resistance.
p0352 = …	**Line resistance**Line resistance in Ω between the inverter and motor for one phase
p0354 = …	**Rotor resistance** (enter in Ω) Specifies the rotor resistance in the motor equivalent circuit (value per phase).
p0356 = …	**Stator stray inductance** (enter in mH) Specifies the stray inductance of the stator in the motor equivalent circuit (value per phase).
p0358 = …	**Rotor stray inductance** (enter in mH) Specifies the stray inductance of the rotor in the motor equivalent circuit (value per phase).
p0360 = …	**Main inductance** (enter in mH) Specifies the main inductance (magnetizing inductance) of the motor equivalent circuit (value per phase).
p0340 = 3	**Calculation of motor parameters** Calculation of closed-loop control data (V/f and vector)

5.3.1.3 Possibilities for identifying motor data

Overview

Identification of the motor and closed-loop control data is selected with p1900 and can take two different forms:

- Identification of motor data at standstill - for V/f and vector control modes
- Identification of motor data while motor is turning - for vector control mode only

Preparation for motor data identification

Parameter	Description
p0625 = …	**Ambient motor temperature** (enter in °C). Ambient temperature at the moment of motor data identification (factory setting: 20 °C). The difference between the motor temperature and the ambient motor temperature (p0625) must lie within a tolerance range of approx. ± 5 K. It may be necessary to allow the motor to cool down.
p0010 = 0	**Commissioning parameter filter** Check that p0010 = 0 (Ready)
p1900 = …	**Select method of motor data identification** 0: Blocked (factory setting) 1: Identification of the motor data at standstill with rotating motor (2,3) 2: Identification of all parameters at standstill 3: Identification of motor data while motor is turning When the motor data identification process is complete, p1900 is reset to 0.

Note

For V/f control (p1300 < 20) the setting p1900 = 1 has the same meaning as p1900 = 2, i.e. identification of all parameters at standstill.

p1900 = 3 is not possible for V/f control.

Identification of motor data at standstill (p1900 = 2)

This function can be selected in inverter state "Ready" (p0010 = 0), e.g. following quick commissioning. It is activated with the next ON command.

Note

Since this function is essential for improving the control quality in vector control mode, it is always activated with the first ON command following quick commissioning for this mode.

Action	Description
ON command	**Start identification of motor data at standstill** As soon as p1900 = 2 is set, alarm A7991 is generated. This alarm notifies the user that motor data identification will start on the next ON command. When the ON command is issued, current flows through the motor and the rotor aligns itself.
OFF1	Once the motor data identification run is completed (A7991 canceled and p1900 = 0), the inverter must be switched to a defined state with an OFF1. The motor data identification process is then finished.

Identification of motor data while motor is turning - for vector control mode only (p1900 = 3)

If a high inverter control quality is required, then the motor data identification can be carried out with the motor rotating. It can be selected in inverter state "Ready" and will then start with the next ON command.

⚠ WARNING

When carrying out the motor data identification with rotating motor, it is not possible to predict in which direction, how long and with which speed the motor will rotate. This is the reason that when this measurement is carried, the load must be uncoupled and secured.

Action	Description
ON command	**Start rotating measurement** As soon as p1900 = 3 is set, alarm A7980 is generated. This alarm notifies the user that rotating measurement will start on the next ON command.
OFF1	Once the rotating measurement is completed (A7980 canceled and p1900 = 0), the inverter must be switched to a defined state with an OFF1. The motor data identification process is then finished.

5.3.2 Application commissioning

Overview

After commissioning the motor and inverter combination using quick commissioning, you then need to make the application-specific settings, for example:

- Calculating or recording the motor temperature
- Setting the command and setpoint sources
- Assigning the I/O functions
- Adapting the ramp-up and ramp-down times

Check that p0010 is set to 1 for these settings.

Further application-specific functions and details of the individual functions can be found in the Function Manual.

General settings

Parameter	Description
p0014 = …	Memory type P0014 = 0: Parameter changes only in the RAM (volatile) - factory setting P0014 = 1: Parameter changes in the RAM are buffered in a non-volatile fashion in the ROM P0014 = 2: Deletes parameter changes that have still not been saved in the ROM and then sets p0014 to 0.
p0210 = …	Supply voltage (enter the voltage in V). You use this parameter to enter the actual supply voltage to which the inverter is connected. Required for vector control when p1254 = 0 (factory setting: p1254 = 1). Required for V/f control when p1294 = 0 (factory setting: p1294 = 0).
p0290 = 2	Inverter overload behavior Specifies the inverter's response to internal overheating. 0: Reduce output frequency 1: Shutdown (F30024) 2: Reduce pulse frequency and output frequency (standard) (not for I2t overload) 3: Reduce pulse frequency, then shut down (F30024) (not for I2t overload)
p0335 = 0	Motor cooling (enter the motor's cooling system) 0: Self-cooling with shaft-mounted fan attached to the motor (default) 1: Separate cooling by means of a separately-driven cooling fan 2: Self-cooling and internal fan 3: Separate cooling and internal fan

Note

With p0971 = 1, the work memory data (RAM) are saved in PS000 in the EEPROM - independent of the setting of p0014. Then p0971 is set to 0.

Data save can take up to 60 seconds. During this time, no entries are possible. After saving, p0971 is reset to 0.

You can display the progress of the data save operation using parameter r3996 (0 no activity, inputs are possible; 1 ... 100 indicates the progress incrementally).

5.3.2.1 Motor temperature inputs

Temperature sensor

Parameter	Description
p0601 = …	**Motor temperature encoder** 0: No sensor 1: PTC warning & timer 2: KTY84 4: Bimetallic NC contact warning & timer
p0604 = …	**Motor overtemperature alarm threshold** (0°C … 240°C; default 130°C) Enter the temperature for the alarm threshold.
p0605 = …	**Motor overtemperature fault threshold** (0°C … 240°C; default 145°C) Enter the temperature for the fault threshold.
p0610 = 2	**Reaction if motor overtemperature I2t** Determines the response when the motor temperature reaches the alarm threshold.) 0: No response, only alarm, no reduction of I_max 1: Alarm with reduction of I_max and fault 2: Alarm and fault, no reduction of I_max

Temperature calculation without encoder

With vector control (p1300 = 20/22), the temperature calculation is possible without an encoder. To do this, the following parameters must be set.

CAUTION

Calculating the motor temperature with V/f control without encoder

Also for V/f control, it is always possible to calculate the motor temperature without encoder. However, in this case, after inverter OFF/ON, independent of the actual temperature, the motor temperature is set to the value defined in p0625. For example, the actual temperature = 100 °C, however, after OFF/ON, the inverter uses the value entered in p0625 for the temperature calculation (factory setting = 20 °C). This can result in a motor overtemperature without a fault response.

Parameters	Description (parameter name and factory setting (if not variable) in bold)
p0621= 1	**Motor temperature measurement after restarting** 0: No temperature measurement (default) 1: Temperature measurement the first time the motor is switched on after OFF/ON 2: Temperature measurement after each "motor ON".
p0622 = …	**Motor magnetizing time for temperature measurement after stator resistance measurement** (0 ... 20 s, default 0 s) This value depends on the recorded rotor time constant. To improve accuracy, this calculation can be carried out more than once.

5.3.2.2 Setting the command sources

Select the command source

It is possible to select either the terminals or the fieldbus interface as command sources for the CU230P-2 DP and the CU230P-2 HVAC.

The terminals are the only possible command source for the CU230P-2.

Note

p0922 = 999 must be set for PROFIBUS Control Units (CU230P-2 DP) to allow the command source to be changed.

Parameters	Description
p0700 = 2/6	**Select the command source** 2: Terminals (factory setting for CU230P-2 HVAC DP / CU230P-2 CAN) 6: Fieldbus (factory setting for CU230P-2)
p0724 = 3	**Debounce time for digital inputs,** 0 ... 20 ms, factory setting 4 ms. Defines the debounce time for the digital inputs.

Pre-assignment of the digital inputs

Table 5- 8 Pre-assignment of the digital inputs 0, 1, and 2

Digital input / terminal	Factory setting			
	CU230P-2 HVAC / CU230P-2 CAN		CU230P-2 DP	
Digital input 0 (DI0), terminal 5	ON/OFF1	p0701 = 1	---	p0701 = 0
Digital input 1 (DI1), terminal 6	Reversal	p0702 = 12	---	p0702 = 0
Digital input 2 (DI2), terminal 7	Acknowledge fault	p0703 = 9	---	p0703 = 9

Table 5- 9 Pre-assignment of the digital inputs 3, 4, and 5

Digital input / terminal	Factory setting	
	CU230P-2 HVAC / CU230P-2 CAN / CU230P-2 DP	
Digital input 3 (DI3), terminal 8	Fixed speed setpoint selection bit 0	p0704 = 15
Digital input 4 (DI4), terminal 16	Fixed speed setpoint selection bit 1	p0705 = 16
Digital input 5 (DI5), terminal 17	Fixed speed setpoint selection bit 2	p0706 = 17

The listed functions are preassigned to the digital inputs. It is also possible to link any other function with the digital input using the relevant binector input parameter.

It is also possible to interconnect each individual digital input via BICO parameterization with any number of signal sinks, so that for example, the ON command can be issued via either the digital input 0 or 1. This also enables you to simultaneously activate multiple functions via one digital input, for example, ON and reverse.

For further details and setting options, please refer to the List Manual and the Function Manual.

5.3.2.3 Setting the frequency setpoint source

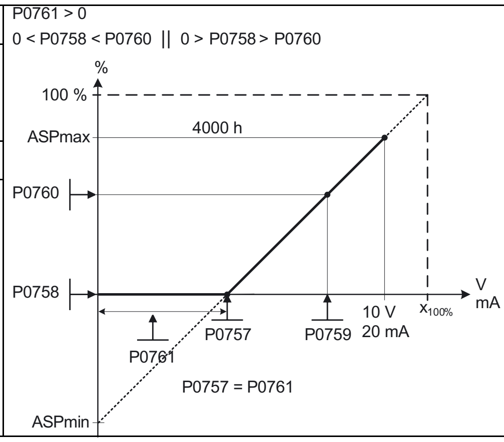
Setting the frequency setpoint

Note

p0922 = 999 must be set for PROFIBUS Control Units (CU230P-2 DP) to allow the setpoint source to be changed.

Parameters	Description
p1000 = ?	**Selection of the frequency setpoint** 0: No main setpoint 1: Motorized potentiometer (p1030 … p1040) 2: Analog setpoint (p0756 … p0762), factory setting for CU230P-2 HVAC and CU230P-2 CAN 3: Fixed speed setpoint (p1001 … p1023) 6: Fieldbus (DP p2037 ... p2079) factory setting for CU230P-2 DP (not for CU230P-2CAN) 7: Analog setpoint 2 10: Supplementary setpoint (1 = MOP) + main setpoint (0 = no main setpoint) 11: Supplementary setpoint (1 = MOP) + main setpoint (1 = MOP) 12: Supplementary setpoint (1 = MOP) + main setpoint (2 = analog setpoint) … 72: Supplementary setpoint (7 = analog setpoint 2) + main setpoint (2 = analog setpoint) 73: Supplementary setpoint (7 = analog setpoint 2) + main setpoint (3 = fixed speed setpoint) ...

Example: Frequency setpoint via analog input (AI) (p1000 = 2)

Parameter	Description	
p0756 = 0	**AI Type** Defines the analog input type and also enables analog input monitoring. 0: Unipolar voltage input (0 V …+10 V) 1: Unipolar voltage input monitored (+2 V …+10 V) 2: Unipolar current input (0 mA …+20 mA) 3: Unipolar current input monitored (+4 mA …+20 mA) 4: Bipolar voltage input (-10 V …+10 V) 6: Ni1000 temperature sensor 7: PT1000 temperature sensor **Note**: Applicable to p0756 … p0760: Index 0: Analog input 0 (AI0), terminals 3 and 4 Index 1: Analog input 1 (AI1), terminals 10 and 11	
p0757 = 0	**Value x1 for AI scaling [V/mA]**	
p0758 = 0.0	**Value y1 of AI-scaling** This parameter represents the amount of x1 as a percentage of p2000 (reference frequency)	
p0759 = 10	**Value x2 for AI scaling [V/mA]**	
p0760 = 100	**Value y2 of AI-scaling** This parameter represents the amount of x2 as a percentage of p2000 (reference frequency)	

Example: Frequency setpoint via fixed frequency (p1000 = 3)

Parameter	Description
p1016 = …	**Fixed frequency mode** 1: Direct selection (factory setting) 2: Binary-coded selection

The signal sources for selecting fixed frequencies are defined in parameters p1020 to p1023. p1020 to p1022 are linked to digital inputs DI3 to DI5 according to the factory setting. Parameter p1023 has no link in the factory setting. Setting p1023 = 722.11 links it to analog input 1 which can be used as the fourth digital input. For wiring see Control terminals of the Control Units (Page 23).

For the purpose of fieldbus control, the fixed frequency selection parameters can be linked to the corresponding control word bits.

15 fixed frequencies can be selected both directly and by means of binary coding. For reasons of simplicity, we will explain the binary coded selection method here. For further details and information about the direct selection option, please refer to the function diagrams in the List Manual.

Table 5- 10 Fixed frequency mode - binary coded selection via terminals

Parameter	Description
p1016 = 2	**Binary coded fixed frequency selection** With binary coded selection, you can directly select any frequency set to one of the parameters between p1001 ... p1015.
p1001 = …	**Fixed speed setpoint 1** (- 650 Hz … 650 Hz)
p1002 = …	**Fixed speed setpoint 2** (- 650 Hz … 650 Hz)
p1003 = …	**Fixed speed setpoint 3** (- 650 Hz … 650 Hz)
p1004 = …	**Fixed speed setpoint 4** (- 650 Hz … 650 Hz)
p1005 = …	**Fixed speed setpoint 5** (- 650 Hz … 650 Hz)
p1006 = …	**Fixed speed setpoint 6** (- 650 Hz … 650 Hz)
p1007 = …	**Fixed speed setpoint 7** (- 650 Hz … 650 Hz)
p1008 = …	**Fixed speed setpoint 8** (- 650 Hz … 650 Hz)
p1009 = …	**Fixed speed setpoint 9** (- 650 Hz … 650 Hz)
p1010 = …	**Fixed speed setpoint 10** (- 650 Hz … 650 Hz)
p1011 = …	**Fixed speed setpoint 11** (- 650 Hz … 650 Hz)
p1012 = …	**Fixed speed setpoint 12** (- 650 Hz … 650 Hz)
p1013 = …	**Fixed speed setpoint 13** (- 650 Hz … 650 Hz)
p1014 = …	**Fixed speed setpoint 14** (- 650 Hz … 650 Hz)
p1015 = …	**Fixed speed setpoint 15** (- 650 Hz … 650 Hz)
p1020 = 722.3	**Fixed speed setpoint - bit 0**, -> DI3 as source for fixed frequency selection bit 0
p1021 = 722.4	**Fixed speed setpoint - bit 1**, -> DI4 as source for fixed frequency selection bit 1
p1022 = 722.5	**Fixed speed setpoint - bit 2**, -> DI5 as source for fixed frequency selection bit 2
p1023 = 722.11	**Fixed speed setpoint - bit 3**, -> AI 0 as DI as source for fixed frequency selection bit 3

Binary selection of fixed frequency via		Parameters for fixed setpoint
DI3	0 0 0 1	p1001
DI4	0 0 1 0	p1002
DI5	0 1 0 0	p1003
AI0	1 0 0 0	p1004
DI4 DI3	0 0 1 1	p1005
DI5 DI3	0 1 0 1	p1006
AI0 DI3	1 0 0 1	p1007
DI5 DI4	0 1 1 0	p1008
AI0 DI4	1 0 1 0	p1009
AI0 DI5	1 1 0 0	p1010
DI5 DI4 DI3	0 1 1 1	p1011
AI0 DI4 DI3	1 0 1 1	p1012
AI0 DI5 DI3	1 1 0 1	p1013
AI0 DI5 DI4	1 1 1 0	p1014
AI0 DI5 DI4 DI3	1 1 1 1	p1015

5.3.2.4 Assigning the digital outputs

Assigning digital outputs

The factory setting links the digital outputs to the following signals:

- DO0: Inverter fault active (p0730 = 52.3)
- DO1: Inverter alarm active (p0731 = 52.7)
- DO2: Operation enabled (p0732 = 52.2)

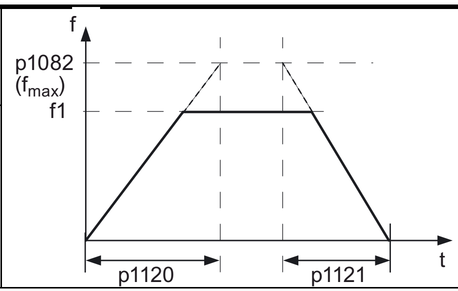
You can invert the digital outputs with p0748.

You can assign different signals to the digital outputs by linking them to the corresponding BO parameter.
Example: Digital output 2 must display whether fixed frequencies are selected. p0732 must be set to 1025.0 (r1025 status fixed frequencies) for this purpose.

For details and a list of all BO parameters, please refer to the List Manual.

5.3.2.5 Setting the ramp times

Ramp times

Parameter	Description	
p1120 = …	**Ramp-up time** Enter the acceleration time in seconds (factory setting 10 s).	
p1121 = …	**Coast-down time** Enter the ramp-down time in seconds (factory setting 10 s).	

Rounding

Parameter	Description	
p1130 = …	**Ramp-function generator initial rounding** (factory setting 0 s) applies to ramp-up and ramp-down	Rounding times are recommended to reduce shock loading of mechanical components to a minimum. The ramp-up and ramp-down times are extended by the initial and final rounding times.
p1131 = …	**Ramp-function generator final rounding** (factory setting 0 s) applies to ramp-up and ramp-down	
p1134 = …	**Rounding type** 0: Continuous smoothing (jolt free), factory setting 1: Discontinuous smoothing **NOTE**: With discontinuous smoothing (p1134 = 1), final rounding at ramp-up (p1131) and initial rounding at ramp-down (p1130) are not carried out after the setpoint has been reduced or after an OFF1 command.	

5.3.3 Completing the individual commissioning

Other parameters that need to be set to conclude individual commissioning

The following parameters must be configured for each application.

Parameter	Description
p1800 = 4	**Pulse frequency** (kHz) The pulse frequency can be changed in increments of 2 kHz within a range of 4 kHz to 16 kHz. The full output current of the inverter at 50 °C is reached at 4 kHz. The maximum output frequency is dependent on the pulse frequency. With a pulse frequency of 4 kHz, the inverter can be operated up to an output frequency of 266 Hz. If a higher output frequency is required, the pulse frequency should also be increased (10 kHz pulse frequency - maximum output frequency of 650 Hz). Where low-noise operation is not a requirement, selecting a lower pulse frequency can reduce the inverter losses and the high-frequency interference emitted by the inverter.
p2000 = …	**Reference frequency** (Hz) The Hertz value entered here is equivalent to 100 %. Factory setting is 50 Hz. This setting should be changed if a maximum frequency higher than 50°Hz has been set in p1082.

Further reference variables can be set in p2001 to p2007. For details, please refer to the List Manual.

Ending application commissioning

Parameter	Description
p0971 = 1	**Transfer data from the RAM to the EEPROM** • 0: Disabled (factory setting) • 1: Start data transfer, RAM → EEPROM All parameter changes are transferred from the RAM (volatile) to the EEPROM (non-volatile). The transfer of data to the EEPROM can take up to three minutes, depending on the volume. During this time, no entries are possible. After the transfer, p0971 is reset to 0. You can display the progress of the save using parameter r3996 (0 no activity, inputs are possible; 1 to 100 indicates the progress incrementally). Existing parameter sets of the same name are overwritten without further notice.

5.4 Further commissioning functions

5.4.1 Series commissioning

Description

The procedure for series commissioning involves the following steps:

1. Individual commissioning of an inverter using IOP or STARTER
2. Save the parameter set of this inverter as a standard data set on the memory card (see below)
3. Once the save process has been completed, remove the memory card from the inverter.
4. Insert the memory card in the other inverter before switching on the supply voltage.
5. Switch on the supply voltage. During ramp-up, the standard data set (PS000) of the memory card is written to the work memory and to the EEPROM of the inverter.

This completes series commissioning for the first inverter. Repeat steps 3 to 5 for the other inverters.

⚠ CAUTION
The user is wholly responsible for ensuring that the standard parameter set on the memory card is correct. Series commissioning between different Control Units or firmware versions is not permitted.

Note

In the case of a failed download, the inverter enters a fault state and the LED RDY flashes.

Details about the fault cause are indicated by the fault numbers in p0947 and p0949.

Saving the standard data set on the memory card

You can save the current settings for your frequency inverter as a standard data set by selecting p0971 = 1.

Note

Saving is indicated by the LED as well as by parameter r3996 (1 to 100). During saving, no entries are possible. When saving has finished, r3996 is set to 0 again and further inputs are possible.

The memory card must not be removed during saving.

5.4.2 Resetting parameters

Overview

The inverter can be reset to a defined state via parameter p0970.

Options:

- p0970 = 1: starts reset to factory settings
- p0970 = 10: Starts to load PS010 into the RAM
- p0970 = 11: Starts to load PS011 into the RAM
- p0970 = 12: Starts to load PS012 into the RAM
- p0970 = 100: starts resetting of BICO interconnections

For p0014 = 1, the actual parameter assignments are saved from the RAM to the ROM as PS000 when a parameter set is loaded (p970 = 1x).

NOTICE
A reset or loading can only be started if p0010 = 30 (parameter reset) has been set in advance. During resetting, no entries are possible. A reset or load operation is indicated by slow flashing of the LEDs on the Control Unit. With control via terminals (p2030 = 0), only the RDY LED flashes. With control via the fieldbus interface, the BF LED also flashes. The progress of the reset or load operation can also be displayed via r3996 (1 … 100) on the IOP or in STARTER. When the reset or load operation is finished, parameters p0010 and p0970 are reset to 0 and other inputs are possible. The actual frequency is indicated on the IOP. Fieldbus communication with the control is interrupted while reset or loading is in progress. **Points to note about communication via PROFIBUS DP** • Communication with Class 1 masters (e.g. S7 PLC) is interrupted • Communication with Class 2 masters (e.g. STARTER) remains functional

Resetting to factory settings or loading a parameter set

Parameter or procedure	Description
p0010 = 30	**Commissioning parameter** 30: Select a reset to factory settings
p0970 = …	**Resetting to factory setting** 1: Start reset to factory settings 10: Start loading of PS010 11: Start loading of PS011 12: Start loading of PS012 100: Start resetting of BiCo links
LEDs flash, Progress in r3996	Once the reset or load operation is finished, p0970 and p0010 are set to 0 and the standard display reappears on the IOP.

Note

Points to note about resetting to factory settings

The reset operation is not applied to parameters p0014, p0100, p0201, p0205 or the communication parameters.

Motor parameters p0300 ... p0311 are preassigned suitably for the power unit.

5.5 Operation with braking resistor

Inverters with PM240 Power Modules can be operated with a braking resistor.

With the exception of frame size FSGX, all PM240 Power Modules have an integrated braking chopper. The installation of a braking chopper in this Power Module is described in the installation manual for the PM240 Power Module.

Note

The Vdc max closed-loop control must be deactivated when operating the inverter with a braking resistor.

The braking resistor is connected at the power unit at terminals DC-P/R1 and R2.

WARNING

It is not permissible to operate Power Modules capable of energy recovery (PM250 or PM260) with a braking resistor.

Communication 6

6.1 Communication interfaces on the CU230P-2

The CU230P-2 provides communication interfaces for PROFIBUS DP, USS, Modbus RTU and CANopen depending on the product variant (see below).

- CU230P-2 DP for PROFIBUS DP
  - Control in cyclical operation using telegrams 1, 20, 352 and 999
  - Parameterization by means of acyclic communication
- CU230P-2 HVAC for USS
  - Control via the process data area (0 to 8 words)
  - Parameterization via the PKW area (0 … variable length)
- CU230P-2 HVAC for Modbus RTU
  - Control and parameterization according to the Modbus register
- CU230P-2 CAN for CANopen
  - Control by means of PDO
  - Parameterization by means of SDO

6.2 Communication via PROFIBUS DP

General information

PROFIBUS is an open, international fieldbus standard that can be used in many applications in the field of production and process automation.

The following standards ensure open, vendor-independent systems:

- International standard EN 50170
- International standard IEC 61158

PROFIBUS is optimized for time-critical, high-speed data transmission at the field level.

Note

PROFIBUS for drive technology is standardized and described in the following document:

Reference: /P5/ PROFIdrive Profile Drive Technology

CAUTION
Before synchronizing to the isochronous PROFIBUS, all of the pulses of the drive objects must be inhibited - also for those drives that are not controlled via PROFIBUS.

For information about connection, please refer to section Connecting CU230P-2 DP to the PROFIBUS DP network (Page 33).

6.2.1 User data structure defined in PROFIdrive Profile 4.1

Introduction

The inverters of the SINAMICS G120 series can be controlled over the cyclic PROFIBUS DP channel or the acyclic DPV1 channel. The user data structure of the cyclic/acyclic channel is specified in the PROFIdrive profile, Version 4.1.

The PROFIdrive profile defines the user data structure for inverters through which a master can access the inverter (slave) via cyclic or acyclic data transfer.

Note

PROFIdrive for drive technology is standardized and described in the following document:
References: /P5/ PROFIdrive Profile Drive Technology

Telegram data structure

The user data structure for cyclic data transmission consists for the CU230P-2 DP of the process data area (PZD). This comprises control words (status information) and setpoints (actual values).

The transferred process data will only be effective when the bits used in the control words, setpoints, status words and actual values have been routed in the inverter.

Process data area (PZD)							
PZD01 STW1 ZSW1	PZD02 HSW HIW	PZD03	PZD04	PZD05	PZD06	PZD07	PZD08
1. word	2. word	3. word	4. word	5. word	6. word	7. word	8. word

PZD: Process data
STW: control word 1
ZSW: status word 1
HSW: Main setpoint
HIW: Main actual value

6.2.2 Communication settings for PROFIBUS DP

PROFIBUS DP parameters

You can fully integrate your inverter with a CU230P-2 DP Control Unit into a PROFIBUS communication system without the need for any PROFIBUS-specific parameter settings when the following preconditions are met:

- The PROFIBUS address is set using a DIP switch
- The communication procedure uses standard telegram 1

If you want to modify the communication settings, the parameters that you can use to make changes are listed in the table below.

Table 6- 1 PROFIBUS DP parameters

Parameter	Contents
p0918	PROFIBUS address (connecting CU230P-2 DP to the PROFIBUS DP network (Page 33))
p0922	Seletion of the PROFIBUS standard telegram
p2038	Selection of the communication profile (PROFIdrive profile 4.1 / VIK NAMUR)
p2042	Selection of the profile code for the controller (PROFIdrive profile 4.1 / VIK NAMUR)
r2050	Setpoint source for process data (BICO)
p2051	Actual values for process data (BICO)
p2030	Fieldbus interface protocol selection
p2037	Processing mode for PROFIdrive
p2044	PROFIdrive fault delay
p2047	Supplementary PROFIBUS monitoring time
r2054	Diagnostics of the communication module
r2055	PROFIBUS diagnosis - standard
r2075	PROFIdrive diagnosis telegram offset - receive PZD
r2076	PROFIdrive diagnosis telegram offset - send PZD
p2079	PROFIdrive PZD telegram selection extended

Selection of the PROFIBUS standard telegram via p0922

This parameter allows you to modify the telegram type. (Factory setting = 1, other possibilities: 20, 352 and 999). For further details, see section "Cyclic communication (Page 72)"

Change setpoint and command sources (r2050 and p2051) via BICO

You can define specific setpoint and actual values for communication using the BICO parameters r2050 and p2051.

Table 6- 2 Parameters for the flexible connection of process data

Telegram	PZD1 STW/ZSW	PZD2 HSW/HIW	PZD3	PZD4	PZD5	PZD6
Connection values for setpoints, master to inverter	r2050.00	r2050.01	r2050.02	r2050.03	r2050.04	r2050.05
Connection parameters for actual values, inverter to master	p2051.00	p2051.01	p2051.02	p2051.03	p2051.04	p2051.05

Note

The setpoint received over the PROFIBUS DP interface can also be output and checked using r2050.

Process data whose complete control word (PZD1) is set to zero is not transferred to the inverter by the PROFIBUS DP interface. This can result in the message F01910 or F01920.

6.2.3 Cyclic communication

Telegrams

Selecting a telegram via p0922 defines, on the drive side, which process data is to be transferred between master and slave.

From the viewpoint of the slave, words can be receive words and send words.

The receive and send words comprise the following elements:

- Receive words - Control words or setpoints
- Send words - Status words or actual values

Telegram types used

The following telegrams can be set via parameter p0922:

Standard telegrams

The standard telegrams are structured in compliance with the PROFIdrive profile. The internal process data connections are automatically set according to the set telegram number.

- Telegram 1, speed control, 2 words
- Telegram 20, speed control, VIK/NAMUR

Manufacturer-specific telegrams

The manufacturer-specific telegrams are structured in accordance with internal company specifications. The internal process data connections are automatically set according to the set telegram number.

- Telegram 352, speed control, PCS7

Device-specific telegrams

Send and receive telegrams can be configured as required by using BICO technology to interconnect the send and receive process data.

- Telegram 999 free connection via BICO (up to 8 data words)

6.2.3.1 Telegram structure

BICO connection

Once a telegram is selected, the appropriate BICO connection parameters become permanent and can no longer be changed (with the exception of p0701 and the following digital inputs). With the setting p0922 = 999, this parameter retains the current BICO connection parameters, but the BICO parameters can now be changed.

Telegram structure and settings of p0922

Table 6- 3 Telegrams from 1 to 999

Telegram	Parameter channel	Process data area (PZD)							
		PZD01 STW1 ZSW1	PZD 02 HSW HIW	PZD 03	PZD 04	PZD 05	PZD 06	PZD 07	PZD 08
Telegram 1 Speed control, 2 words	-	STW1	NSOLL_A	⇐ Receive telegram from PROFIBUS					
		ZSW1	NIST_A	⇒ Send telegram to PROFIBUS					
Telegram 20 Speed control, VIK/NAMUR	-	STW1	NSOLL_A						
		ZSW1	NIST_A_ GLATT	IAIST	MIST	PIST	MELD_ NAMUR	<1>	
Telegram 352 Speed control, PCS7	-	STW1	NSOLL_A	<3>	<3>	<3>	<3>		
		ZSW1	NIST_A_ GLATT	IAIST	MIST	FAULT_ CODE	WARN_ CODE		
Telegram 999 Free interconnection via BICO	-	STW1 <1>	Telegram length is max. 8 words on receipt. This can be freely selected using the central PROFIBUS configuration, e.g. HW Config (universal module in GSD) <2>						
		ZSW1 <1>	Telegram length is max. 8 words on sending. This can be freely selected using the central PROFIBUS configuration, e.g. HW Config (universal module in GSD) <2>						
<1> For compliance with the PROFIdrive profile, PZD1 must be used as STW1 or ZSW1. <2> Structure as for a standard telegram <3> Dummy for PCS7 process data									
STW1/2 (r2090/r2091) ZSW1/2 (r0052/r0053) NSOLL_A *) (p1070) NIST_A *) (r0021) IA_IST (r0027) MIST (r0031) PIST (r0032) M_LIM (p1522) FAULT_CODE (r2131) WARN_CODE (r2132) MELD_NAMUR		Control word 1/2 Status word 1/2 Speed setpoint A (16 bit) Speed actual value A (16 bit) Output current Torque actual value Active power Torque limit Fault code Alarm code NAMUR message bit bar (r3113)							
*) NSOLL_A, NIST_A are FSOLL and FIST for SINAMICS G120									

6.2.3.2 VIK/NAMUR telegram structure

Description

Table 6- 4 VIK/NAMUR telegram structure

Telegram	Parameter channel	Process data area (PZD)							
		PZD01 STW1 ZSW1	**PZD 02 HSW HIW**	**PZD 03**	**PZD 04**	**PZD 05**	**PZD 06**	**PZD 07**	**PZD 08**
Telegram 20 Speed control, VIK/NAMUR	-	STW1	NSOLL_ A						
	-	ZSW1	NIST_A_ GLATT	IAIST	MIST	PIST	MELD_ NAMUR	<1>	
<1> For compliance with the PROFIdrive profile, PZD1 must be used as STW1 or ZSW1. <4> NAMUR message bit bar (r3113)									
STW1/2 (r2090/r2091) ZSW1/2 (r0052/r0053) NSOLL_A *) (p1070) NIST_A *) (r0021) IA_IST (r0027) MIST (r0031) PIST (r0032) MELD_NAMUR		Control word 1/2 Status word 1/2 Speed setpoint A (16 bit) Speed actual value A (16 bit) Output current Torque actual value Active power NAMUR message bit bar (r3113)							
*) NSOLL_A, NIST_A are FSOLL and FIST for SINAMICS G120									

If p0922 = 20 is used to select the VIK/NAMUR telegram, parameter p2038 "Select current profile" is automatically set to VIK/NAMUR.

It is also necessary to specify the identification number (GSD) using parameter p2042:

- **SIMATIC Object Manager (Slave OM)**

  For drives that were configured using the SIMATIC Object Manager (Slave OM), parameter p2042 is set to 0 (factory setting).

- **VIK/NAMUR GSD (device master data file)**

  For any other configuration, the setting for p2042 must be changed to 1 (NAMUR). This means that the VIK/NAMUR ID of PROFIBUS International will be sent to the PLC.

Note

When the PROFIBUS standard telegram is changed to the NAMUR telegram or vice-versa (parameters: p0922, p2038 and p2042), the inverter ID is changed. The change in inverter ID is only effective after the CU240S Control Unit has been switched off and on again.

When changing back from VIK/NAMUR to the standard telegram, p0922 = 999 (free BICO connection) must be set and then p2038 and p2042 must be reset to the PROFIdrive profile. With the setting p0922 = 999, this parameter retains the current BICO connection parameters, but the BICO parameters can now be changed.

6.2.3.3 Inverter behavior when switching over the communication telegram

Overview

When switching over the communication telegram via p0922, the following listed parameters are pre-assigned again. The individual parameters can be subsequently set back to all of the values permissible in the particular parameters.

Parameters		Pre-assignment when switching over to p0922 =		
		1	20	352
p0701[0]	Digital input 0	0	0	0
p0702[0]	Digital input 1	0	0	0
p0703[0]	Digital input 2	9	9	9
p0704[0]	Digital input 3	15	15	15
p0705[0]	Digital input 4	16	16	16
p0706[0]	Digital input 5	17	17	17
p0712[0]	Digital input 6	0	0	0
p0713[0]	Digital input 7	0	0	0
p0840[0]	BI: ON/OFF1	r2090.0	r2090.0	r2090.0
p0844[0]	BI: 1. OFF2	r2090.1	r2090.1	r2090.1
p0848[0]	BI: 1. OFF3	r2090.2	r2090.2	r2090.2
p0852[0]	BI: Enable operation	r2090.3	r2090.3	r2090.3
p1140[0]	BI: Ramp-function generator	r2090.4	r2090.4	r2090.4
p1141[0]	BI: Start ramp-function generator	r2090.5	r2090.5	r2090.5
p1142[0]	BI: Enable speed setpoint	r2090.6	r2090.6	r2090.6
p2103[0]	BI: 1. Acknowledge faults	r2090.7	r2090.7	r2090.7
p0854[0]	BI: Master control by PLC	r2090.10	r2090.10	r2090.10
p1113[0]	BI: Setpoint inversion	r2090.11	r2090.11	r2090.11
p1035[0]	BI: Motorized potentiometer, setpoint, raise	r2090.13	---	r2090.13
p1036[0]	BI: Motorized potentiometer, setpoint, lower	r2090.14	---	r2090.14
p2080.0	BI: Binector-connector converter, status word 1	p0899.0	p0899.0	p0899.0
p2080.1	BI: Binector-connector converter, status word 1	p0899.1	p0899.1	p0899.1
p2080.2	BI: Binector-connector converter, status word 1	p0899.2	p0899.2	p0899.2
p2080.3	BI: Binector-connector converter, status word 1	p2139.3	p2139.3	p2139.3
p2080.4	BI: Motorized potentiometer, setpoint, lower	p0899.4	p0899.4	p0899.4
p2080.5	BI: Binector-connector converter, status word 1	p0899.5	p0899.5	p0899.5
p2080.6	BI: Binector-connector converter, status word 1	p0899.6	p0899.6	p0899.6
p2080.7	BI: Binector-connector converter, status word 1	p2139.7	p2139.7	p2139.7

Parameters		Pre-assignment when switching over to p0922 =		
		1	20	352
p2080.8	BI: Binector-connector converter, status word 1	p2197.7	p2197.7	p2197.7
p2080.9	BI: Binector-connector converter, status word 1	p0899.9	p0899.9	p0899.9
p2080.10	BI: Motorized potentiometer, setpoint, lower	p2199.1	p2199.1	p2199.1
p2080.11	BI: Binector-connector converter, status word 1	p1407.7	p1407.7	p1407.7
p2080.12	BI: Binector-connector converter, status word 1	p0899.12	---	p0899.12
p2080.13	BI: Binector-connector converter, status word 1	p2135.14	p2135.14	p2135.14
p2080.14	BI: Binector-connector converter, status word 1	p2197.3	p2197.3	p2197.3
p2080.15	BI: Binector-connector converter, status word 1	p2135.15		p2135.15
p2051.0	CI: PROFIdrive PZD send word	p2089.0	p2089.0	p2089.0
p2088	Binector-connector converter, invert status word	43008	43008	43008
p2038	PROFIdrive STW/ZSW interface node	0	2	0
p1020[0]	BI: Fixed speed setpoint selection bit 0	r0722.3	r0722.3	r0722.3
p1021[0]	BI: Fixed speed setpoint selection bit 1	r0722.4	r0722.4	r0722.4
p1022[0]	BI: Fixed speed setpoint selection bit 2	r0722.5	r0722.5	r0722.5
p2220[0]	BI: Technology controller fixed value selection bit 0	r0722.3	r0722.3	r0722.3
p2221[0]	BI: Technology controller fixed value selection bit 1	r0722.4	r0722.4	r0722.4
p2222[0]	BI: Technology controller fixed value selection bit 2	r0722.5	r0722.5	r0722.5

"---" no change

6.2.3.4 Control and status words

Description

The control and status words fulfill the specifications for the PROFIdrive profile, Version 4.1 for "Closed-loop speed control" mode.

Control word 1 (STW1)

Control word 1 (bits 0 to 10 according to PROFIdrive profile and VIK/NAMUR; bits 11 to 15 specifically for SINAMICS G120).

Table 6- 5 Assignment of control word 1

Bit	Value	Meaning and comment	Interconnection parameter	p0922 = 20 (VIK/ NAMUR)	p0922 = 1 / 352 (PROFIdrive profile)
STW1.0	1 0	**ON** (↑), inverter goes into "ready for operation", direction of rotation via bit 11. **OFF1,** shutdown with ramp-down along the RFG ramp, pulse cancellation when f < f_{min}, inverter goes into "ready for switching on".	p0840 =	2090.0	2090.0
STW1.1	1 0	**No OFF2,** pulse enable possible **OFF2,** immediate pulse suppression, coast down to standstill and switching on inhibited.	p0844 =	2090.1	2090.1
STW1.2	1 0	**No OFF3,** pulse enable possible **OFF3 - fast stop,** deacceleration along the OFF3 ramp (p1135) then pulse cancellation and closing lockout	p0848 =	2090.2	2090.2
STW1.3	1 0	**Enable operation,** enable closed-loop control and inverter pulses. **Inhibit operation,** closed-loop control is blocked, pulse suppression	p0852 =	2090.3	2090.3
STW1.4	1 0	**Operating condition,** ramp-function generator enable possible **Reset ramp-function generator,** RFG output is set to 0 (quickest possible deceleration); inverter remains in the ON state.	p1140 =	2090.4	2090.4
STW1.5	1 0	**Enable ramp-function generator** **Block ramp-function generator,** the setpoint currently provided by the ramp-function generator is "frozen".	p1141 =	2090.5	2090.5
STW1.6	1 0	**Enable setpoint,** the value selected on the ramp-function generator input is enabled. **Deactivate setpoint,** the value selected on the ramp-function generator input is set to 0 (zero).	p1142 =	2090.6	2090.6
STW1.7	1	↑ **Fault acknowledgement**, inverter goes into the state "Start inhibit"	p1024 =	2090.7	2090.7
STW1.8		Reserved		----	----
STW1.9		Reserved		----	----
STW1.10	1 0	**PLC control,** control via interface, process data valid **No PLC control,** process data invalid "Sign-of-life" expected.		----	----
STW1.11	1 0	**Invert setpoint,** motor rotates counter-clockwise as response to a positive setpoint. **No setpoint inversion,** motor rotates clockwise as response to a positive setpoint.	p1113 =	2090.11	----
STW1.12		Reserved		----	----
STW1.13	1	**Motorized potentiometer UP**	p1035 =	No change	2090.13
STW1.14	1	**Motorized potentiometer DOWN**	p1036 =	No change	2090.14
STW1.15		Reserved		----	----

Example:

In remote operation mode, the commands and target values will be transferred from a higher-level control system to the inverter over PROFIBUS. By switching over to local operation, the command and target value sources are switched over; operation is performed from now on using the digital inputs and the analog target values.

On-site operation = Command data set 0: In this case, the command code for the terminal strip is p0700 Index 0 = 2 and the frequency target value is the analog target value p1000 Index 0 = 2.

Remote operation = Command data set 1: In this case, the command code corresponds to the control word (word 0) that is received from PROFIBUS p0700 Index 1 = 6; and the frequency target value corresponds to control word 1 that is received from PROFIBUS p1000 Index 1 = 6.

Standard assignment of control word 2 (STW2)

Control word 2 is assigned by default as follows. This can be changed by using BICO.

Table 6- 6 Pre-assignment of control word 2 (not defined for VIK/NAMUR)

Bit	Value	Meaning	p0922 = 1	p0922 = 352
STW2.0	1	Fixed frequency selection bit 0	p1020 depends on p070x	p1020 depends on p070x
STW2.1	1	Fixed frequency selection bit 1	p1021 depends on p070x	p1021 depends on p070x
STW2.2	1	Fixed frequency selection bit 2	p1022 depends on p070x	p1022 depends on p070x
STW2.3	1	Fixed frequency selection bit 3	p1023 depends on p070x	p1023 depends on p070x
STW2.4		Reserved	--	--
STW2.5		Reserved	--	--
STW2.6		Reserved	--	--
STW2.7		Reserved	--	--
STW2.8	1	Enable technology controller	0.0	0.0
STW2.9	1	Enable DC brake	0.0	0.0
STW2.10		Reserved	--	--
STW2.11	1	Speed controller droop enable	Enable droop	Enable droop
STW2.12	1 0	Torque control Speed control	0.0	0.0
STW2.13	0	External fault 1	1.0	1.0
STW2.14		Reserved	--	--
STW2.15		Reserved	--	--

Status word 1 (ZSW1)

Status word 1 (bits 0 to 10 according to PROFIdrive profile and VIK/NAMUR; bits 11 to 15 specifically for SINAMICS G120).

Table 6- 7 Bit assignments of status word 1 (for all PROFIdrive and VIK/NAMUR telegrams)

Bit	Value	Meaning, remarks
ZSW1.0	1 0	**Ready for switching on,** power supply is switched on, electronics is initialized, pulses are inhibited. **Not ready for switching-on**
ZSW1.1	1 0	**Ready for operation,** inverter is switched on (ON command is present); no fault is active; inverter can run as soon as the "Enable operation" command is issued. See control word 1, bit 0. **Not ready for operation**.
ZSW1.2	1 0	**Operation enabled,** drive follows setpoint. See control word 1, bit 3. **Operation inhibited**
ZSW1.3	1 0	**Fault present,** drive faulted. A fault is present in the drive; it is, therefore, not operating and will switch back to the "Start-up inhibit" state once the fault has been successfully eliminated and acknowledged. **No fault**
ZSW1.4	1 0	**"Coast down to standstill" not activated** **"Coast down to standstill" activated,** command "Coast down to standstill" (OFF 2) present.
ZSW1.5	1 0	**"Fast stop" not activated** **Fast stop activated,** fast stop command (OFF 3) present.
ZSW1.6	1 0	**Switch-on inhibited,** the drive is then only brought into the "Switched on" state again when the "No coast down" AND "No quick stop" commands, followed by "ON", are issued. **Switch-on not inhibited**
ZSW1.7	1 0	**Alarm present,** drive still operational; alarm in the service/maintenance parameter; no acknowledgement; see alarm parameter r2110. **No alarm,** no alarm is present or the alarm has disappeared again.
ZSW1.8	1 0	**Speed deviation within the tolerance range,** setpoint-actual value deviation within the tolerance range. **Speed deviation outside the tolerance range**
ZSW1.9	1 0	**Master control requested,** automation system is requested to take over the control. **No control requested,** presently, the master control is not the master.
ZSW1.10	1 0	**Maximum frequency reached or exceeded,** inverter output frequency is greater than or equal to the maximum frequency. **Highest frequency is not reached**
ZSW1.11	0	**Alarm: Motor current/torque limit reached**
ZSW1.12	1	**Motor holding brake active,** signal can be used to control a holding brake.
ZSW1.13	0	**Motor overload,** motor data displays overload status.
ZSW1.14	1 0	**Clockwise rotation** **Counter-clockwise rotation**
ZSW1.15	0	**Inverter overload,** e.g. current or temperature.

Status word 2 (ZSW2)

The standard assignment of status word 2 is as follows: This can be changed by using BICO.

Table 6- 8 Pre-assignment of status word 2 (not defined for VIK/NAMUR)

Bit	Value	Meaning	Description
ZSW2.0	1	DC brake active	
ZSW2.1	1	|n_act| > p1226 (n_standstill)	Absolute value of the actual frequency > shutdown limit value
ZSW2.2	1	n_act ≧ p1080	Actual frequency ≥ minimum frequency
ZSW2.3	1	i_act ≧ p2170	Current ≥ limit
ZSW2.4	1	n_act > p2155	Actual frequency > reference frequency
ZSW2.5	1	n_act ≦ p2155	Actual frequency < reference frequency
ZSW2.6	1	Speed setpoint reached	Actual frequency ≥ setpoint
ZSW2.7	1	DC link voltage ≤ p2172	Vdc_act ≦ Vdc threshold value
ZSW2.8	1	DC link voltage > p2172	Vdc_act > Vdc threshold value
ZSW2.9	1	Speed ramp terminated	
ZSW2.10	1	Technology controller output ≦ p2292	PI frequency ≦ threshold value
ZSW2.11	1	Technology controller output > p2291	PI saturation
ZSW2.12		Reserved	
ZSW2.13		Reserved	
ZSW2.14		Reserved	
ZSW2.15		Reserved	

6.2.4 Acyclic communication

Overview of acyclic communication

The contents of the transferred data set corresponds to the structure of the acyclic parameter channel according to the PROFIdrive profile, Version 4.1 (http://www.profibus.com/organization.html).

The acyclic data transfer mode generally allows:

- The transfer of large volumes of user data (up to 240 bytes). A parameter request/response must fit into a data set (max. 240 bytes). The requests/responses are no longer distributed over several data sets.
- Transfer of complete fields or field parts or the complete parameter description.
- Transfer of different parameters in one access (multiple request).
- Reading of profile-specific parameters over an acyclic channel
- Acyclic data transfer in parallel with cyclic data transfer.

Only one parameter request is processed at a time (no pipelining). No spontaneous messages are transferred.

Acyclic communication over PROFIBUS DP (DPV1)

The PROFIBUS DP expansions DPV1 comprise the definition of acyclic data exchange.

It supports concurrent access by other PROFIBUS masters (masters of Class 2, e.g. commissioning tool).

Implementation of the extended PROFIBUS DP functions

Suitable channels are provided in the inverters of the SINAMICS G120 series for the different masters/different data transfer types:

- Acyclic data exchange with the same Class 1 master using the DPV1 functions READ and WRITE (with data set 47 (DS47)).
- Acyclic data exchange with the help of a SIEMENS startup tool (master of Class 2, e.g. STARTER). The startup tool can acyclically access parameters and process data in the inverter.
- Acyclic data exchange with a SIMATIC HMI (Human Machine Interface) (second master of Class 2). The SIMATIC HMI can acyclically access parameters in the inverter.
- Instead of a SIEMENS startup tool or a SIMATIC HMI, it is also possible for an external master (master of Class 2) as defined in the acyclic parameter channel according to the PROFIdrive profile, Version 4.1 (with DS47), to access the inverter.

6.2.4.1 Parameter channel (data block 47 - PROFIdrive)

Characteristics of the parameter channel

- One 16-bit address for each parameter number and subindex.
- Concurrent access by several masters (e.g. commissioning tool).
- Transfer of different parameters in one access (multiple request).
- Entire fields or parts of fields can be transferred.
- Only one parameter request is processed at a time (no pipelining).
- A parameter request/response must fit into a data set (max. 240 bytes to DPV1).
- The header of the task or the response comprises user data.

Structure of parameter request and parameter response

Each parameter request consists of three parts:

- Request header

  ID for the request and number of parameters being accessed.

- Parameter address

  Addressing a parameter. If multiple parameters are being accessed there will be a corresponding number of parameter addresses. The parameter address appears in the request and not in the response.

- Parameter value

  For each parameter addressed there is a segment for the parameter values. Depending on the request ID, the parameter values appear either only in the request or in the response.

Table 6- 9 Parameter request

	Word	
	Byte	**Byte**
Request header	Request reference	Request ID
	Drive object ID	No. of parameters
1st parameter address	Attribute	No. of elements
	Parameter number (PNU)	
	Subindex	
…		
nth parameter address	Attribute	No. of elements
	Parameter number (PNU)	
	Subindex	
1st parameter value(s) (only in the case of "change parameters" request)	Format	No. of values
	Values	
	…	
…		
nth parameter value(s)	Format	No. of Values
	Values	
	…	

Table 6- 10 Parameter response

	Word	
	Byte	**Byte**
Response header	Request reference mirrored	Request ID
	Drive object ID mirrored	No. of parameters
1st parameter value(s) (only in the case of "request" request)	Format	No. of Values
	Values or error values	
	…	
…		
nth parameter value(s)	Format	No. of Values
	Values or error values	
	…	

Field description for parameter requests and responses

Table 6- 11 Field description for parameter requests

Field	Data type	Values	Remark
Request reference	Unsigned 8	0x01 … 0xFF	
	Unambiguous identification of the request/response pair for the master. The master changes the request reference with each new request. The slave mirrors the request reference in its response.		
Request ID	Unsigned 8	0x01 0x02	Read request, write request
	Specifies the type of request. In the event of a write request the changes are saved to volatile memory (RAM). A save operation is required in order to transfer the data to non-volatile memory (p0971).		
Drive object ID	Unsigned 8	0x00 … 0xFF	For the CU230P-2, 1 must always be entered as drive object ID.
	Setting for the drive object number with a drive unit with more than one drive object. Different drive objects with separate parameter number ranges can be accessed over the same connection.		
No. of parameters	Unsigned 8	0x01 … 0x27	No. 1 to 39 Limited by the length of the telegram
	In the case of multiple requests, defines the number of adjacent areas for the parameter address and/or the parameter value for multiple parameter requests. The number of parameters = 1 for individual requests.		
Attribute	Unsigned 8	0x10 0x20 0x30	Value Description Text (not implemented)
	Type of parameter element accessed.		
No. of elements	Unsigned 8	0x00 0x01 … 0x75	Special function No. 1 to 117 Limited by the length of the telegram
	Number of field elements being accessed.		
Parameter number	Unsigned 16	0x0001 … 0xFFFF	No. 1 to 65535
	Number of parameter being accessed.		
Subindex	Unsigned 16	0x0000 … 0xFFFF	No. 0 to 65535
	Addresses the first field element of the parameter to be accessed.		
Format	Unsigned 8	0x02 0x03 0x04 0x05 0x06 0x07 0x08 Other values 0x40 0x41 0x42 0x43 0x44	Data type Integer 8 Data type Integer 16 Data type Integer 32 Data type Unsigned 8 Data type Unsigned 16 Data type Unsigned 32 Data type Floating Point See PROFIdrive profile Zero (no values, as positive sub-response to a write request) Byte Word Double word Error

Field	Data type	Values	Remark
	Format and number specify the neighboring area in the telegram that contains values. Data types compatible with the PROFIdrive profile are preferred for write access. Bytes, words and double words may also be used as alternatives.		
No. of Values	Unsigned 8	0x00 … 0xEA	No. 0 to 234 Limited by the length of the telegram
	Specifies the number of subsequent values.		
Values	Unsigned 16	0x0000 … 0x00FF	
	The values of the parameters for read or write access. A zero byte is appended if the values result in an odd number of bytes. This ensures integrity of the word structure in the telegram.		

Table 6- 12 Field descriptions for parameter responses

Field	Data type	Values	Remark	
Request reference	See the table above			
Request ID	Unsigned 8	0x01 0x02 0x81 0x82	Read request (+) Write request (+) Read request (–) Write request (–)	Request positive -> status OK Request negative -> error state
	Mirrors the request ID and indicates whether the execution of the request has been positive or negative. Negative means: Part or all of the request cannot be executed. Instead of the values for every subresponse, the error values are transferred.			
Drive object ID	See the table above			
No. of parameters	See the table above			
Format	See the table above			
No. of Values	See the table above			
Values	See the table above			
Error values	Unsigned 16	0x0000 … 0x00FF	Meaning of the error values: See the next table	
	The error values in the case of a negative response. A zero byte is appended if the values result in an odd number of bytes. This ensures integrity of the word structure in the telegram.			

Note

The drive ES SIMATIC provides function blocks within the standard block libraries for writing/reading parameters, including a few examples.

Error values in DPV1 parameter responses

Table 6- 13 Explanation of error values in parameter responses

Fault value	Description	Remarks	Additional inf.
0x00	Illegal parameter number	Access to a parameter which does not exist.	-
0x01	Parameter value cannot be changed	Modification access to a parameter value which cannot be changed.	Subindex
0x02	Lower or upper value limit exceeded	Modification access with value outside value limits.	Subindex
0x03	Invalid subindex	Access to a subindex which does not exist.	Subindex
0x04	No array	Access with subindex to an unindexed parameter.	-
0x05	Wrong data type	Modification access with a value which does not match the data type of the parameter.	-
0x06	Illegal set operation (only reset allowed)	Modification access with a value not equal to 0 in a case where this is not allowed.	Subindex
0x07	Description element cannot be changed	Modification access to a description element which cannot be changed.	Subindex
0x09	No description data	Access to a description which does not exist (the parameter value exists).	-
0x0B	No operating priority	Modification access with no operating priority.	-
0x0F	No text array exists	Access to a text array which does not exist (the parameter value exists).	-
0x11	Request cannot be executed due to operating status	Access is not possible temporarily for unspecified reasons.	-
0x14	Illegal value	Modification access with a value which is within the limits but which is illegal for other permanent reasons (parameter with defined individual values).	Subindex
0x15	Response too long	The length of the present response exceeds the maximum transfer length.	-
0x16	Illegal parameter address	Impermissible or unsupported value for attribute, number of elements, parameter number, subindex or a combination of these.	-
0x17	Illegal format	Write request: illegal or unsupported parameter data format	-
0x18	No. of values inconsistent	Write request: a mismatch exists between the number of values in the parameter data and the number of elements in the parameter address.	-
0x19	Drive object does not exist	You have attempted to access a drive object that does not exist.	-
0x20	The parameter text element cannot be modified.	Attempt to modify a read only parameter text element.	
0x21	Invalid request	–	
0x22	Multi-parameter accessing is not supported		
0x65	Presently deactivated.	You have tried to access a parameter that, although available, does not currently perform a function (e.g. n control set and access to a V/f control parameter).	-

Fault value	Description	Remarks	Additional inf.
0x6B	Parameter %s [%s]: no write access with enabled controller	-	-
0x6C	Parameter %s [%s]: unit unknown	-	-
0x6D	Parameter %s [%s]: Write access only in the commissioning state, encoder (p0010 = 4).	-	-
0x6E	Parameter %s [%s]: Write access only in the commissioning state, motor (p0010 = 3).	-	-
0x6F	Parameter %s [%s]: Write access only in the commissioning state, power unit (p0010 = 2).	-	-
0x70	Parameter %s [%s]: Write access only in the quick commissioning mode (p0010 = 1).	-	-
0x71	Parameter %s [%s]: Write access only in the ready state (p0010 = 0).	-	-
0x72	Parameter %s [%s]: Write access only in the commissioning state, parameter reset (p0010 = 30).	-	-
0x73	Parameter %s [%s]: Write access only in the commissioning state, Safety (p0010 = 95).	-	-
0x74	Parameter %s [%s]: Write access only in the commissioning state, tech. application/units (p0010 = 5).	-	-
0x75	Parameter %s [%s]: Write access only in the commissioning state (p0010 not equal to 0).	-	-
0x76	Parameter %s [%s]: Write access only in the commissioning state, download (p0010 = 29).	-	-
0x77	Parameter %s [%s] may not be written in download.	-	-
0x78	Parameter %s [%s]: Write access only in the commissioning state, drive configuration (device: p0009 = 3).	-	-
0x79	Parameter %s [%s]: Write access only in the commissioning state, define drive type (device: p0009 = 2).	-	-
0x7A	Parameter %s [%s]: Write access only in the commissioning state, data set basis configuration (device: p0009 = 4).	-	-
0x7B	Parameter %s [%s]: Write access only in the commissioning state, device configuration (device: p0009 = 1).	-	-

Fault value	Description	Remarks	Additional inf.
0x7C	Parameter %s [%s]: Write access only in the commissioning state, device download (device: p0009 = 29).	-	-
0x7D	Parameter %s [%s]: Write access only in the commissioning state, device parameter reset (device: p0009 = 30).	-	-
0x7E	Parameter %s [%s]: Write access only in the commissioning state, device ready (device: p0009 = 0).	-	-
0x7F	Parameter %s [%s]: Write access only in the commissioning state, device (device: p0009 not equal to 0).	-	-
0x81	Parameter %s [%s] may not be written in download.	-	-
0x82	Transfer of master control is inhibited by BI: p0806.	-	-
0x83	Parameter %s [%s]: requested BICO interconnection not possible	BICO output does not supply float values. The BICO input, however, requires a float value.	-
0x84	Parameter %s [%s]: Parameter change inhibited (refer to p0300, p0400, p0922)	-	-
0x85	Parameter %s [%s]: access method not defined.	-	-
0xC8	Below the valid values.	Modification request for a value that, although within "absolute" limits, is below the currently valid lower limit.	-
0xC9	Above the valid values.	Modification request for a value that, although within "absolute" limits, is above the currently valid upper limit (e.g. governed by the current inverter rating).	-
0xCC	Write access not permitted.	Write access is not permitted because an access key is not available.	-
0xFF	Successful read/write procedure	The value has been successfully read or written	-
255	Request performed successfully	-	-

6.3 USS communication

General information

Communication using the USS protocol takes place over the RS485 interface with a maximum of 31 slaves. The following character frame applies for the USS telegram:

1 Bit Start	8 bits of data	1 Bit P-even	1 Bit stop

For information about connection, please refer to section Connecting the CU230P-2 HVAC via the RS485 interface (Page 30).

6.3.1 Structure of a USS telegram

Description

The structure of a typical USS telegram is shown in the figure below.

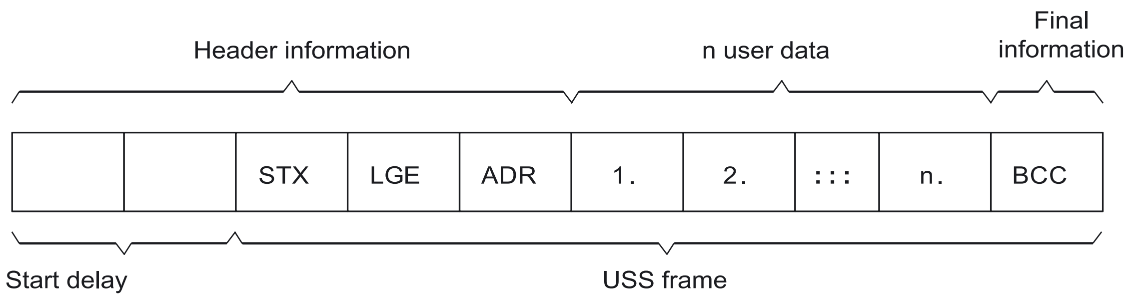


Figure 6-1 Structure of a USS telegram

Telegrams with both a variable and fixed length can be used. This can be selected using parameters p2022 and p2023 to define the length of the PZD and the PKW.

The most common application using a fixed length is shown below:

STX	1 byte	
LGE	1 byte	
ADR	1 byte	
Net data	PKW	8 bytes (4 words: PKE + IND + PWE1 + PWE2)
	PZD	4 bytes (2 words: PZD1 + PZD2)
BCC	1 byte	
Total:	16 bytes (LGE indicates 14 bytes, because STX and LGE were not counted in LGE)	

Start delay

The duration of the start delay must at least be as long as the time for two characters and depends on the baud rate.

Table 6- 14 Duration of the start delay

Baud rate in bits/s	Transmission time per character (= 11 bits)	Transmission time per bit	Min. start delay
9600	1.146 ms	104.170 µs	> 2.291 ms
19200	0.573 ms	52.084 µs	> 1.146 ms
38400	0.286 ms	26.042 µs	> 0.573 ms
57600	0.191 ms	17.361 µs	> 0.382 ms
115200	0.059 ms	5.340 µs	> 0.117 ms

Note: The time between two characters must be shorter than the start delay.

STX

The STX block is a single-byte ASCII STX character (0x02) and indicates the beginning of the message.

LGE

LGE is a single-byte block and specifies the number of bytes that follow in the telegram. It is defined as the sum of

- User data characters (quantity n)
- Address byte (ADR)
- Block check character (BCC)

The actual overall telegram is, of course, two bytes longer because STX and LGE are not counted in LGE.

ADR

The ADR range is a single byte which contains the address of the slave node (e.g. inverter). The individual bits in the address byte are addressed as follows:

7	6	5	4	3	2	1	0
Special	Mirror	Send	5 address bits				

- Bit 5 is the broadcast bit.

> **Note**
>
> The Broadcast function is not supported in the current software version.

- Bit 6 = 1 indicates a mirror telegram.
  The node address is evaluated and the addressed slave returns the telegram to the master unchanged.

Bit 5 = 0, bit 6 = 0 and bit 7 = 0 indicate normal data communication for devices. The node address (bit 0 to bit 4) is evaluated.

BCC

BCC stands for Block Check Character. It is an exclusive OR checksum (XOR) over all telegram bytes with the exception of the BCC itself.

6.3.2 User data range of the USS telegram

Basic parameters for communication with USS protocol via the RS485 interface

p2030	Fieldbus protocol selection: (0 no protocol, 1 USS, 2 Modbus)
p2020	USS baud rate: 2400 … 115200 baud
p2021	USS slave address: 0 … 30
p2022	USS PZD length: 0 … 2 … 8 words
p2023	USS parameter channel length: [0 no PKW component, 3 (3 words), 4 (4 words), 127 (variable length)]
r2029	Display receive errors on the fieldbus interface
p2040	USS telegram timeout: 0 … 65535 ms. 0 = no monitoring
r2050	Received PZD
p2051	BICO selection for PZD which must be sent
r2053	PROFIdrive diagnosis send PZD (word)
r2080 … r2089	Status word 1
r2090 … r2099	Control word 1

Structure of the user data

The user data range of the USS protocol is used to transmit application data. This comprises the parameter channel data and the process data (PZD).

The user data occupy the bytes within the USS frame (STX, LGE, ADR, BCC). The size of the user data can be configured using parameters p2023 and p2022. The structure and sequence of the parameter channel and process data (PZD) are shown in the figure below.

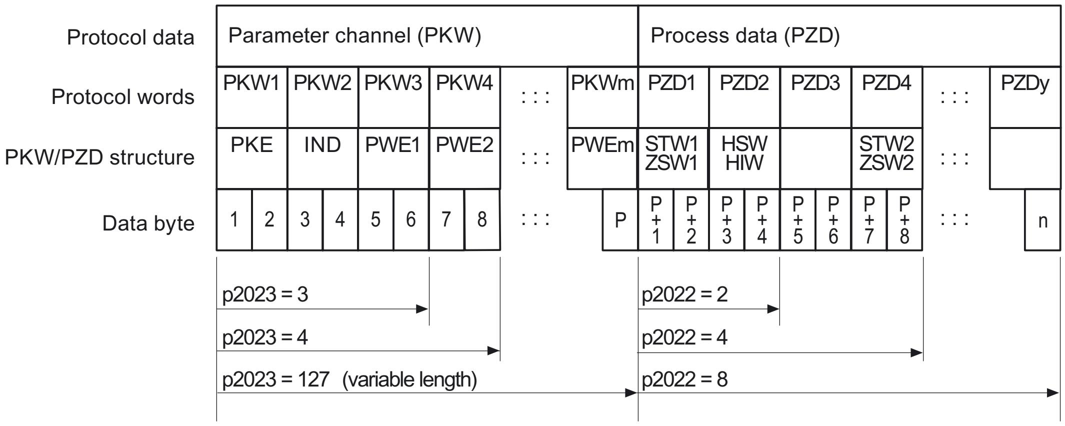


Figure 6-2 USS user data structure

The length for the parameter channel is determined by parameter p2023 and the length for the process data is specified by parameter p2022. If neither the parameter channel nor the PZD are required, the relevant parameters can be set to zero ("PKW only" or "PZD only").

It is not possible to transfer "PKW only" and "PZD only" alternatively. If both channels are required, they must be transferred together.

6.3.3 Data structure of the USS parameter channel

Description

The USS protocol defines for inverters the user data structure via which a master can access the slave inverter. The parameter channel can be used to monitor and change any parameters in the inverter.

Parameter channel

Process data can be edited and monitored (written/read) via the parameter channel, as described below. The parameter channel can be set to a fixed length of 3 or 4 data words or to a variable length.

The first data word always contains the parameter identifier (PKE) and the second contains the parameter index.

The third, fourth and subsequent data words contain parameter values, texts and descriptions.

Parameter identifier (PKE), first word

The parameter identifier (PKE) is always a 16-bit value.

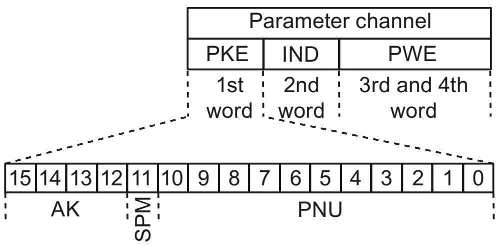


Figure 6-3 PKE structure

- Bits 0 to 10 (PNU) contain the remainder of the parameter number (value range 1 to 61999).

For parameter numbers ≥ 2000, an offset must be added that is defined using the upper bits of the IND byte.

- Bit 11 (SPM) is reserved and always = 0.
- Bits 12 to 15 (AK) contain the request or response identifier.

The significance of the request identifier for request telegrams (master → inverter) is described in the following table.

Table 6- 15 Request identifier (master → inverter)

Request identifier	Description	Response identifier	
		Positive	Negative
0	No request	0	7
1	Request parameter value	1 / 2	7
2	Change parameter value (word)	1	7
3	Change parameter value (double word)	2	7
4	Request descriptive element 1)	3	7
6	Request parameter value 1) 2)	4 / 5	7
7	Change parameter value (word) 1) 2)	4	7
8	Change parameter value (double word) 1) 2)	5	7

1) The required element of the parameter description is specified in IND (second word).

2) Identifier 1 is identical to identifier 6, ID 2 is identical to 7, and 3 is identical to 8. We recommend that you use identifiers 6, 7, and 8.

The significance of the response identifier for response telegrams (inverter → master) is described in the following table. The request identifier determines which response identifiers are possible.

Table 6- 16 Response identifier (inverter → master)

Response identifier	Description
0	No response
1	Transfer parameter value (word)
2	Transfer parameter value (double word)
3	Transfer descriptive element 1)
4	Transfer parameter value (field, word) 2)
5	Transfer parameter value (field, double word) 2)
6	Transfer number of field elements
7	Request cannot be processed, task cannot be performed (with error number)

1) The required element of the parameter description is specified in IND (second word).

2) The required element of the indexed parameter is specified in IND (second word).

If the response identifier is 7 (request cannot be processed), one of the error numbers listed in the following table will be saved in parameter value 2 (PWE2).

Table 6- 17 Error numbers for the response "Request cannot be processed"

No.	Description	Remarks
0	Impermissible parameter number (PNU)	Parameter does not exist
1	Parameter value cannot be changed	The parameter can only be read
2	Minimum/maximum not achieved or exceeded	–
3	Wrong subindex	–
4	No field	An individual parameter was addressed with a field request and subindex > 0
5	Wrong parameter type / wrong data type	Confusion of word and double word
6	Setting is not permitted (only resetting)	Index is outside the parameter field[]
7	The descriptive element cannot be changed	Description cannot be changed
11	Not in "master control" mode	Change request without "master control" mode (see p0927)
12	Keyword missing	–
17	Request cannot be processed on account of the operating state	The current inverter status is not compatible with the received request
101	Parameter number is currently deactivated	Dependent on the operating mode of the inverter
102	Channel width is insufficient	Communication channel is too small for response
104	Illegal parameter value	The parameter can only assume certain values
106	Request not included / task is not supported	After request identifier 5,11,12,13,14,15
200/201	Changed minimum/maximum not achieved or exceeded	The maximum or minimum can be limited further during operation
204	The available access authorization does not cover parameter changes	–

Parameter index (IND) second word

The field subindex is simply referred to as "subindex" in the PROFIdrive profile.

Data transfer structure

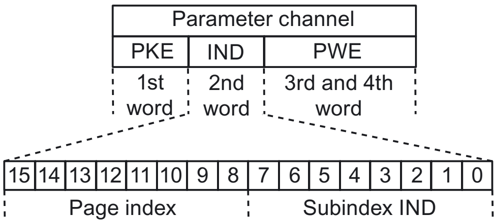


Figure 6-4 IND structure

- The field subindex is an 8-bit value that is transferred in the low-value byte (bits 0 to 7) of the parameter index (IND).
- The task of selecting parameter pages for additional parameters is performed in this case by the higher-value byte (bits 8 to 15) of the parameter index. This structure meets the requirements of the USS specification.

  Example: Coding a parameter number in PKE and IND for "p2029, Index 5"

PKE		IND		PWE1		PWE2	
xx	1D	80	05				

Rules for the parameter range

The bit for selecting the parameter page functions as follows:

When it is set to 1, an offset of 2000 is applied in the inverter to the parameter number (PNU) in the parameter channel request before transfer.

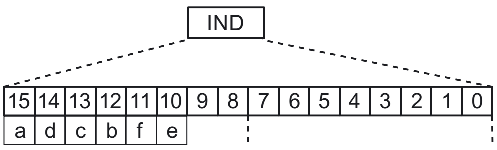


Figure 6-5 IND page index

Table 6- 18 Rules for setting PNU

Parameter range	Page index						Bit		Hex value	+ PNU
	a	d	c	b	f	e	9	8		
0000 … 1999	0	0	0	0	0	0	0	0	0x00	0 – 7CF
2000 … 3999	1	0	0	0	0	0	0	0	0x80	0 – 7CF
4000 … 5999	0	0	0	1	0	0	0	0	0x10	0 – 7CF
6000 … 7999	1	0	0	1	0	0	0	0	0x90	0 – 7CF
8000 … 9999	0	0	1	0	0	0	0	0	0x20	0 – 7CF
…	…	…	…	…	…	…	…	…	…	…
32.000 … 33.999	0	0	0	0	0	1	0	0	0x04	0 – 7CF
…	…	…	…	…	…	…	…	…	…	…
64.000 … 65.999	0	0	0	0	1	0	0	0	0x08	0 – 7CF

Table 6- 19 Coding example for a parameter number in PKE and IND for p2029, index 5

	PKE		IND	
Decimal	xx	29	128	05
Hex	xx	1D	80	05

Parameter value (PWE)

With communication over USS, the number of PWEs can vary. One PWE is required for 16-bit values. Two PWEs are required if 32-bit values are exchanged.

Note

U8 data types are transferred as U16 with the upper byte set to zero. U8 fields therefore require one PWE per index.

A parameter channel of 3 words represents a typical data telegram for exchanging 16-bit data or alarms. The mode with a fixed word length of 3 is used when p2013 = 3.

A parameter channel for 4 words is a typical data telegram for exchanging 32 bit data variables and requires p2013 = 4.

A parameter channel with a variable length is used with p2013 = 127. The telegram length between the master and slave can vary in terms of the number of PWEs.

When the length of the parameter channel is fixed (P2013 = 3 or 4), the master must always transmit either 3 or 4 words on the parameter channel accordingly. Otherwise the slave will not respond to the telegram. The response of the slave will also comprise 3 or 4 words. For a fixed length, 4 should be used because 3 is insufficient for many parameters (i.e. double words). For a variable length of parameter channel (p2013 = 127), the master will only send the number of words necessary for the task in the parameter channel. The response telegram is also no longer than necessary.

Rules for editing requests/responses

- A request or a response can only be referred to one parameter.
- The master must constantly repeat a request until it receives a suitable response.
- The master recognizes the response to a request that it sent by:
  - Evaluating the response identifier
  - Evaluating the parameter number (PNU)
  - Evaluating the parameter index (IND), if necessary, or
  - Evaluating the parameter value PWE, if necessary.
- The complete request must be sent in a telegram. Request telegrams cannot be subdivided. The same applies to responses.
- If response telegrams contain parameter values, the drive always returns the current parameter value when it repeats response telegrams.

6.3.4 Time-out and other errors

Telegram timeouts

The character runtime is important for timeout monitoring:

Table 6- 20 Character runtime

Baud rate in bits/s	Transmission time per character (= 11 bits)	Transmission time per bit	Character runtime
9600	1.146 ms	104.170 us	1.146 ms
19200	0.573 ms	52.084 us	0.573 ms
38400	0.286 ms	26.042 us	0.286 ms
115200	0.059 ms	5.340 us	0.059 ms

The figure below shows the meaning of "Residual runtime":

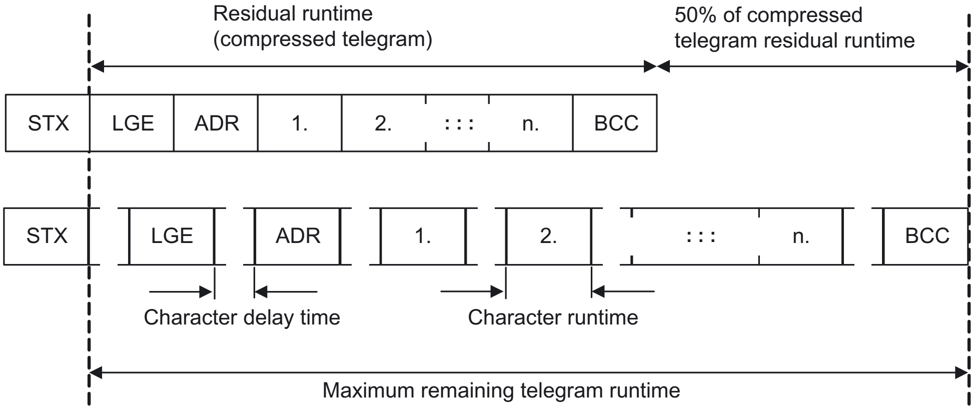


Figure 6-6 Residual runtime and character delay time

The character delay time can be zero, but it must always be lower than the start delay time.

The figure below shows the different delay times and transmission times:

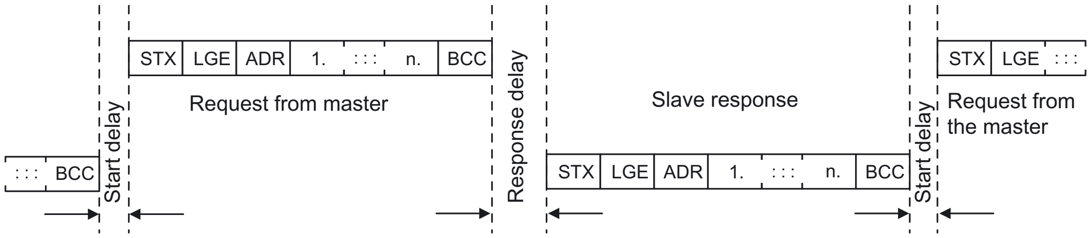


Figure 6-7 Start delay and response delay

Character delay time	Off time between characters; must be less than 2x character runtime, but can also be zero
Start delay	Off time between USS messages; must be > 2 * character runtime.
Response delay	Processing time of the slave; must be < 20 ms, but larger than the start delay.
Residual runtime	< 1.5 * (n + 3) * character runtime (whereby n = number of data bytes)
"Slave transfer"/ "Master transfer"	Sum of "Start delay", "Response delay" and "Residual runtime"

The master must check the following times:

- "Response delay" Response time of the slave to a USS request
- "Residual runtime" Transmission time of the response telegram sent from the slave

The slave must check the following times:

- "Start delay" Off time between USS messages
- "Residual runtime" Transmission time for a request telegram coming from the master

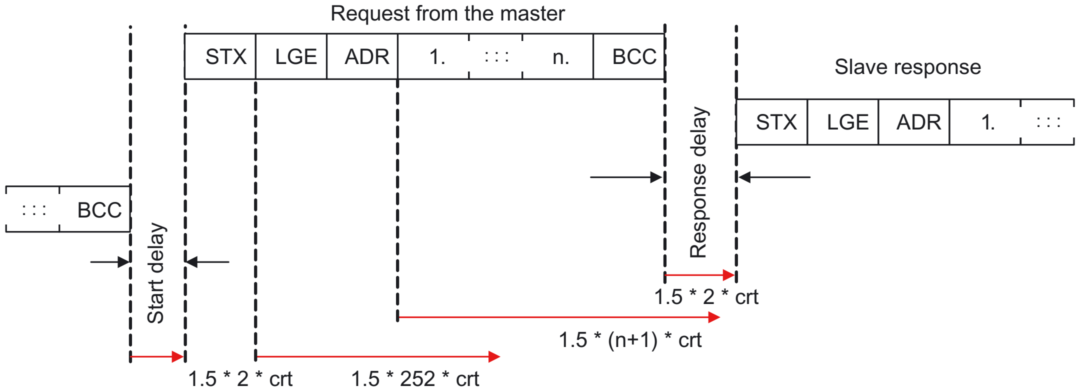


Figure 6-8 Timeout checks on the USS slave

The figure above shows the timeout ranges that have been verified on the USS slave. "crt" means "Character run time". The maximum range is a factor of 1.5. "Start delay" and minimum "Response delay" are values that are predefined in the software. The "Residual runtimes" monitor values that can cause a timeout if they are exceeded on the receipt of characters.

Process timeouts

Parameter p2040 defines the timeout in ms. The check for a timeout is suppressed by the value zero. Parameter p2040 checks the cyclic update of bit 10 in control word 1.

If the USS is configured as a command source for the drive and p2040 is not zero, bit 10 of the received control word 1 is checked. If the bit is not set, an internal timeout counter is increased. If the threshold defined in p2040 is reached, the drive sets a process timeout error.

6.3.5 USS process data channel (PZD)

Description

In this area of the telegram, process data (PZD) is continuously exchanged between the master and slave. Depending on the direction of transmission, the process data channel contains either request data for the USS slave or response data for the USS master. The request contains control words and setpoints for the slaves and the response contains status words and actual values for the master.

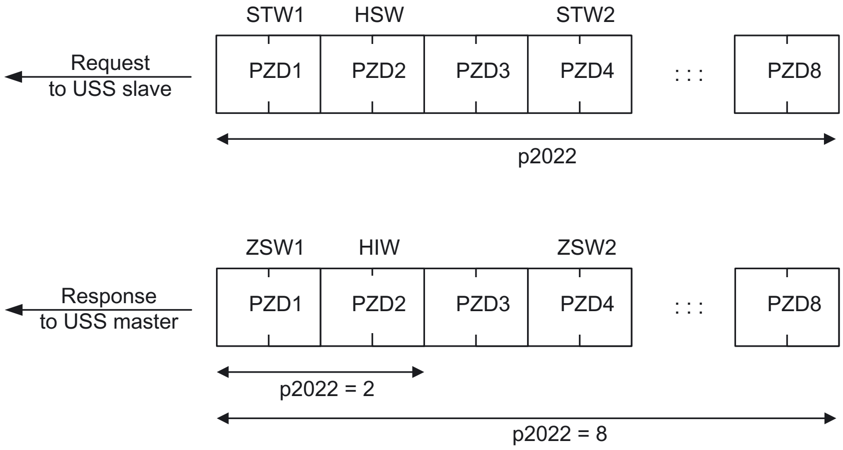


Figure 6-9 Process data channel

The number of PZD words in a USS telegram is defined by parameter p2022. The first two words are:

- Control word 1 (STW1) and main setpoint (HSW)
- Status word 1 (ZSW1) and main setpoint (HIW)

If P2022 is greater than or the same as 4, the additional control word (STW2) is transferred as the fourth PZD word (default setting).

The sources of all other PZD are defined by parameter p2051 for an RS485 interface.

6.4 Communication over Modbus RTU

Overview

The Modbus protocol is a communication protocol with linear topology based on a master/slave architecture.

Modbus offers three transmission modes:

- Modbus ASCII
  Data is transferred in ASCII code. The data can therefore be read directly by humans, however, the data throughput is lower in comparison to RTU.
- Modbus RTU
  Modbus RTU (RTU: Remote Terminal Unit): Data is transferred in binary format and the data throughput is greater than in ASCII code.
- Modbus TCP
  This type of data transmission is very similar to RTU, except that TCP/IP packages are used to send the data. TCP port 502 is reserved for Modbus TCP. Modbus TCP is currently undergoing definition as a standard (IEC PAS 62030 (pre-standard)).

CU230P-2 HVAC supports Modbus RTU as slave with even parity.

1 Bit Start	8 bits of data	1 Bit P-even	1 Bit stop

Communication interface

Communication using Modbus RTU takes place over the RS485 interface with a maximum of 247 slaves.

For information about connection, please refer to section Connecting the CU230P-2 HVAC via the RS485 interface (Page 30).

6.4.1 Parameters for Modbus communication settings

Parameters for setting communication via Modbus

P no.	Parameter name	Remark
p2030	Fieldbus telegram selection	0: No protocol 1: USS 2: Modbus 3: Profibus 4: CAN
p0700	Command source selection	2: Via terminals 6: Via fieldbus
p1000	Speed setpoint selection	0: No main setpoint 1: Via motorized potentiometer 2: Via analog setpoint 3: Via fixed speed setpoint 6: Via fieldbus 7: Via analog setpoint 2
p2020	Fieldbus baud rate	Baud rates from 4800 bit/s to 19200 bit/s can be set for communication, the factory setting is 19200 bit/s.
p2021	Fieldbus address	Selectable slave address from 1 to 247. Default value is 1.
p2024	Modbus timing (see Section "Baud rates and mapping tables (Page 106)")	**Index 0: Maximum slave telegram processing time:** The time after which the slave must have sent a response to the master. **Index 1: Character delay time:** Character delay time: Maximum permissible delay time between the individual characters in the Modbus frame. (Modbus standard processing time for 1.5 bytes). **Index2: Inter-telegram delay:** Maximum permissible delay time between Modbus telegrams. (Modbus standard processing time for 3.5 bytes).
p2029	Fieldbus error statistics	Displays receive errors on the fieldbus interface
p2040	Process data monitoring time	Determines the time after which an alarm is generated if no process data are transferred. Note: This time must be adapted depending on the number of slaves and the baud rate set for the bus (factory setting = 100 ms).

Possible causes of a timeout

Alarm No.	Parameter name	Remark
A1910	Setpoint timeout	The alarm is generated when p2040 ≠ 0 ms and one of the following causes is present: • The bus connection is interrupted • The MODBUS master is switched off • Communication error (CRC, parity bit, logical error) • An excessively low value for the fieldbus monitoring time (p2040)

6.4.2 Modbus RTU telegram

Description

For Modbus, there is precisely one master and up to 247 slaves. Communication is always triggered by the master. The slaves can only transfer data at the request of the master. Slave-to-slave communication is not possible. The CU230P-2 always operates as slave.

The following figure shows the structure of a Modbus RTU telegram.

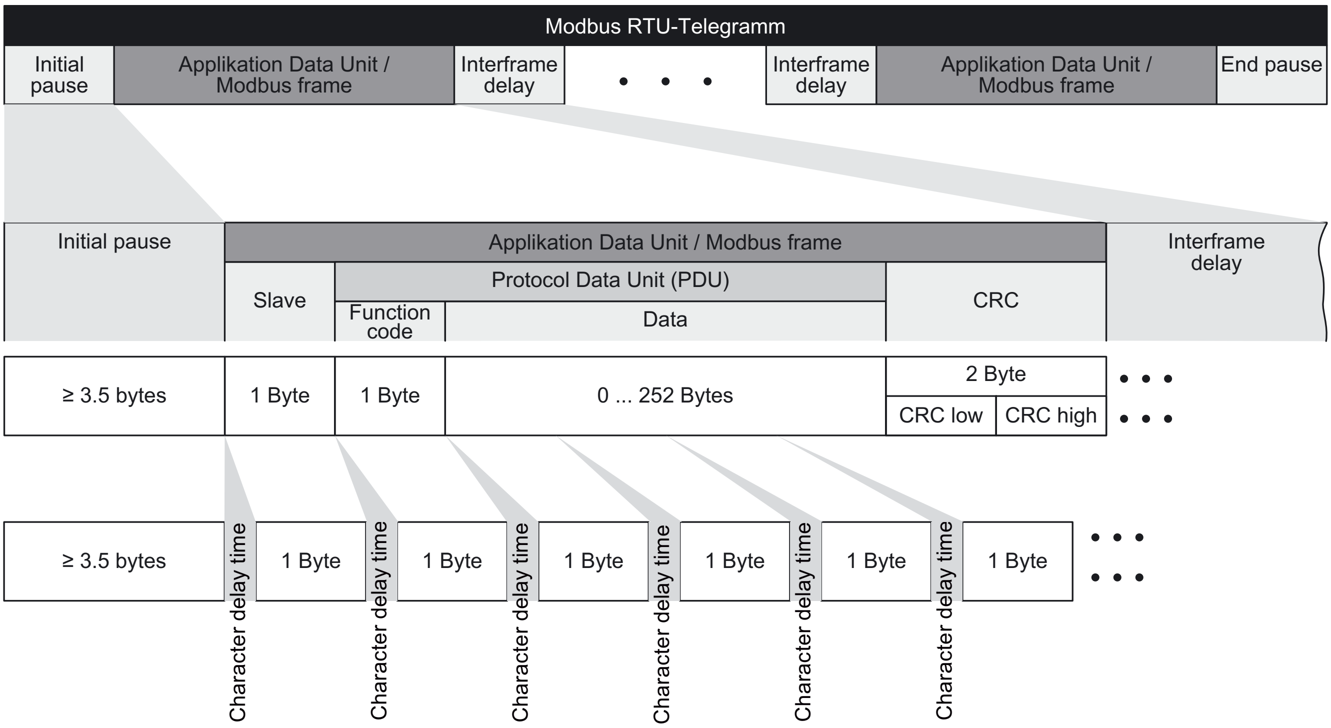


Figure 6-10 Modbus with delay times

The data area of the telegram is structured according to the mapping tables.

6.4.3 Baud rates and mapping tables

Permissible baud rates and telegram delay

The Modbus RTU telegram requires a pause for the following cases:

- Start detection
- Between the individual frames
- End detection

Minimum duration: Processing time for 3.5 bytes (can be set via p2024[2]).

A character delay time is also permitted between the individual bytes of a frame. Maximum duration: Processing time for 1.5 bytes (can be set via p2024[1]).

Table 6- 21 Baud rates, transmission times, and delays

Baud rate in bit/s (p2020)	Transmission time per character (11 bits)	Minimum pause between two telegrams (p2024[2])	Maximum pause between two bytes (p2024[1])
4800	2.292 ms	≥ 8.022 ms	≤ 3.4380 ms
9600	1.146 ms	≥4.011 ms	≤ 2.1900 ms
19200 (factory setting)	0.573 ms	≥ 2.0055 ms	≤ 0.8595 ms

Note

The factory setting for p2024[1] and p2024[2] is 0. The particular values are pre-assigned depending on the protocol selection (p2030) or the baud rate.

Modbus register and CU230P-2 parameters

Since the Modbus protocol can only handle register or bit numbers for addressing the memory, assignment to the appropriate control words, status words and parameters is performed on the slave side.

To ensure compatibility with the Micromaster MM436, two address areas are supported.

- MM436 40001 … 40065
- CU230P-2 HVAC from 40100 to 40522

The valid holding register address area extends from 40001 to 40522. Access to other holding registers generates the fault "Exception Code".

As a user, you can use both the register from the MM436 area and from the CU230P-2 HVAC area.

The registers 40100 to 40111 are described as process data. A telegram monitoring time can be activated in p2040 for these registers.

Note

R"; "W"; "R/W" in the column Modbus access stands for read (with FC03); write (with FC06); read/write.

Table 6- 22 Assigning the Modbus register to the parameters of the CU230P-2

Modbus Reg. No.	Description	Modbus access	Unit	Scaling factor	On/Off text or value range		Data / parameter
Process data							
Control data							
40100	Control word	R/W	--	1			Process data 1
40101	Main setpoint	R/W	--	1			Process data 2
Status data							
40110	Status word	R	--	1			Process data 1
40111	Main actual value	R	--	1			Process data 2
Parameter data							
Digital outputs							
40200	DO 0	R/W	--	1	HIGH	LOW	p0730, r747.0, p748.0
40201	DO 1	R/W	--	1	HIGH	LOW	p0731, r747.1, p748.1
40202	DO 2	R/W	--	1	HIGH	LOW	p0732, r747.2, p748.2
Analog outputs							
40220	AO 0	R	%	100	-100.0 … 100.0		r0774.0
40221	AO 1	R	%	100	-100.0 … 100.0		r0774.1
Digital inputs							
40240	DI 0	R	--	1	HIGH	LOW	r0722.0
40241	DI 1	R	--	1	HIGH	LOW	r0722.1
40242	DI 2	R	--	1	HIGH	LOW	r0722.2
40243	DI 3	R	--	1	HIGH	LOW	r0722.3
40244	DI 4	R	--	1	HIGH	LOW	r0722.4
40245	DI 5	R	--	1	HIGH	LOW	r0722.5
Analog inputs							
40260	AI 0	R	%	100	-300.0 … 300.0		r0755 [0]
40261	AI 1	R	%	100	-300.0 … 300.0		r0755 [1]
40262	AI 2	R	%	100	-300.0 … 300.0		r0755 [2]
40263	AI 3	R	%	100	-300.0 … 300.0		r0755 [3]
Inverter identification							
40300	Powerstack number	R	--	1	0 … 32767		r0200
40301	CU firmware	R	--	0.0001	0.00 … 327.67		r0018
Inverter data							
40320	Rated power of the power unit	R	kW	100	0 … 327.67		r0206
40321	Current Limit	R/W	%	10	10.0 … 400.0		p0640
40322	Rampup time	R/W	s	100	0.00 … 650.0		p1120
40323	Ramp-down time	R/W	s	100	0.00 … 650.0		p1121
40324	Reference speed	R/W	Hz	100	1.00 … 650.0		p2000
Inverter diagnostics							
40340	Speed setpoint	R	Hz	100	-327.68 … 327.67		r0020
40341	Actual speed value	R	RPM	1	-16250 … 16250		r0022
40342	Output frequency	R	Hz	100	- 327.68 … 327.67		r0024

Modbus Reg. No.	Description	Modbus access	Unit	Scaling factor	On/Off text or value range		Data / parameter
40343	Output voltage	R	V	1	0 … 32767		r0025
40344	DC link voltage	R	V	1	0 … 32767		r0026
40345	Actual value of current	R	A	100	0 … 163.83		r0027
40346	Actual torque value	R	Nm	100	- 325.00 … 325.00		r0031
40347	Actual active power	R	kW	100	0 … 327.67		r0032
40348	Energy consumption	R	kWh	1	0 … 32767		r0039
40349	Control priority	R	--	1	HAND	AUTO	r0807
Fault diagnostics							
40400	Fault number, Index 0	R	--	1	0 … 32767		r0947 [0]
40401	Fault number, Index 1	R	--	1	0 … 32767		r0947 [1]
40402	Fault number, Index 2	R	--	1	0 … 32767		r0947 [2]
40403	Fault number, Index 2	R	--	1	0 … 32767		r0947 [3]
40404	Fault number, Index 3	R	--	1	0 … 32767		r0947 [4]
40405	Fault number, Index 4	R	--	1	0 … 32767		r0947 [5]
40406	Fault number, Index 5	R	--	1	0 … 32767		r0947 [6]
40407	Fault number, Index 6	R	--	1	0 … 32767		r0947 [7]
40408	Alarm number	R	--	1	0 …32767		r2110 [0]
40499	PRM ERROR code	R	--	1	0 …99		--
Technology controller							
40500	Technology controller enable	R/W	--	1	0 … 1		p2200, r2349.0
40501	Technology controller MOP	R/W	%	100	-200.0 … 200.0		p2240
Technology controller adjustment							
40510	Time constant for actual value filter of the technology controller	R/W	--	100	0.00 … 60.0		p2265
40511	Scaling factor for actual value of the technology controller	R/W	%	100	0.00 … 500.00		p2269
40512	Proportional amplification of the technology controller	R/W	--	1000	0.000 … 65.000		p2280
40513	Integral time of the technology controller	R/W	s	1	0 … 60		p2285
40514	Time constant D-component of the technology controller	R/W	--	1	0 … 60		p2274
40515	Max. limit of technology controller	R/W	%	100	-200.0 … 200.0		p2291
40516	Min. limit technology controller	R/W	%	100	-200.0 … 200.0		p2292
PID diagnostics							
40520	Effective setpoint acc. to internal technology controller MOP ramp-function generator	R/W	%	100	-100.0 … 100.0		r2250
40521	Actual value of technology controller after filter	R/W	%	100	-100.0 … 100.0		r2266
40522	Output signal technology controller	R/W	%	100	-100.0 … 100.0		r2294

6.4.4 Write and read access via FC 3 and FC 6

Function codes used

For data exchange between the master and slave, predefined function codes are used for communication via Modbus.

The CU230P-2 uses the Modbus function code 03, FC 03, (read holding registers) for reading and the Modbus function code 06, FC 06, (preset single register) for writing.

Structure of a read request via Modbus function code 03 (FC 03)

All valid register addresses are permitted as a start address. If a register address is invalid, exception code 02 (invalid data address) is returned. An attempt to read a write-only register or a reserved register is replied to with a normal telegram in which all values are set to 0.

Using FC 03, it is possible to address more than 1 register with one request. The number of addressed registers is contained in bytes 4 and 5 of the read request.

Number of registers

If more than 125 registers are addressed, exception code 03 (Illegal data value) is returned. If the start address plus the number of registers for an address are outside of a defined register block, exception code 02 (invalid data address) is returned.

Table 6- 23 Structure of a read request for slave number 17

Example		
	Byte	Description
11 h	0	Slave address
03 h	1	Function code
00 h	2	Register start address "High" (register 40110)
6D h	3	Register start address "Low"
00 h	4	No. of registers "High" (2 registers: 40110; 40111)
02 h	5	Number of registers "Low"
xx h	6	CRC "Low"
xx h	7	CRC "High"

The response returns the corresponding data set:

Table 6- 24 Slave response to the read request

Example		
	Byte	Description
11 h	0	Slave address
03 h	1	Function code
04 h	2	No. of bytes (4 bytes are returned)
11 h	3	Data of first register "High"
22 h	4	Data of first register "Low"
33 h	5	Data of second register "High"
44 h	6	Data of second register "Low"
xx h	7	CRC "Low"
xx h	8	CRC "High"

Structure of a write request via Modbus function code 06 (FC 06)

The start address is the holding register address. If an incorrect address is entered (a holding register address does not exist), exception code 02 (invalid data address) is returned. An attempt to write to a read-only register or a reserved register is replied to with a Modbus error telegram (Exception Code 4 - device failure). In this instance, the detailed internal error code that occurred on the last parameter access via the holding registers can be read out via holding register 40499.

Using FC 06, precisely one register can always be addressed with one request. The value which is to be written to the addressed register is contained in bytes 4 and 5 of the write request.

Table 6- 25 Structure of a write request for slave number 17

Example		
	Byte	Description
11 h	0	Slave address
06 h	1	Function code
00 h	2	Register start address "High" (write register 40100)
63 h	3	Register start address "Low"
55 h	4	Register data "High"
66 h	5	Register data "Low"
xx h	6	CRC "Low"
xx h	7	CRC "High"

The response returns the register address (bytes 2 and 3) and the value (bytes 4 and 5) that was written to the register.

Table 6- 26 Slave response to the write request

Example		
	Byte	Description
11 h	0	Slave address
06 h	1	Function code
00 h	2	Register start address "High"
63 h	3	Register start address "Low"
55 h	4	Register data "High"
66 h	5	Register data "Low"
xx h	6	CRC "Low"
xx h	7	CRC "High"

6.4.5 Communication procedure

Procedure for communication in a normal case

Normally, the master sends a telegram to a slave (address range 1 ... 247). The slave sends a response telegram to the master. This response telegram mirrors the function code, and the slave enters its own address in the telegram, which enables the master to assign the slave.

The slave only processes orders and telegrams which are directly addressed to it.

Communication errors

If the slave detects a communication error on receipt (parity, CRC), it does not send a response to the master (this can lead to "setpoint timeout").

Logical error

If the slave detects a logical error within a request, it responds to the master with an "exception response". In the response, the highest bit in the function code is set to 1. If the slave receives, for example, an unsupported function code from the master, the slave responds with an "exception response" with code 01 (Illegal function code).

Table 6- 27 Overview of exception codes

Exception code	Modbus name	Remark
01	Illegal function code	An unknown (not supported) function code was sent to the slave.
02	Illegal Data Address	An invalid address was requested.
03	Illegal data value	An invalid data value was detected.
04	Server failure	Slave has terminated during processing.

Maximum processing time, p2024[0]

For error-free communication, the slave response time (time within which the Modbus master expects a response to a request) must have the same value in the master and the slave (p2024[0] in the inverter).

Process data monitoring time (setpoint timeout), p2040

The alarm "Setpoint timeout" (F1910) is issued by the Modbus if p2040 is set to a value > 0 ms and no process data are requested within this time period.

The alarm "Setpoint timeout" only applies for access to process data (40100, 40101, 40110, 40111). The alarm "Setpoint timeout" is not generated for parameter data (40200 … 40522).

Note

This time must be adapted depending on the number of slaves and the baud rate set for the bus (factory setting = 100 ms).

6.5 Communication over CANopen

General information

For information about connection, please refer to section Connecting CU230P-2 CAN to CAN bus (Page 36).

6.5.1 Objects to access SINAMICS parameters

Description

All inverter parameters can be addressed via the SDO parameter channel in the 2000 hex to 470F hex range of the object directory.

The parameter numbers of the inverter must be converted to hex values and an offset of 2000 hex must also be applied. This figure is the object number contained in the SDO request for the parameter number of the inverter.

There is no need to distinguish between write and display parameters (p and r parameters).

Table 6- 28 Examples of conversions of inverter parameters to manufacturer-specific CANopen objects

Inverter parameter	CANopen object number	Name of inverter parameter
P0010	200A hex	Commissioning parameter filter
r0062	203E hex	Speed setpoint
r0947	23B3 hex	Fault number

Setting accessing options

The access settings are entered in parameter p8630[0…2]

p8630[0]: Access to virtual CANopen objects

- p8630[0] = 0: No access to virtual CANopen objects
- p8630[0] = 1: Access to virtual CANopen objects
- p8630[0] = 2: Not relevant for G120 inverters

p8630[1]: Selection of index range of inverter parameters

A maximum total of 255 indices can be transferred in a CANopen object. Should it be necessary to transfer parameters with more indices, an additional CANopen object must be set up. It is possible to transfer a maximum of 1024 indices.

- p8630[1] = 0: 0 … 255
- p8630[1] = 1: 256 … 511
- p8630[1] = 2: 512 … 767
- p8630[1] = 3: 768 … 1023

P8630[2] Parameter range selection

- P8630[2] = 0: 1 ... 9999

Note

Since all parameter numbers of the G120 inverter are in the < 10000 range, it is only necessary to set P8630[2] = 0.

If the inverter were to use parameter numbers ≥ 10000, it would be possible to set ranges from 10000 … 39999 using indices 1 … 3.

6.5.2 CANopen functionality of the CU230P-2 CAN

Introduction

CANopen is a CAN-based communication protocol with linear topology that operates on the basis of communication objects (COB).

The communication objects can be subdivided as follows:

- Service data objects (SDO) for reading and changing parameters
- Process data objects (PDO) for transferring process data
- Network management objects (NMT) for controlling CANopen communication and for monitoring the individual nodes on the basis of a master-slave relationship.
- Additional objects such as synchronize (SYNC), time stamp and error messages (EMCY).

The CU230P-2 CAN supports the following communication objects:

- NMT
- PDO
- SDO
- SYNC
- EMCY

The CU230P-2 CAN works with communication objects from the following profiles:

- CANopen communication profile DS 301 version 4.0
- Device profile DSP 402 (drives and motion control) version 2.0
- Indicator profile DR303-3 version 1.0.

The communication monitoring can be used via both node guarding and heartbeat protocol (heartbeat producer).

Note

Communication status after CAN controller state "Bus off" (inverter fault F8700, fault value 1)

If this fault is acknowledged by an OFF/ON, the bus off state is also canceled and communication is re-established after the next boot.

If the fault is acknowledged by means of DI2, or directly with p3981, the inverter remains in the bus off state. To initialize the communication links, p8608 = 1 must be set in this case.

WARNING

If a bus fault has occurred, the CAN communication links are not restored if the fault is acknowledged by means of digital input DI2 (p3981 = 1).

If p8641 = 0 is set (inverter does not switch to fault status after a bus fault), it means that the motor cannot be stopped via the control system unless communication has been restored beforehand with p8608 = 1.

6.5.3 General CANopen functions

6.5.3.1 COB-ID

Introduction

Each COB (communication object) can be uniquely identified by means of an identifier (COB ID), which is a part of the COB. CAN specification 2.0A supports up to 2048 COBs, which are identified by means of 11-bit identifiers.

A list of COB identifiers, which contains all the COBs that can be accessed via CAN, is available in the object directory for the relevant SINAMICS inverter.

The COB ID can be used to prioritize the communication objects. This means in particular that different processing methods (cyclic, event-controlled, on request or synchronized) can be defined for different process data.

COB IDs for SINAMICS

The following table contains the COB IDs for transmit and receive telegrams defined for the "Predefined Connection Set" for SINAMICS inverters (drive objects). The object directory index (OD index) starts at 1800 for the TPDO and at 1400 for the RPDO.

Table 6- 29 Identifier assignment

Communication objects	Function code		Resulting COB ID		OD index (hex)
	dec	bin	hex	Explanation	
TPDO	3	0011	181–1FF	180 hex + node ID	1800
RPDO	4	0100	201–27F	200 hex + node ID	1400

6.5.3.2 Network management (NMT service)

Introduction

Network management (NMT) is node-oriented and has a master-slave structure.

The NMT services can be used to initialize, start, monitor, reset, or stop nodes. All NMT services have the COB ID = 0. This cannot be changed.

The SINAMICS inverter is an NMT slave and can adopt the following statuses in CANopen:

- Initialization
  The inverter passes through this state after Power On. In the factory setting, the inverter then enters the "Pre-Operational" state, which also corresponds to the CANopen standard. The setting can be changed using p8684 as follows:
  - p8684 = 4 Stopped
  - p8684 = 5 Operational
  - p8684 = 127 Pre-Operational (factory setting)
- Pre-Operational
  In this state, the node cannot process any process data (PDO). It can, however, be parameterized or operated via SDOs, which means that setpoints can also be specified using SDO.
- Operational
  In this state, the node can process both SDO and PDO.
- Stopped
  In this state, the node cannot process either PDO or SDO. Stopped mode is exited by specifying one of the following commands:
  - Enter Pre-Operational
  - Start Remote Node
  - Reset Node
  - Reset Communication

The NMT recognizes the following transitional states:

- Start Remote Node:
  command for switching from the "Pre-Operational" communication status to "Operational". The drive can only transmit and receive process data (PDO) in "Operational" status.
- Stop Remote Node
  command for switching from "Pre-Operational" or "Operational" to "Stopped". The node can only process NMT commands in the "Stopped" status.
- Enter Pre-Operational
  command for switching from "Operational" or "Stopped" to "Pre-Operational". In this state, the node cannot process any process data (PDO). It can, however, be parameterized or operated via SDOs, which means that setpoints can also be specified.

- Reset Node:
  command for switching from "Operational", "Pre-Operational", or "Stopped" to "Initialization". When the Reset Node command is issued, all the objects (1000 hex - 9FFF hex) are reset to the status that was present after "Power On".
- Reset Communication:
  command for switching from "Operational", "Pre-Operational", or "Stopped" to "Initialization". When the Reset Communication command is issued, all communication objects (1000 hex - 1FFF hex) are reset to the status that was present after "Power On".

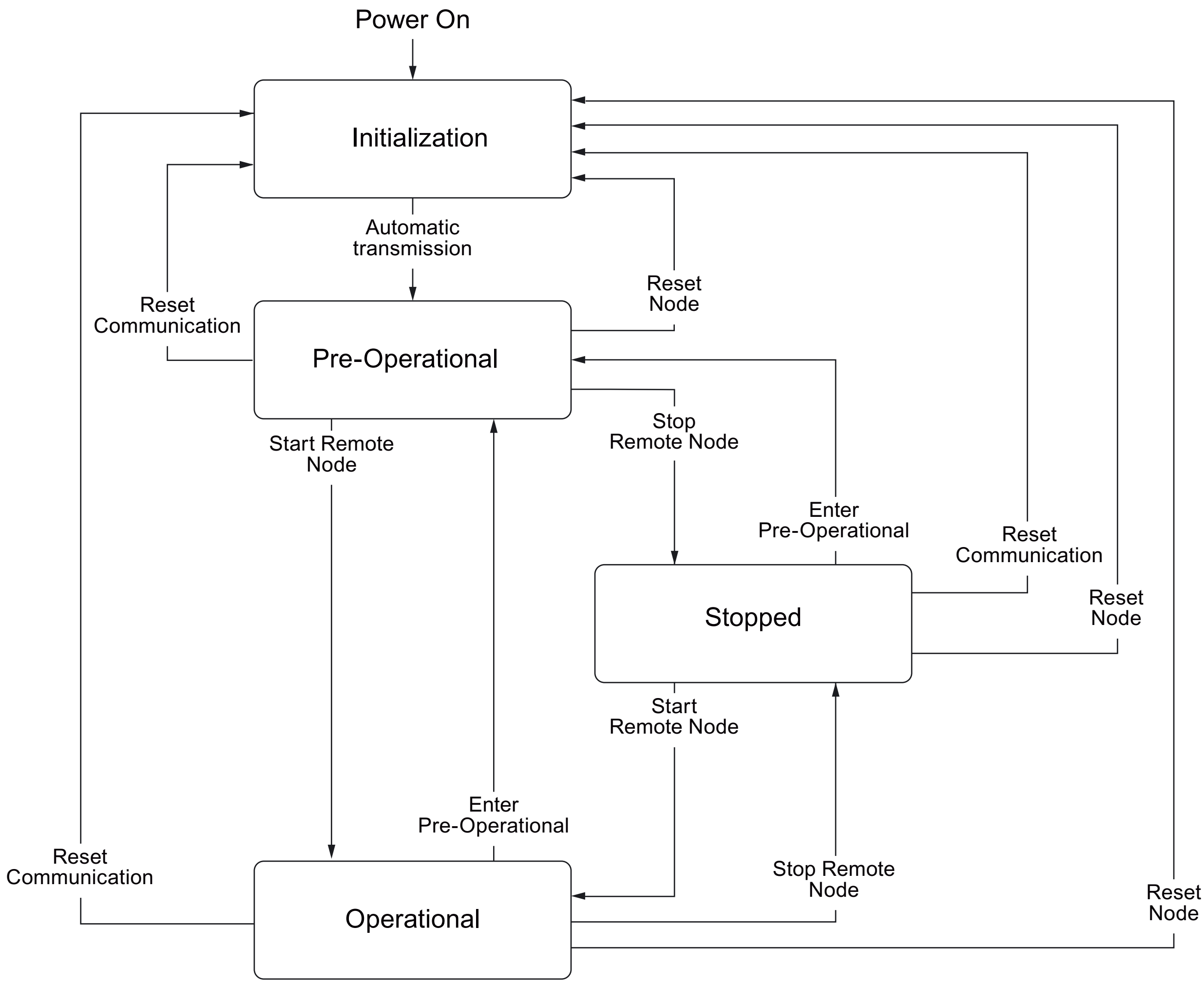


Figure 6-11 CANopen status diagram

The transition states and addressed nodes are displayed using the command specifier and the Node_ID:

Table 6- 30 Overview of NMT commands

NMT Master Request ----> NMT Slave message		
Command	**Byte 0 (command specifier, CS)**	**Byte 1**
Start	1 (01hex)	Node ID of the addressed node
Stop	2 (02hex)	Node ID of the addressed node
Enter Pre-Operational	128 (80hex)	Node ID of the addressed node
Reset Node	129 (81hex)	Node ID of the addressed node
Reset Communication	130 (82 hex)	Node ID of the addressed node

The NMT master can simultaneously direct a request to one or more slaves. The following is applicable:

- Requirement of a slave:
  The slave is addressed using its node ID (1 - 127).
- Requirement for all slaves:
  Node ID = 0

The current state of the node is displayed via p8685. It can also be changed directly using this parameter:

- p8685 = 0 Initializing (cannot be changed)
- p8685 = 4 Stopped
- p8685 = 5 Operational
- p8685 = 127 Pre-Operational (factory setting)
- p8685 = 128 Reset Node
- p8685 = 129 Reset Communication

6.5.3.3 PDO and PDO services

PDO

The real-time data transfer for CANopen takes place using "Process Data Objects (PDO)".

In the configuration, the PDOs are linked with objects from the object directory that require real-time data transfer (PDO mapping).

The number of PDOs and their mapping structure is transferred to the device via SDO services (during configuration of the device).

The following variants of PDO exist:

- TPDO (Transmit PDO): Transmits data
- RPDO (Receive PDO): Receives data.

CANopen devices with TPDO are known as PDO producers and CANopen devices with RPDO are known as PDO consumers.

The PDO is defined by the PDO communication parameter and the PDO mapping parameter. The structure of these two parameters is listed in the following tables.

Table 6- 31 PDO communication parameters 1400h ff (RPDO), 1800h ff (TPDO)

Subindex	Name	Data type
00h	Highest subindex that is supported	UNSIGNED8
01h	COB ID	UNSIGNED32
02h	Transfer mode	UNSIGNED8
03h	Inhibit time (only for TPDO)	UNSIGNED16
04h	Reserved (only for TPDO)	UNSIGNED8
05h	Event timer (only for TPDO)	UNSIGNED16

Table 6- 32 PDO mapping parameters 1600h ff (RPDO), 1A00h ff (TPDO)

Subindex	Name	Data type
00h	Number of objects mapped to the PDO (max. 4)	UNSIGNED8
01h	First mapped object	UNSIGNED32
02h	Second mapped object	UNSIGNED32
03h	Third mapped object	UNSIGNED32
04h	Fourth mapped object	UNSIGNED32

Note

PDO communication parameter

- For receive telegrams: p8700 to p8707,
- For transmit telegrams: p8720 to p8727.

PDO mapping parameter

- For receive telegrams: p8710 to p8717,
- For transmit telegrams: p8730 to p8737.

Transmission modes for process data objects (PDO)

The following types of transmission are possible for PDO:

- Synchronous data transmission
- Asynchronous data transmission

Synchronous data transmission

In order for the devices on the CANopen bus to remain synchronized during transmission, a synchronization object (SYNC object) must be transmitted at periodic intervals.

Each PDO that is transferred as a synchronous object must be assigned a transmission type 1 ... n. The following is applicable:

- Transmission type 1: The PDO is transferred in every SYNC cycle.
- Transmission type n: The PDO is transferred in every nth SYNC cycle.

The following diagram shows the principle of synchronous and asynchronous transmission:

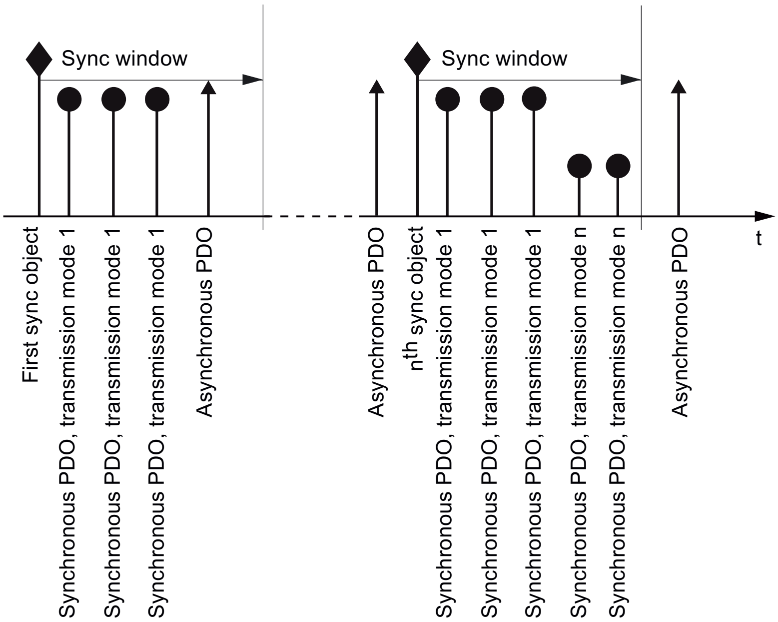


Figure 6-12 Principle of synchronous and asynchronous transmission

For synchronous TPDOs, the transmission mode also identifies the transmission rate as a factor of the SYNC object transmission intervals. Here, transmission type "1" means that the message will be transmitted in every SYNC object cycle. Transmission type "n" means that the message will be transmitted in every nth SYNC object cycle.

Data from synchronous RPDOs that are received after a SYNC signal is not transmitted to the application until after the next SYNC signal.

Note

The SYNC signal does not synchronize the applications in the SINAMICS drive, only the communication on the CANopen bus

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Asynchronous data transmission

Asynchronous PDOs are transferred - cyclically or acyclically - without reference to the SYNC signal.

PDO services

The PDO services can be subdivided as follows:

- Write PDO
- Read PDO
- SYNC service

Write PDO

The "Write PDO" service is based on the "push" model. The PDO has exactly one producer. There can be no consumer, one consumer, or multiple consumers.

Via Write PDO, the producer of the PDO sends the data of the mapped application object to the individual consumer.

Read PDO

The "Read PDO" service is based on the "pull" model. The PDO has exactly one producer. There can be one consumer or multiple consumers.

Via Read PDO, the consumer of the PDO receives the data of the mapped application object from the producer.

SYNC service

The SYNC object is periodically sent from the SYNC producer. The SYNC signal represents the basic network cycle. The time interval between two SYNC signals is determined in the master by the standard parameter "Communication cycle time".

In order to ensure CANopen accesses in real-time, the SYNC object has a high priority, which is defined using the COB ID. This can be changed via p8602 (factory setting = 80hex). The service runs unconfirmed.

Note

The COB ID of the Sync object must be set to the same value for all nodes of a bus that respond to the SYNC telegram from the master.

The COB ID of the SYNC object is defined in the object 1005h.

6.5.3.4 PDO mapping

Introduction

PDO mapping is used to link (map) standardized drive objects (process data, e.g. setpoints or actual values) and "free objects" from the object directory for the PDO service to telegrams.

The PDO transfers the data values for these objects.

For this purpose, a maximum of 8 receive and 8 transmit PDOS are available.

A CAN telegram can transfer up to 8 bytes of user data. The user uses the mapping to decide which user data is to be transferred in a PDO.

Example

The following diagram shows an example of PDO mapping (values are hexadecimal (e.g. object size 10 hex = 16 bits)):

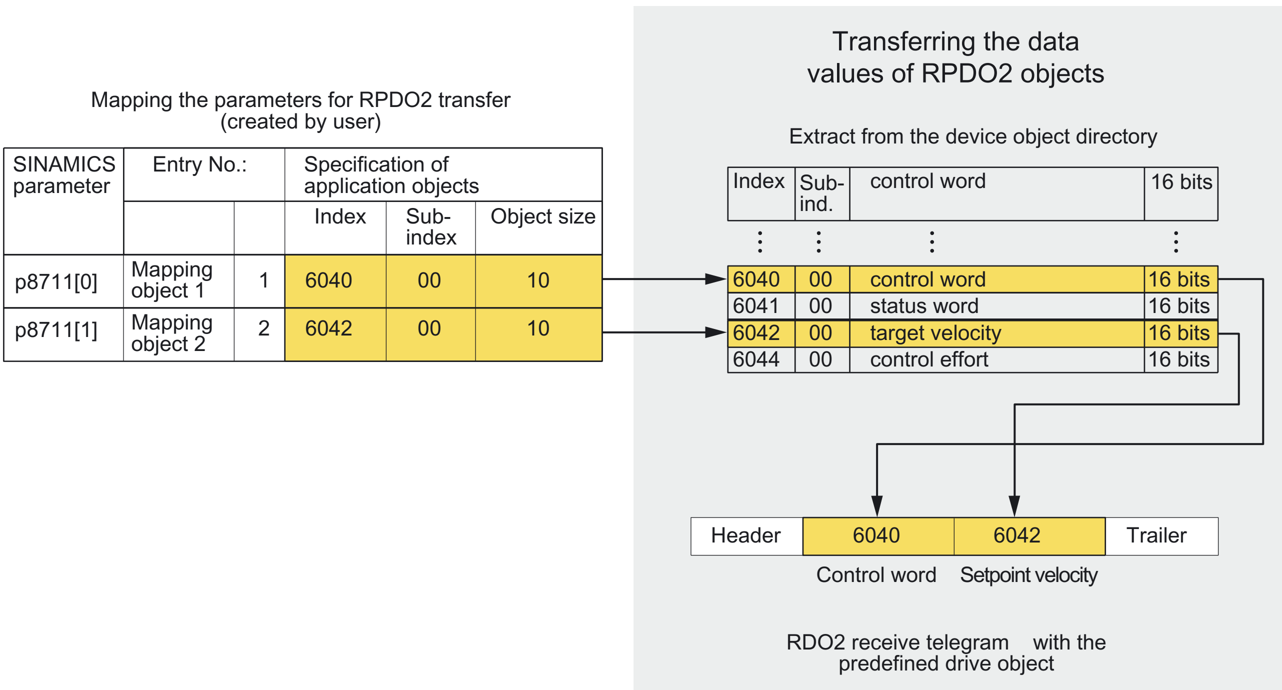


Figure 6-13 PDO mapping for control word and setpoint velocity

6.5.3.5 SDO services

Introduction

SDO services allow you to access the object directory for the connected drive unit. An SDO connection is a peer-to-peer connection between an SDO client and a server.

The drive unit with its object directory is an SDO server.

The identifiers for the first SDO channel of a drive unit are defined according to CANopen as follows:

Receiving:	Server <= Client:	COB ID = 600 hex + node ID
Transmitting:	Server => Client:	COB ID = 580 hex + node ID

Properties

The SDOs have the following properties:

- Confirmed transfer of objects
- Transmission always asynchronous (corresponds to acyclic data exchange with PROFIBUS DB)
- Transmission of values > 4 bytes (normal transfer)
- Transmission of values ≤ 4 bytes (expedited transfer)
- All drive unit parameters can be addressed via SDO.

Structure of the SDO protocols

The SDO services use the appropriate protocol depending on the task. The most important are:

- SDO Protocol Write
- SDO Protocol Read
- SDO Protocol Cancel Transfer Protocol

SDO Protocol Write

This protocol is used to write data to the drive unit. Request by "Write Request", confirmation by "Write Response".

Table 6- 33 SDO Protocol Write

Byte 0	Bytes 1 and 2	Byte 3	Bytes 4 ... 7
Write Request (request from the CANopen master to the inverter to write data)			
cs = 2Fhex	index	subindex	data byte 4
cs = 2Bhex	index	subindex	data byte 4-5
cs = 27hex	index	subindex	data byte 4-6
cs = 23hex	index	subindex	data byte 4-7
Write Response (confirmation from the inverter to the master after successful writing)			
Cs = 60	index	subindex	Reserved

SDO Protocol Read

This protocol is used to read data from the drive unit. Read request by "Read Request", confirmation by "Read Response".

Table 6- 34 SDO Protocol Read

Byte 0	Bytes 1 and 2	Byte 3	Bytes 4 ... 7
Read Request (request from the CANopen master to the inverter to read data and send it to the master)			
cs = 40	index	subindex	Reserved
Read Response (data from the inverter is sent to the master)			
cs = 4Fhex	index	subindex	data byte 4
cs = 4Bhex	index	subindex	data byte 4-5
cs = 47hex	index	subindex	data byte 4-6
cs = 43hex	index	subindex	data byte 4-7

SDO Protocol Cancel Transfer Protocol

This protocol is used to perform the SDO service "Cancel Transfer Protocol".

Table 6- 35 SDO Protocol Cancel Transfer Protocol

Master -> Slave / Slave -> Master			
Byte 0	Bytes 1 and 2	Byte 3	Bytes 4 ... 7
Error Response			
cs = 80	index	subindex	SDO abort code (unsigned 32)

SDO abort codes

Table 6- 36 SDO abort codes

Abort code	Description
0503 0000h	Toggle bit not alternated
0504 0000h	SDO protocol timed out
0504 0001h	Client/server command specifier not valid or unknown
0504 0002h	Invalid block size (block mode only)
0504 0003h	Invalid sequence number (block mode only)
0504 0004h.	CRC error (block mode only)
0504 0005h	Out of memory
0601 0000h	Unsupported access to an object
0601 0001h	Attempt to read a write-only object
0601 0002h	Attempt to write a read-only object
0602 0000h	Object does not exist in the object dictionary
0604 0041h	Object cannot be mapped to the PDO
0604 0042h	The number and length of the objects to be mapped would exceed PDO length
0604 0043h	General parameter incompatibility reason
0604 0047h	General internal incompatibility in the device
0602 0000h	Object does not exist in the object dictionary
0604 0041h	Object cannot be mapped to the PDO
0604 0042h	The number and length of the objects to be mapped would exceed PDO length
0604 0043h	General parameter incompatibility reason
0604 0047h	General internal incompatibility in the device
0606 0000h	Access failed due to a hardware error
0607 0010h	Data type does not match, length of service parameter does not match.
0607 0012h	Data type does not match, length of service parameter too high.
0607 0013h	Data type does not match, length of service parameter too low.
0609 0011h	Subindex does not exist
0609 0030h	Value range of parameter exceeded (only for write access)
0609 0031h	Value of parameter written too high
0609 0032h	Value of parameter written too low
0609 0036h	Maximum value is less than minimum value
0800 0000h	General error
0800 0020h	Data cannot be transferred to or stored in the application
0800 0021h	Data cannot be transferred to or stored in the application because of local control
0800 0022h	Data cannot be transferred to or stored in the application because of the current device state.
0800 0023h	Object dictionary dynamic generation failed or no object dictionary is present (e.g. object dictionary is generated from file and generation fails because of a file error).

6.5.4 Communication objects

6.5.4.1 Overview

Section content

This section lists the objects (data values) that are used in SINAMICS for communication via CANopen. The individual objects are:

- Configuration objects
- Manufacturer-specific objects
- Objects in drive profile DSP402

The objects are stored in the object directory of the inverter.

6.5.4.2 Configuration objects

Introduction

Eight PDOs per drive can be parameterized for transmitting and receiving.

For each PDO, the following configuration objects are available:

- Communication parameters and
- Mapping parameters (max. 8 bytes)

Rule

The "predefined connection set" column contains the predefined values for the "predefined connection set".

Communication parameters and indices for the configuration objects of the receive PDO

The following table lists the communication parameters together with the indices for the individual configuration objects of the receive PDOs:

Table 6- 37 Configuration objects of receive PDOs - communication parameters

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Transmission	Data type	Predefined connection set	Read/ write
1400		Receive PDO 1 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8700.0	SDO	Unsigned32	200 hex + node ID	R/W
	2	Transmission type	p8700.1	SDO	Unsigned8	FE hex	R/W
1401		Receive PDO 2 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8701.0	SDO	Unsigned32	300 hex + node ID	R/W
	2	Transmission type	p8701.1	SDO	Unsigned8	FE hex	R/W
1402		Receive PDO 3 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8702.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8702.1	SDO	Unsigned8	FE hex	R/W
1403		Receive PDO 4 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8703.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8703.1	SDO	Unsigned8	FE hex	R/W
1404		Receive PDO 5 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8704.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8704.1	SDO	Unsigned8	FE hex	R/W
1405		Receive PDO 6 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8705.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8705.1	SDO	Unsigned8	FE hex	R/W
1406		Receive PDO 7 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8706.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8706.1	SDO	Unsigned8	FE hex	R/W
1407		Receive PDO 8 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	2	R
	1	COB ID used by PDO	p8707.0	SDO	Unsigned32	8000 06DF hex	R/W
	2	Transmission type	p8707.1	SDO	Unsigned8	FE hex	R/W

Table 6- 38 Configuration objects receive PDO - mapping parameters

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Transmission	Data type	Predefined connection set	Read/ write
1600		Receive PDO 1 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	1	R
	1	PDO mapping for the first application object to be mapped	p8710.0	SDO	Unsigned32	6040 hex	R/W
	2	PDO mapping for the second application object to be mapped	p8710.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8710.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8710.3	SDO	Unsigned32	0	R/W
1601		Receive PDO 2 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	2	R
	1	PDO mapping for the first application object to be mapped	p8711.0	SDO	Unsigned32	6040 hex	R/W
	2	PDO mapping for the second application object to be mapped	p8711.1	SDO	Unsigned32	6042 hex	R/W
	3	PDO mapping for the third application object to be mapped	p8711.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8711.3	SDO	Unsigned32	0	R/W
1602		Receive PDO 3 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8712.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8712.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8712.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8712.3	SDO	Unsigned32	0	R/W
1603		Receive PDO 4 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8713.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8713.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8713.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8713.3	SDO	Unsigned32	0	R/W

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Transmission	Data type	Predefined connection set	Read/ write
1604		Receive PDO 5 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8714.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8714.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8714.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8714.3	SDO	Unsigned32	0	R/W
1605		Receive PDO 6 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8715.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8715.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8715.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8715.3	SDO	Unsigned32	0	R/W
1606		Receive PDO 7 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8716.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8716.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8716.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8716.3	SDO	Unsigned32	0	R/W
1607		Receive PDO 8 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8717.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8717.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8717.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8717.3	SDO	Unsigned32	0	R/W

Communication parameters and indices for the configuration objects of the send PDO

The following table lists the communication parameters together with the indices for the individual configuration objects of the transmit PDOs:

Table 6- 39 Configuration objects of transmit PDOs - communication parameters

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Trans-mission	Data type	Predefined connection set	Read/ write
1800		Transmit PDO 1 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8720.0	SDO	Unsigned32	180 hex + node ID	R/W
	2	Transmission type	p8720.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8720.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8720.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8720.4	SDO	Unsigned16	0	R/W
1801		Transmit PDO 2 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8721.0	SDO	Unsigned32	280 hex + node ID	R/W
	2	Transmission type	p8721.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8721.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8721.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8721.4	SDO	Unsigned16	0	R/W
1802		Transmit PDO 3 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8722.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8722.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8722.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8722.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8722.4	SDO	Unsigned16	0	R/W
1803		Transmit PDO 4 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8723.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8723.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8723.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8723.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8723.4	SDO	Unsigned16	0	R/W
1804		Transmit PDO 5 communication parameter					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8724.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8724.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8724.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8724.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8724.4	SDO	Unsigned16	0	R/W

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Trans-mission	Data type	Predefined connection set	Read/ write
1805		**Transmit PDO 6 communication parameter**					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8725.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8725.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8725.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8725.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8725.4	SDO	Unsigned16	0	R/W
1806		**Transmit PDO 7 communication parameter**					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8726.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8726.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8726.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8726.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8726.4	SDO	Unsigned16	0	R/W
1807		**Transmit PDO 8 communication parameter**					
	0	Largest subindex supported		SDO	Unsigned8	5	R
	1	COB ID used by PDO	p8727.0	SDO	Unsigned32	C000 06DF hex	R/W
	2	Transmission type	p8727.1	SDO	Unsigned8	FE hex	R/W
	3	Inhibit time	p8727.2	SDO	Unsigned16	0	R/W
	4	Reserved	p8727.3	SDO	Unsigned8	---	R/W
	5	Event timer	p8727.4	SDO	Unsigned16	0	R/W

Table 6- 40 Configuration objects of transmit PDOs - mapping parameters

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Trans-mission	Data type	Predefined connection set	Read/ write
1A00		Transmit PDO 1 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	1	R
	1	PDO mapping for the first application object to be mapped	p8730.0	SDO	Unsigned32	6041 hex	R/W
	2	PDO mapping for the second application object to be mapped	p8730.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8730.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8730.3	SDO	Unsigned32	0	R/W
1A01		Transmit PDO 2 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	2	R
	1	PDO mapping for the first application object to be mapped	p8731.0	SDO	Unsigned32	6041 hex	R/W
	2	PDO mapping for the second application object to be mapped	p8731.1	SDO	Unsigned32	6044 hex	R/W
	3	PDO mapping for the third application object to be mapped	p8731.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8731.3	SDO	Unsigned32	0	R/W
1A02		Transmit PDO 3 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8732.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8732.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8732.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8732.3	SDO	Unsigned32	0	R/W
1A03		Transmit PDO 4 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8733.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8733.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8733.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8733.3	SDO	Unsigned32	0	R/W

OD Index (hex)	Sub-Index (hex)	Object name	SINAMICS parameter	Trans-mission	Data type	Predefined connection set	Read/ write
1A04		Transmit PDO 5 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8734.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8734.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8734.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8734.3	SDO	Unsigned32	0	R/W
1A05		Transmit PDO 6 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8735.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8735.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8735.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8735.3	SDO	Unsigned32	0	R/W
1A06		Transmit PDO 7 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8736.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8736.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8736.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8736.3	SDO	Unsigned32	0	R/W
1A07		Transmit PDO 8 mapping parameter					
	0	Number of mapped application objects in PDO		SDO	Unsigned8	0	R
	1	PDO mapping for the first application object to be mapped	p8737.0	SDO	Unsigned32	0	R/W
	2	PDO mapping for the second application object to be mapped	p8737.1	SDO	Unsigned32	0	R/W
	3	PDO mapping for the third application object to be mapped	p8737.2	SDO	Unsigned32	0	R/W
	4	PDO mapping for the fourth application object to be mapped	p8737.3	SDO	Unsigned32	0	R/W

6.5.4.3 Free objects

OD index (hex)	Description	Data type per PZD	Default values	Can be written to/ read
5800 to 580F	16 freely-interconnectable receive process data	Integer16	0	R/W
5810 to 581F	16 freely-interconnectable transmit process data	Integer16	0	R

You can interconnect any process data objects using receive/transmit words/double words of the receive and transmit buffer.

Scaling the process data of the free objects:

- 16-bit (word): 4000hex corresponds to 100 %

If the process data is a temperature value, the scaling of the free objects appears as follows:

- 16-bit (word): 4000hex corresponds to 100 °C

6.5.4.4 Objects in drive profile DSP402

Overview

The following table lists the object directory with the index of the individual objects for the drives. The "SINAMICS parameter" column contains the SINAMICS parameter number.

Table 6- 41 Objects in drive profile DSP402

OD index (hex)	Sub-index (hex)	Object name	SINAMICS parameter	Transmission	Data type	Default values	Read/ write
Predefinitions							
67FF		Single device type		SDO			
Common entries in the object dictionary							
6007		Abort connection option code	p8641	SDO	Integer32	0	R/W
6502		Supported drive modes		SDO	Integer32		
6504		Drive manufacturer		SDO	String		
Device control							
6040		Control word	p8890	PDO/SDO	Unsigned16	–	R/W 1)
6041		Status word	r8784	PDO/SDO	Unsigned16	–	R
6060		Modes of operation	p1300	SDO	Integer16	–	R/W
6061		Modes of operation display	p1300	SDO	Integer16	–	R/W
Profile torque mode							
6071		Target torque Set torque	p1513[0]	SDO/PDO	Integer16	–	R/W 1)
6072		Max. torque	p1520/p1521	SDO	0	0	0
6074		Torque demand value Actual torque	r0080	SDO/PDO	Integer16	–	R
Velocity mode							
6042	0	vl target velocity	r0060	SDO/PDO	Integer16	0	R/W
6044	0	vl control effort	r0063	SDO/PDO	Integer16	-	R

1) SDO access is only possible after mapping the objects and the BICO interconnection to display parameters.

6.5.5 Terminology

The following list provides explanations for the most important terms used in this manual in conjunction with the CAN communication bus. You can find more information about CANopen on the website pages. (http://www.canopensolutions.com/index.html)

CANopen

A CiA-defined communication model based on the CAN bus and CAL. To make it easier to use devices produced by different manufacturers on a bus, a subset of CAL functions for automation applications has been defined with the CANopen communication profile CiA DS 301. Other profiles are also defined for certain device types (e.g. drives).

COB (communication object)

On the CAN bus, data is transferred in packages known as communication objects (COB) or CAN messages.

Devices connected to the CAN bus can transmit and receive COBs.

COB-ID (COB identifier)

Each COB can be uniquely identified by means of an identifier, which is part of the COB. CAN specification 2.0A supports up to 2048 COBs, which are identified by means of 11-bit identifiers. In this Commissioning Manual, COB IDs are always specified as hexadecimal values.

A list of COB identifiers, which contains all the COBs that can be accessed via CAN, is available in the object directory for the relevant drive unit.

Heartbeat protocol

The inverter (producer) transmits its communication status cyclically to the master application.

Channel

With the SINAMICS drive line-up, up to 24 receive PDOs can be received.

One channel in the CAN controller is assigned to each activated receive PDO. Transmit PDOs are transmitted via two predefined channels.

Transmit PDOs always use two predefined channels.

NMT (network management)

A part of CAL used for initialization, configuration, and troubleshooting purposes.

Node guarding

The inverter waits for a certain length of time (node lifetime) for telegrams from the master application and permits a specific number (lifetime factor) of failures within a specified time interval (node guard time).

The node lifetime is calculated by multiplying the node guard time by the lifetime factor.

Node ID (node identification)

Uniquely identifies a device in the CANopen network. For this reason, all the devices must have a unique node ID (bus address). The default distribution (standard setting) of the COB IDs is derived from the node ID. In this Commissioning Manual, node IDs are always specified as hexadecimal values.

OD (object directory)

A "database" – or object directory – containing all the objects supported by a drive is defined for each drive unit. The object directory contains:

- Type, description, and serial number of the device
- Name, format, description + index for each object
- Lists of PDOs and SDOs
- The data that is assigned to the PDOs
- The time at which the PDOs are transmitted (SYNC, change in object, etc.)
- The time at which emergency messages are transmitted
- ...

All the drive unit variables are accessed via objects. The SDO and PDO communication services access the object directory of the drive unit.

Process Data Object (PDO)

The PDO is used for real-time access to selected process data.

PDOs either consist of predefined telegrams (predefined connection set) or freely compiled process data (free PDO mapping).

The SDO is used to access all the other variables.

RPDO (receive PDO)

PDO is received by the device (contains the final position, for example).

Profile torque mode

This mode allows you to specify torques.

Service Data Object (SDO)

The SDO provides access to all variables in a CANopen node (in the case of drives: drive and CANopen variables).

An SDO connection is a peer-to-peer connection between an SDO client and a server. The inverter, together with its object directory, is an SDO server.

The SDO is generally used for configuration and parameter assignment. PDOs provide fast, real-time access to selected variables.

SYNC (synchronization)

SYNC is a special telegram that synchronizes the CAN devices with each other. This telegram has a very high priority.

TPDO (transmit PDO)

PDO transmitted by the drive (contains the actual position value, for example).

Variable

All the drive and CANopen functions can be accessed via variables.

Variables can be accessed via SDOs or PDOs.

Operation 7

7.1 Operating states indicated on LEDs

Operating states of the Control Unit displayed by LEDs

The Control Unit is equipped with two LEDs, i.e. RDY (Ready) and BF (Bus Fault) which indicate the inverter status by showing a red or green steady or flashing light.

There are two main inverter states, i.e. power-up and operation.

- Power-up
  During power-up, the inverter progresses through various states which are indicated by red flickering or flashing for brief periods by the Ready and the Bus Fault LEDs. It is not possible to read the inverter's current state from these displays.
- Operation
  The relevant inverter state in operation is indicated by the LED displays described below.

Possible LED states

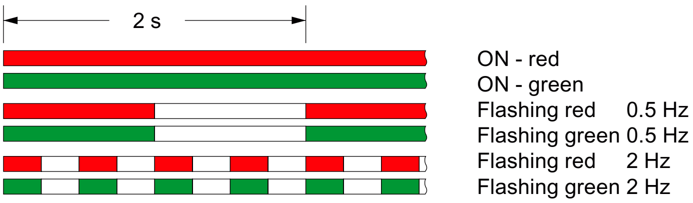


Figure 7-1 Possible LED displays for inverter and communication

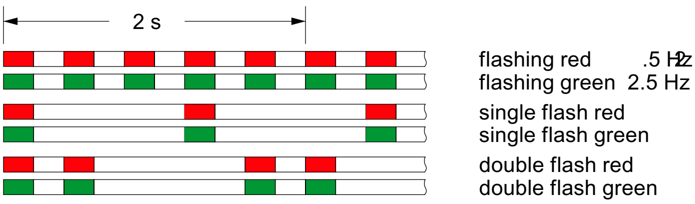


Figure 7-2 Possible status displays for the BF LED with CANopen

Table 7- 1 LED display of inverter states

LED		Explanation
RDY	BF	
ON - green	---	Ready for operation (no active fault)
2 Hz - red	---	General fault
0.5 Hz - green	OFF	Commissioning/reset to factory settings

Table 7- 2 LED status display for communication over PROFIBUS DP

BF LED	Explanation
OFF	Cyclic data exchange (or PROFIBUS not used, p2030 = 0)
0.5 Hz - red	Bus fault - configuration error
2 Hz - red	Bus fault - no data exchange - baud rate search - no connection

Table 7- 3 LED status display for communication via RS485 (USS and Modbus RTU)

BF LED	Explanation
OFF	Receive process data
0.5 Hz - red	Bus active - no process data
2 Hz - red	No bus activity

Table 7- 4 LED status display for communication over CANopen

BF LED	Explanation
ON - red	No bus
ON - green	Bus state "Operational"
2.5 Hz - green	Bus state "Pre-Operational" (flashing)
Single flash - green	Bus state "Stopped"
Single flash - red	Alarm - limit reached
Double flash - red	Error event in control (Error Control Event)

Table 7- 5 LED status display for firmware update

LED		Explanation
RDY	BF	
ON - red	OFF	Firmware update in progress
0.5 Hz - red	0.5 Hz - red	Firmware update ended
2 Hz - red	ON - red	Error during firmware update
2 Hz - red	2 Hz - red	Incompatible firmware / incorrect memory card

Detailed information about faults/errors is displayed by parameters r0947 and r0949.

7.2 ON/OFF commands

In most applications, frequency inverters are implemented as machine components, and commands and setpoints are input either from a higher-level controller or using the command elements of the respective machine.

The following description is therefore limited to the various OFF functions of the frequency inverter.

Other functions, such as data set changeover of command and drive data sets are described in the Function Manual in the section "Data sets".

Overview of the ON/OFF commands

The inverter offers the following ON/OFF commands for switching on and off:

- ON / OFF1 (standard function for On/Off)
- OFF2 (coast down to standstill)
- OFF3 (quick stop)

ON/OFF1

The function ON/OFF1 is a combined ON/OFF command. If the ON signal is reset, OFF1 is activated and the frequency setpoint is coasted down to the value for standstill detection (p1226) by means of the set ramp function (p1121). A timer then starts (p1227) for setpoint and actual value comparison.

The pulses are inhibited

- As soon as the actual value undershoots the value for standstill detection (standard case)
- When the timer for standstill detection has elapsed and the actual value is greater than the value for standstill detection (for error case, see OFF2)

Note

If the inverter loses the motor orientation, it will not be possible to switch off the motor with OFF1 or OFF3. In this case, an OFF2 command can be introduced or the pulses can be deactivated with p0054.3.

OFF2

With this OFF command, the pulses are immediately inhibited and the drive responds according to its load; i.e. it can be accelerated (generating load) as well as braked (motoring load).

The OFF2 command is linked to a switching on inhibited, i.e. the inverter cannot be switched on while an OFF2 is active (r0054.01 = 0).

OFF3

In the case of OFF3, the frequency setpoint is coasted down to the value for standstill detection (p1226) by means of the set ramp function (p1135). A timer then starts (p1227) for setpoint and actual value comparison.

The pulses are inhibited

- as soon as the actual value undershoots the value for standstill detection (standard case)
- When the timer for standstill detection has elapsed and the actual value is greater than the value for standstill detection (for error case, see OFF2)

Like OFF2, the OFF3 command is linked to a switching on inhibited, i.e. the inverter cannot be switched on while an OFF3 is active (r0054.02 = 0).

Note

A positive edge must be applied via ON/OFF1 to switch on after an OFF2 or OFF3. The switching on inhibited must also be canceled (r0054.01 = r0054.02 = 1).

7.3 Power-up behavior of the inverter

Overview of the power-up behavior of the inverter

For the power-up behavior of the inverter, the following situations must be considered:

- Power-up behavior with memory card card (memory card is inserted when switching on)
- Power-up behavior without memory card (memory card is not inserted when switching on). A memory card inserted after switching on is ignored. However, it should be noted that after it has been switched off and switched on again, the inverter behaves like an inverter with memory card.

Power-up behavior of the inverter with memory card

After the inverter is switched on, the Control Unit powers up and retrieves the parameter set for the work memory either from the EEPROM of the CU or from the memory card.

If the memory card contains a standard parameter set (PS000), this will always be used. It is written to the EEPROM and downloaded into the work memory.

If the memory card does not contain a standard parameter set, the standard parameter set from the EEPROM is loaded to the work memory and also stored as a standard parameter set on the memory card.

Initial commissioning must be carried out when the inverter is switched on for the first time.

Power-up behavior of the inverter without memory card

After the inverter is switched on, the Control Unit powers up and retrieves the parameter set for the work memory from the EEPROM of the CU.

Initial commissioning must be carried out when the inverter is switched on for the first time.

CAUTION

Inserting a memory card when the inverter is switched on.

If a memory card is inserted during inverter operation, this does not affect the operational behavior and the data on the memory card remains unchanged.

If this memory card contains a standard parameter set (PS000) with the correct parameter settings, the inverter accepts the memory card data when it is next switched on - as described above in subsection "Power-up behavior of the inverter with memory card".

If the memory card contains a standard parameter set with incorrect data (for example, from another application) the data from the work memory must be written with p0971 =1 to the memory card before the inverter is switched off, otherwise the next time the inverter is switched on, it would load the incorrect data from the memory card to the work memory of the Control Unit.

Missing or incompatible standard parameter set

If the Control Unit fails to power up successfully three times in succession because the standard parameter set is missing or incompatible, the inverter is reset to its factory settings and fault F01018 is generated.

7.4 Saving several parameter sets

Overview

Multiple parameter sets can be created with parameters p0971 and p0802 to p0804. These can then be saved to the inverter's EEPROM or to a memory card.

CAUTION
The memory card must not be removed while parameter sets are being copied or saved.

Note

Saving or downloading a parameter set via p0971 or p0970 is indicated by the LED and also displayed using parameter r3996 (1 … 100). No data can be input while these operations are in progress. When the process is complete, r3996 is set to 0 again and further inputs are possible.

Existing parameter sets of the same name are overwritten without further notice.

A complete parameter set always consists of three or (with CANopen) five files which are named according to the following convention:

AAYYYXXX.ACX, the short name is PSYYY

The file name elements and their variants are explained below:

AA Internal drive designations with the following forms:

- PS: for all Control Units
- CA: for CANopen modules only
- CC: for CANopen modules only

YYY Serial number for parameter set (0 ... 9), defined in p0802, or p0803 / p0971

XXX Internal drive designations with the following forms:

- 000: Internal drive numbering for parameter file 1
- 001: Internal drive numbering for parameter file 2
- 099: Internal drive numbering for parameter file 3

ACX File type (identical for all files)

Examples of parameter sets

Parameter set 11 for Profibus Control Unit, short **PS011**

- PS011000.acx
- PS011001.acx
- PS011099.acx

Parameter set 12 for CANopen Control Unit, short **PS012**

- PS012000.acx
- PS012001.acx
- PS012099.acx
- CA012001.acx
- CC012001.acx

PS000 (standard parameter set), the parameter set which is written during power-up to the work memory of the inverter from either the EEPROM or the memory card. It therefore consists of PS000000.acx, PS000001.acx, PS000099.acx (CA000001, CC000001.acx).

7.4.1 Saving parameter changes using p0014

Description

Depending on the setting in p0014, parameter changes made in the work memory (RAM) are also stored in the EEPROM.

Possible settings:

- p0014 = 0:

  Parameter changes only in the RAM (volatile - data is lost when the power fails or OFF/ON). When switching over from p0014 = 0 -> 1, data is written from the RAM into the PS000 in the EEPROM.

- p0014 = 1:

  Parameter changes in the RAM and ROM (parameter changes are buffered in the ROM in a non-volatile fashion and are kept even for a power failure or OFF/ON). When switching over p0014 = 1 -> 0, the buffered data are written into PS000 in the EEPROM.

- p0014 = 2:

  Deletes parameter changes that have still not been saved in the ROM and then sets p0014 to 0.

7.4.2 Saving parameter sets using p0971

Description

Parameter p0971 is used for selectively saving a parameter set - for EEPROM and memory card. A setting of p0971 ≠ 0 triggers a work memory data save to the selected parameter set. When the data have been saved, p0971 is reset to 0.

Possible settings:

- p0971 = 0: Deactivated (factory setting)
- p0971 = 1: Save retentively under PS000 (standard parameter set for power-up)
- p0971 = 10: Save retentively as PS010
- p0971 = 11: Save retentively as PS011
- p0971 = 12: Save retentively as PS012

After saving, p0971 is reset to 0.

Tip

Creating parameter sets on the memory card

Parameter sets PS000, PS010, PS011 and PS012 can be saved to the EEPROM and memory card with settings p0971 = 1, 10, 11 and 12. All other parameter sets can only be set up by copying using parameters p0802 … p0804.

7.4.3 Saving and copying parameter sets using p0802 to p0804

Description

Parameters p0802, p0803 and p0804 can be used to copy parameter sets between the EEPROM on the inverter and the memory card.

Note

When PS000 is copied using p0802 to p0804, the changes in the work memory are ignored. In order to save the changes in the work memory to PS000, you must set p0971 = 1 before you copy the parameter set.

Copying a parameter set from EEPROM to memory card

You can save up to 101 parameter sets on the memory card.

1. Use p0803 to select the parameter set on the EEPROM that you wish to copy to the memory card.
2. Use p0802 to define the name under which the parameter set is to be written to the memory card.
3. Start the copy operation by selecting p0804 = 2.
   p0804 is again reset to 0 when the parameter set has been copied.

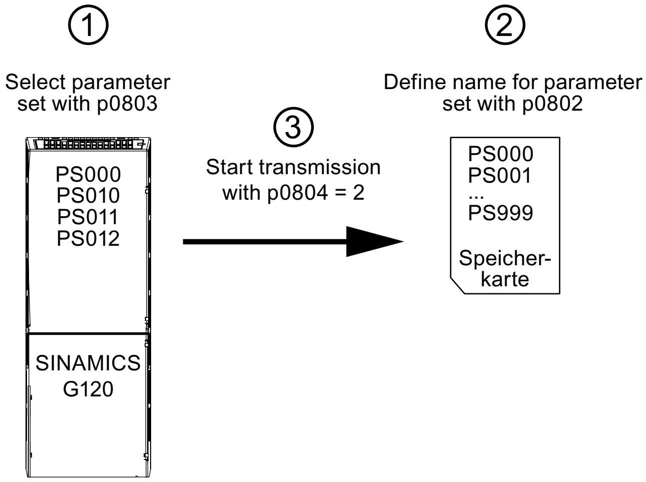


Figure 7-3 Copying parameters from EEPROM to memory card

Copying parameter sets from the memory card to the EEPROM

You can save four parameter sets to the EEPROM on the CU.

1. Use p0802 to select the parameter set on the memory card that you wish to copy to the EEPROM.
2. Use p0803 to define the name under which the parameter set is to be written to the EEPROM.
3. Start the copy operation by selecting p0804 = 1.
   p0804 is again reset to 0 when the parameter set has been copied.

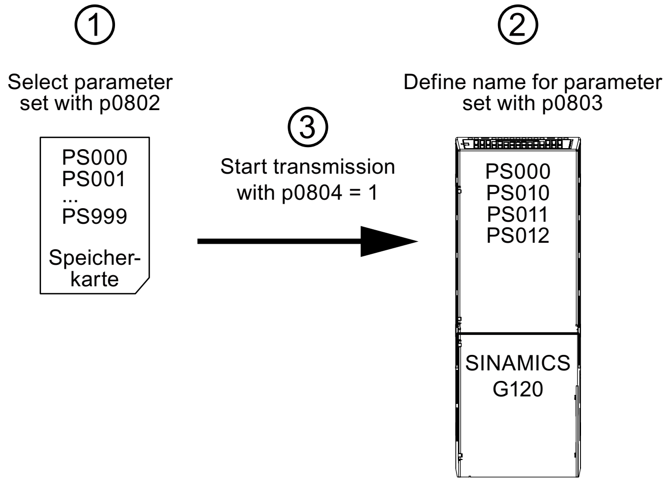


Figure 7-4 Copying parameters from the memory card to the EEPROM

Example: Copy parameter set PS012 from EEPROM to memory card as PS066

Check whether p0010 is set to 0 or 30.

Use p0803 to select the parameter set to be copied from the EEPROM, in this case p0803 = 12.

Use p0802 to select the name under which the parameter set is written to the memory card, in this case p0802 = 66.

Use p0804 to start copying from the EEPROM -> memory card - in this case p0804 = 2.

Parameter set cannot be saved

NOTICE
In the event that the parameter set cannot be written using p0804, the parameter is not set to 0 but specifies the following information: • p0804 = 1001: Cannot open file on memory card (parameter set does not exist on memory card or memory card has incorrect data format) • p0804 = 1002: Cannot open file in the device memory (parameter set does not exist in the work memory of the inverter) • p0804 = 1003: Memory card not found (no memory card has been inserted) • p0804 = 1100: File transfer not possible (memory card has incorrect data format)

Loading a parameter set from the EEPROM or memory card to the work memory

A parameter set which is stored, for example, as PS012 in the EEPROM, can be loaded to the work memory on the inverter with the following commands:

p0010 = 30 (change parameter settings)

p0970 = 12 (load PS012 to the work memory) In this case, the standard parameter set PS000 in the EEPROM or memory card is not overwritten, i.e. the old standard parameter set PS000 is loaded to the work memory after OFF/ON.

In order to write the data set of the work memory, also as standard parameter set into the EEPROM or the memory card, before switching-off, p0971 must be set to 1.

8 Alarms, faults and system messages

Description

The states detected by the inverter are displayed with messages. The messages are categorized into faults and alarms.

For detailed descriptions of the faults and alarms, see Chapter 3 of the List Manual. You will also find the following function diagrams there:

- Fault buffer
- Alarm buffer
- Fault trigger
- Fault configuration

8.1 Alarms and faults

Alarms

- are identified by Axxxxx
- have no direct influence on the inverter
- disappear again when the cause has been remedied and do not need to be acknowledged
- Status display in status word 1.7
- Three external alarms can be output by means of p2112, p2116 and p2117

The numbers of alarms which have been activated are stored in a total of 64 indices in parameter r2110. p2111 displays the number of alarms and r2132 shows the currently active alarm (0 = no alarm is currently active). For details, please refer to the Parameter List.

Faults

- are identified by Fxxxxx.
- can lead to a fault reaction.
- must be acknowledged once the cause has been remedied.
- Status via Control Unit and LED RDY.
- Status display status word 1.3

The numbers of faults which have been activated are stored in a total of 64 indices in parameter r0947. The fault time is displayed in r2130 (system runtime in days) and in r0948 (in milliseconds on the day the fault occurred). r0949 displays the fault value of the currently active fault (0 = no fault) which is important for internal fault diagnostics. The number of faults is stored in p0952. For details, please refer to the Parameter List.

Note

The inverter does not enter the "ready for switching on" state until all active faults have been eliminated and the faults acknowledged.

General fault acknowledgement

Faults can generally be acknowledged as follows:

- By switching the inverter off and on again

  (switch off the main power supply and the external 24 V supply for the Control Unit and switch it on again).
- At the IOP
- Via DI 2
- Set bit 7 in control word 1 (r0054) for Control Units with a fieldbus interface

Special fault acknowledgement

The following faults can only be acknowledged by switching off and on again:

- F01000 Internal software error
- F01004 Internal software error
- F01015 Internal software error
- F01016 Firmware changed
- F01018 Power-up aborted more than once
- F01040 Back up parameters and perform a POWER ON
- F01044 CU: Error in description data
- F01105 CU: Insufficient memory
- F01250 CU: Error in CU EEPROM read-only data
- F01512 BICO: No normalization
- F01662 Internal communication error
- F30052 Error in EEPROM data
- F30662 Internal communication error
- F30664 Fault during power-up
- F30850 Power unit: Internal software error

Motor failure without a fault code or alarm message

if the motor does not start up after the ON command has been entered:

- Check that p0010 = 0.
- Check the inverter state in r0052
- Check the command and setpoint source (p0700 and p1000).
- Check that the motor data and inverter data "Load range" and "Voltage" match.

Note

For fault rectification purposes, refer to the "Installation checklist" in the "Installation" section of this manual.

8.2 Change message type

Change message type

The inverter allows you to change the message type for specific alarms and faults, i.e. you can change a message defined as a fault into an alarm and vice versa. Messages of this type are identified in the following way:

- A***** (F) Alarm can be changed to a fault
- F***** (A) Fault can be changed to an alarm.

Procedure:

Enter message number in p2118[n] (n = 0 … 19)

Enter message type in the same index in 2119[n], possible reactions
1: Fault
2: Alarm
3: No message

Example:

- p2118[5] = 01016
- P2119[5] = 1

This has changed alarm message "Firmware changed" to a fault.

8.3 Diagnostics display

Overview

The G120 inverter provides the following diagnostics displays:

- LEDs on the Control Unit

  For a detailed overview of the LED states, see the section "Operating states indicated by LEDs (Page 141)" in this manual.
- Alarm and fault numbers

  The alarm and fault numbers are used for fault rectification with the IOP and STARTER. For further details about fault rectification using STARTER, see the relevant online help or refer to the List Manual.
- Diagnostic parameters

  Fault rectification with a higher-level control system

8.4 Fault rectification with the IOP

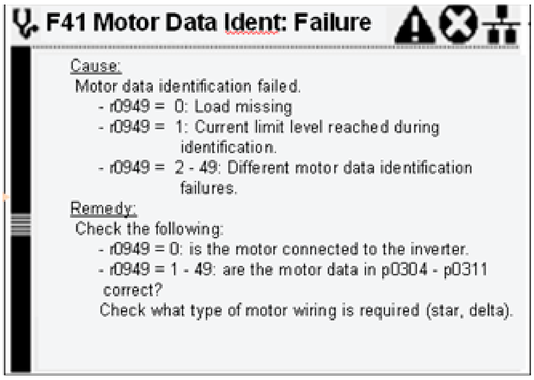
Diagnostics using alarms and faults

For the diagnosis of alarms and faults, the IOP must be connected to the Control Unit (directly on the CU, with a handheld kit or door mounting set). When an alarm or fault occurs, the IOP displays the corresponding alarm or fault number.

In the case of an alarm, the alarm message is displayed and the inverter continues to operate.

In the case of a fault, the fault code is displayed and the inverter pulses are inhibited.

Example of a fault code:



8.5 Fault rectification with the control system

Diagnostics using diagnostic parameters

The following diagnostic parameters are set by a higher-level control system. This function is available with Control Units that communicate over PROFIBUS.

Standard diagnostics

Parameter r2054 supplies the following diagnostic information via the PROFIBUS-DP interface:

r2054 = 0	Status of PROFIBUS DP:
r2054 = 1	No connection (search for baud rate)
r2054 = 2	Connection OK (baud rate found)
r2054 = 3	Cyclic connection with master (Data Exchange)
r2054 = 4	Cyclic data OK

Parameter accessing error

If parameter accessing errors occur, the corresponding message is transferred in DS47.

9 Service and maintenance

9.1 Service and support information

I DT Technical Support

Throughout the world, three main centers provide a 24-hour service of technical support.

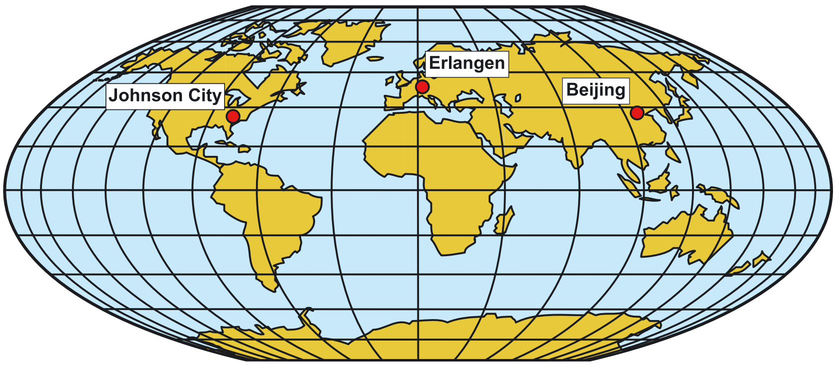


Figure 9-1 I DT Service and Support worldwide

Service and support online

If you have a servicing requirement, your first point of contact is the regional sales, marketing or service organization responsible for your country.

In the case of technical support, it is recommended that you submit a support request via the following link: (http://www.siemens.com/automation/support-request)

Europe / Africa (Erlangen)

Internet support request: Technical support

Phone: +49 (180) 5050 222	0.14 €/min. from German landlines (mobile call charges may differ)
Fax: +49 (180) 5050 223	

e-mail: (support.automation@siemens.com)

Americas (Johnson City)

Internet support request: Technical support

Phone: +1 (423) 262 2552
Fax: +1 (423) 262 2589
e-mail: (support.usa.automation@siemens.com)

Asia/Pacific (Beijing)

Internet support request: Technical support

Phone: +86 (1064) 757 575
Fax: +86 (1064) 747 474
e-mail: (support.asia.automation@siemens.com)

Contact address

If you have any questions or problems regarding these operating instructions, please contact the following Siemens address:

Siemens AG
Automation & Drives
I DT SD SPA PM4
PO Box 3269
D-91050 Erlangen

e-mail: (documentation.standard.drives@siemens.com)

Regional contacts

For questions regarding service, prices and conditions for technical support, please contact your local Siemens partner.

9.2 Replacing components

Overview

The SINAMICS G120 frequency inverters are designed to allow replacement of the Power Module or the Control Unit where necessary without recommissioning the drive provided the boundary conditions described below are adhered to.

Reference potential

As soon as the Control Unit is snapped onto the power unit, it is connected to the reference potential of the Power Module.

⚠ WARNING
As soon as the Control Unit is detached from the Power Module, the connection with the module's potential is interrupted. For this reason, all 230 V cables on the Control Unit (DO0 and DO1) must be disconnected from the supply voltage before the Control Unit is detached from the Power Module.

Further requirements

Where the following requirements are fulfilled, inverter components (Control Units and Power Modules) can be replaced without the need to recommission the drive:

- Inverter is operating with a memory card
- Components are always replaced by identical parts (identical MLFB and firmware version)
- Depending on whether parameter changes in the work memory must be transferred to buffer memory, either
  - parameter p0014 = 1, changes in work memory are transferred to buffer
  - parameter p0014 = 0, changes in work memory are not transferred to buffer

⚠ DANGER
If these conditions are not fulfilled when components are replaced, the customer must bear full responsibility for the accuracy of the parameter settings. In this instance, we would advise you to recommission the drive.

Replacement constraints of Power Modules

Power Modules can also be replaced subject to compliance with the following conditions:

- Same type (e.g. PM240), different power ->
  Following replacement, inverter is ready for operation with some possible restrictions in performance Adjust the motor and performance data if necessary.
- **If a Power Module is replaced by a Power Module of a different type (e.g. PM240 by PM250), "Reset to factory settings" is performed automatically and initial commissioning will need to be carried out.**

Constraints to replacing Control Units with memory card

If you replace a CU230P-2 by a different CU type (e.g. CU230P-2 DP by CU230P-2 HVAC), the inverter will display alarm A01028 "Configuration error". You can acknowledge this alarm with p0971 = 1 and thus transfer the settings of the new Control Unit as the standard parameter set (PS000). In this case, check the parameter settings and recommission the drive.

Constraints to replacing Control Units without memory card

If you replace a CU230P-2 by a different CU type (e.g. CU230P-2 DP by CU230P-2 HVAC), the inverter does not signal an alarm and the standard parameter set of the new CU is written to the work memory. In this case, check the parameter settings and commission the drive if necessary.

9.2.1 Replacing the Power Module

Procedure for replacing a Power Module

To replace a Power Module, proceed as follows:

1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply to the Control Unit.
2. After switching off the line voltage, wait 5 minutes until the device has discharged itself.
3. Unplug the network connections
4. Remove the Control Unit from the power unit
5. Replace the old Power Module with the new one, following the instructions in the Hardware Installation Manual.
6. Snap the Control Unit onto the new Power Module
7. Connect the new Power Module correctly
8. Switch on the supply voltage
9. If necessary, adjust the motor and performance data or commission the drive.

9.2.2 Replacing the Control Unit

Procedure for replacing a Control Unit with memory card with valid parameter set

1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply to the Control Unit.
2. Remove the control cables of the Control Unit
3. Remove the defective CU from the Power Module
4. Insert a new CU on the Power Module.
5. Remove the memory card from the old Control Unit and insert this in the new Control Unit.
6. Reconnect the control cables of the Control Unit.
7. Connect up the line voltage again.
8. The inverter powers up, PS000 is written from the memory card into the EEPROM and into the work memory. Data possibly available in the buffer of the Control Unit is deleted. The inverter goes into the "ready-to-switch-on" state.
9. For Control Units of the same type, you can switched-on the inverter without any additional commissioning.
   Alarm A01028 will be output for Control Units of different types. This alarm indicates that the parameter settings of PS000 are not compatible with the CU. In this case, acknowledge the fault using p0971 = 1 and re-commission the drive.

Procedure for replacing a Control Unit without a memory card

1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply to the Control Unit.
2. Remove the control cables of the Control Unit
3. Remove the defective CU from the Power Module
4. Insert a new CU on the Power Module.
5. Reconnect the control cables of the Control Unit.
6. Connect up the line voltage again.
7. The inverter powers up (PS000 is written from the EEPROM to the work memory) and switches over to the "ready for switching on" state.
8. If you are exchanging identical Control Units and PS000 contains the correct settings (e.g. with Series commissioning (Page 63)), you can switch the inverter on. In all other cases, we would recommend that you commission the drive.

10 Technical data

Characteristic	Data
Operating voltage	24 V DC from the Power Module or an external 24 V DC supply (20.4 V to 28.8 V, 0.5 A) over control terminals 31 and 32
Open-loop/closed-loop control procedure	V/f control, output frequency between 0 Hz and 650 Hz: • Linear V/f control, • Linear V/f control with FCC, • Linear V/f control with ECO mode, • Quadratic V/f control, • Multipoint V/f control, • V/f control for applications in the textile industry, • V/f control with FCC for applications in the textile industry, • V/f control with independent voltage setpoint, Vector control, output frequency between 0 Hz and 200 Hz: • Speed control without encoder • Torque control without encoder
Fixed frequencies	16, parameterizable
Skip frequencies	4, parameterizable
Setpoint resolution	0.01 Hz digital; 0.01 Hz serial; 10 bit analog (motorized potentiometer 0.1 Hz [0.1% in PID mode])
Digital inputs (dependent on the CU type)	Up to 6 digital inputs, isolated; SIMATIC-compatible, Low < 5 V, High > 10 V, maximum input voltage 30 V Switchable via terminals • PNP: Bridge terminal 69 with terminal 9 • NPN: Bridge terminal 69 with terminal 28
Analog inputs	• 2 switchable (current/voltage), both configurable as additional digital inputs. 0 V to 10 V, 0 mA to 20 mA and -10 V to +10 V (AI0) 0 V to 10 V and 0 mA to 20 mA (AI1) • 1 switchable (current / Ni1000) (AI2) 0/4 mA … 20 mA NI1000: - 50°C … 150 °C PT1000: - 50 °C … 250 C • 1 fixed (NI1000, PT1000) (AI3) NI1000: - 50°C … 150 °C PT1000: - 50 °C … 250 C
Relay outputs	3, parameterizable • DO1: 30 V DC / 0 A to 5 A (resistive load), 250 V AC / 2 A • DO2: 30 V DC / 0 A to 0,5 A (resistive load) • DO3: 30 V DC / 0 A to 5 A (resistive load), 250 V AC / 2 A
Analog outputs	2, parameterizable: 0 mA … 20 mA, 0 V … 10 V (AO0, AO1)
Dimensions (WxHxD)	73 mm x 199 mm x 65.5 mm
Weight	0.61 kg
Operating temperature	- 10°C … +60 °C (possible restrictions as a result of the Power Module should be observed)
Storage temperature	- 40°C … +70 °C
Humidity	< 95 % RH, non-condensing

Note

As far as the environmental requirements are concerned, possible restrictions resulting from the permissible values for the Power Module must be taken into account.

The control terminals on the Control Unit are galvanically isolated from the supply voltage (PELV).

Accessories

11

Intelligent Operator Panel (IOP)

The IOP is a parameterization tool that is connected directly to the SINAMICS G120 inverter via the option port. It can be used to upload and download parameter sets. A detailed description can be found in the accompanying description on the Manual Collection.

Memory card

The memory card is used for backing up parameter settings, series commissioning and for quick commissioning when replacing a CU.

Up to 101 parameter sets can be saved on the memory card. Details are provided in Section Saving and copying parameter sets using p0802 to p0804 (Page 149)

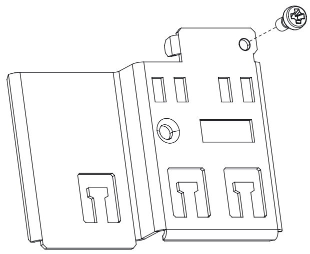
The inverter can be operated with or without a memory card inserted.

Note

For software version 4.1, we recommend the use of the memory card tested by us, order number 6SL3254-0AM00-0AA0. If you use a different memory card, you must ensure that it is formatted as FAT16 and that it supports the SPI bus mode.

Screening kit for the Control Unit

The Screening Kit allows easy bonding of the control cable shields to the potential of the Control Unit.



Shield connection kit

Note

The shield connection kit of the Control Unit must not be used to connect the protective conductor. All of the protective conductors must be connected through the Power Module.

PC inverter connection kit

The PC connection kit comprises a USB cable (3 m), which is used to connect the inverter to the PC, and a DVD to install the "STARTER" commissioning software on a PC. The STARTER software offers the possibility of

- parameterizing (commissioning, optimization),
- monitoring (diagnostics)
- controlling (e.g. for test purposes during the commissioning) the inverter

Appendix

A

A.1 EMC classification in accordance with product standard EN618000-3

Environment 1

An environment that contains residential areas and facilities which are directly connected to a public low-voltage supply system without an intermediate transformer.

Note

Example: Residential buildings, apartments, businesses or offices within a residential area.

Environment 2

An environment that contains trading areas and facilities which are not directly connected to a public low-voltage supply system.

Note

Example: Constructed trading units and other utility areas that are supplied through a dedicated transformer.

Class C1

Power Drive System (PDS) with a rated voltage less than 1000 V for use in the first environment.

Class C2

Power Drive System (PDS) with a rated voltage of less than 1000 V which may not be equipped with a plug-in connection nor is designed for (simple) transport when used in the first environment. System must be installed and commissioned by a properly qualified person.

Note

A qualified person is a person or organization that is capable of installing and/or commissioning a Power Drive System (PDS) including its EMC aspects.

Class C3

Power Drive System (PDS) with a rated voltage less than 1000 V for use in the second environment.

A.2 Overall behavior as regards EMC

The SINAMICS G120 drives have been tested in accordance with the EMC product standard EN 61800-3:2004.

Emitted electromagnetic interference

The SINAMICS G120 drives were tested in accordance with the requirements regarding emitted interference for environments of Class C2 (residential areas).

Table A- 1 Conductor-related and radiated emitted interference

EMC impact	Standard	Step
Conducted emissions	EN 55011	Class A
Radiated emissions	EN 55011	Class A

Note

To comply with this behavior, the factory-set switching frequency must not be exceeded.

A series of external PM240 filter options are available for compliance with conducted emissions in accordance with EN 55011 Class B.

Whether emissions are complied with in accordance with EN 55011 Class B depends largely on whether the drive is correctly installed in a metal housing. The limit values are not complied with when the drive is not built into a housing and is not installed taking the electromagnetic compatibility into account.

Harmonic currents

The emissions of the harmonic currents of the SINAMICS G120 drives are as follows:

Table A- 2 Harmonic currents

Rating	Typical harmonic current (% of the rated input current)							
	5.	7.	11.	13.	17.	19.	23.	25.
FSA to FSF (400 V, 370 W to 110 kW)	73	52	25	23	22	15	12	10

Note

Connection approval is required for the public low-voltage supply network for units which are installed in an environment of Class C2 (residential). Please contact your local supply system operator in this case.

No connection approval is required for units which are installed in an environment of Class C3 (commercial).

EMC immunity

The SINAMICS G120 drives were tested in accordance with the interference immunity requirements for environments of Class C3 (commercial).

Table A- 3 EMC immunity

EMC impact	Standard	Step	Performance criterion
Electrostatic discharge (ESD)	EN 61000-4-2	4 kV discharge by contact	A
		8 kV discharge in air	
Electromagnetic high frequency field	EN 61000-4-3	80 MHz to 1000 MHz 10 V/m	A
Amplitude modulated		80% AM at 1 kHz	
Transient overvoltages	EN 61000-4-4	2 kV at 5 kHz	A
Voltage surge	EN 61000-4-5	1 kV inverse mode (L-L)	A
1.2/50 µs		2 kV common mode (L-E)	
Conducted	EN 61000-4-6	0.15 MHz to 80 MHz 10 V/rms	A
High frequency common mode		80% AM at 1 kHz	
System interruptions and voltage dips	EN 61000-4-11	100% voltage dip for 3 ms	A
		30% voltage dip for 10 ms	B
		60% voltage dip for 100 ms	C
		95% voltage dip for 5000 ms	D
Voltage distortion	EN 61000-2-4 Class 3	10% THD	A
Voltage asymmetry	EN 61000-2-4 Class 3	3% inverse reactance	A
Frequency fluctuation	EN 61000-2-4 Class 3	Rated value 50 Hz or 60 Hz (± 4%)	A
Commutation notches	EN 60146-1-1 Class B	Depth = 40%	A
		Area = 250% x degrees	

Note

The interference immunity requirements apply equally for devices with and without filters.

A.3 Standards and Directives

The devices fulfill the requirements for the standards and directives listed below.

European Low-Voltage Directive

The SINAMICS G120 product series meets the requirements of Low-Voltage Directive 2006/95/EU. The devices are certified as complying with the following standards:

EN 61800-5-1 - Semiconductor power converters - General requirements and line-commutated converters

EN 60204-1 - Safety of machinery - Electrical equipment of machines

European Machinery Directive

The SINAMICS G120 inverter series does not fall within the area covered by the machinery directive. However, the use of the products in a typical machine application has been fully assessed for compliance with the main regulations in this directive concerning health and safety. A declaration regarding the general acceptance is available upon request.

European EMC Directive

When installed in accordance with the recommendations specified in this manual, the SINAMICS G120 complies with all regulations of the EMC directive according to the definition provided by EN 61800-3 "EMC Product Standard for Power Drive Systems".

Underwriters Laboratories

The device is listed by UL and CUL for power conversion in an environment with the pollution degree 2.

Note: UL certification is currently being processed.

ISO 9001

Siemens AG uses a quality management system that meets the requirements of ISO 9001.

Certificates can be downloaded from the Internet under the following link:
http://support.automation.siemens.com/WW/view/en/22339653/134200
(http://support.automation.siemens.com/WW/view/en/22339653/134200)

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